

ASHG ANNUAL MEETING MONTRÉAL
OCTOBER 20-24, 2026

Fine-mapping and colocalization with structural variants

Practical workflow to find causal SVs using GWAS and multi-omic summary statistics

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Stanford University

Financial Disclosure

Tanner has financial relationships to report within the past 24 months with ACCME defined ineligible companies.

Presenter	Name of Company	Nature of Relationship
Tanner Jensen	MyOme	Paid Contractor
Yann Le Guen	Nothing to disclose	.

Yann does not have any relationships to report within the last 24 months with ACCME defined ineligible companies.

Workshop outline

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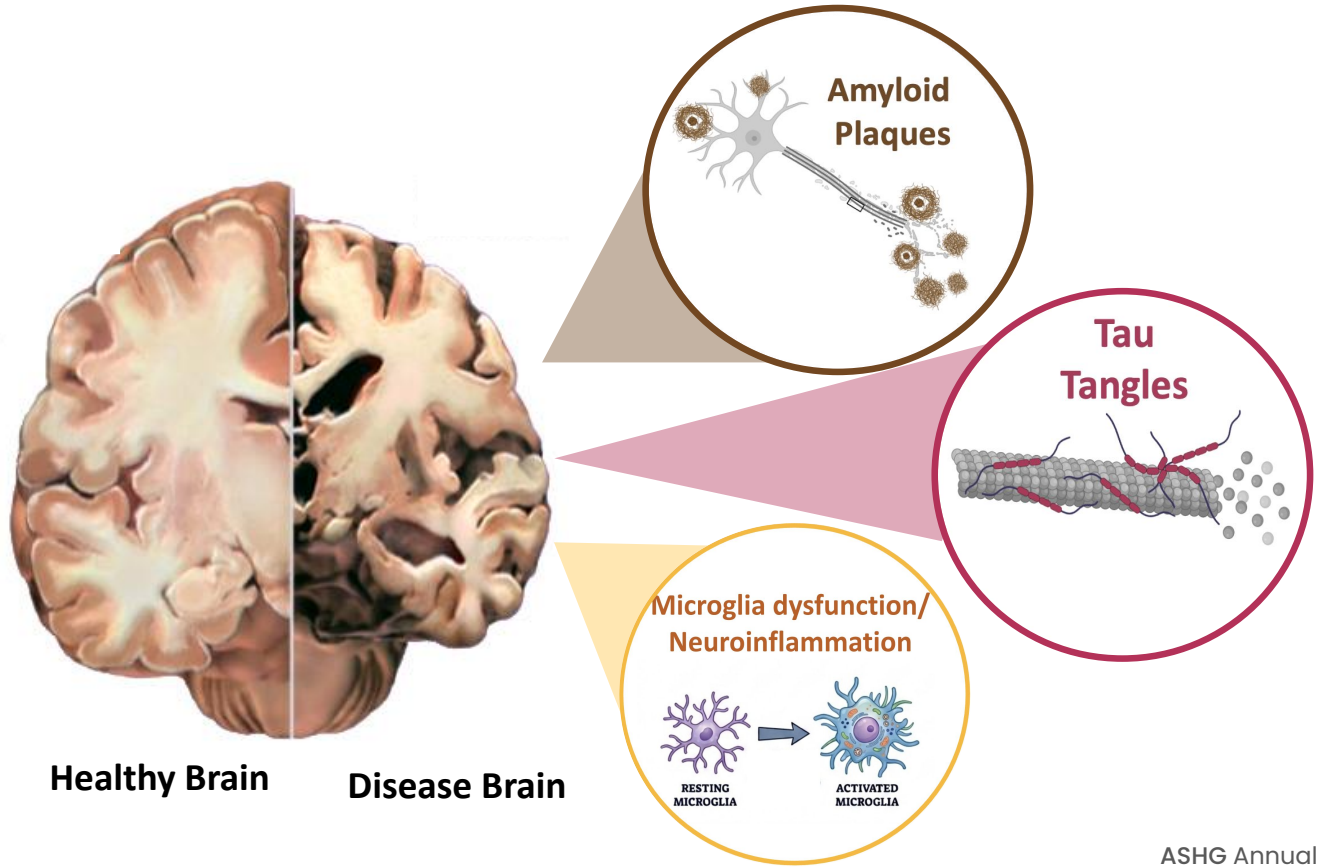
1. Background and Key Concepts
2. ADRC/SAMS long-read and multi-omics resource
3. Interactive workbook: Google Co-lab code to finemap and colocalize Alzheimer's and Parkinson's risk SVs
   https://github.com/Junkkkk/sv-finemapping-coloc-tutorial
4. Rare structural variant prioritization
5. Conclusion and Takeaways

01

Background and Key Concepts

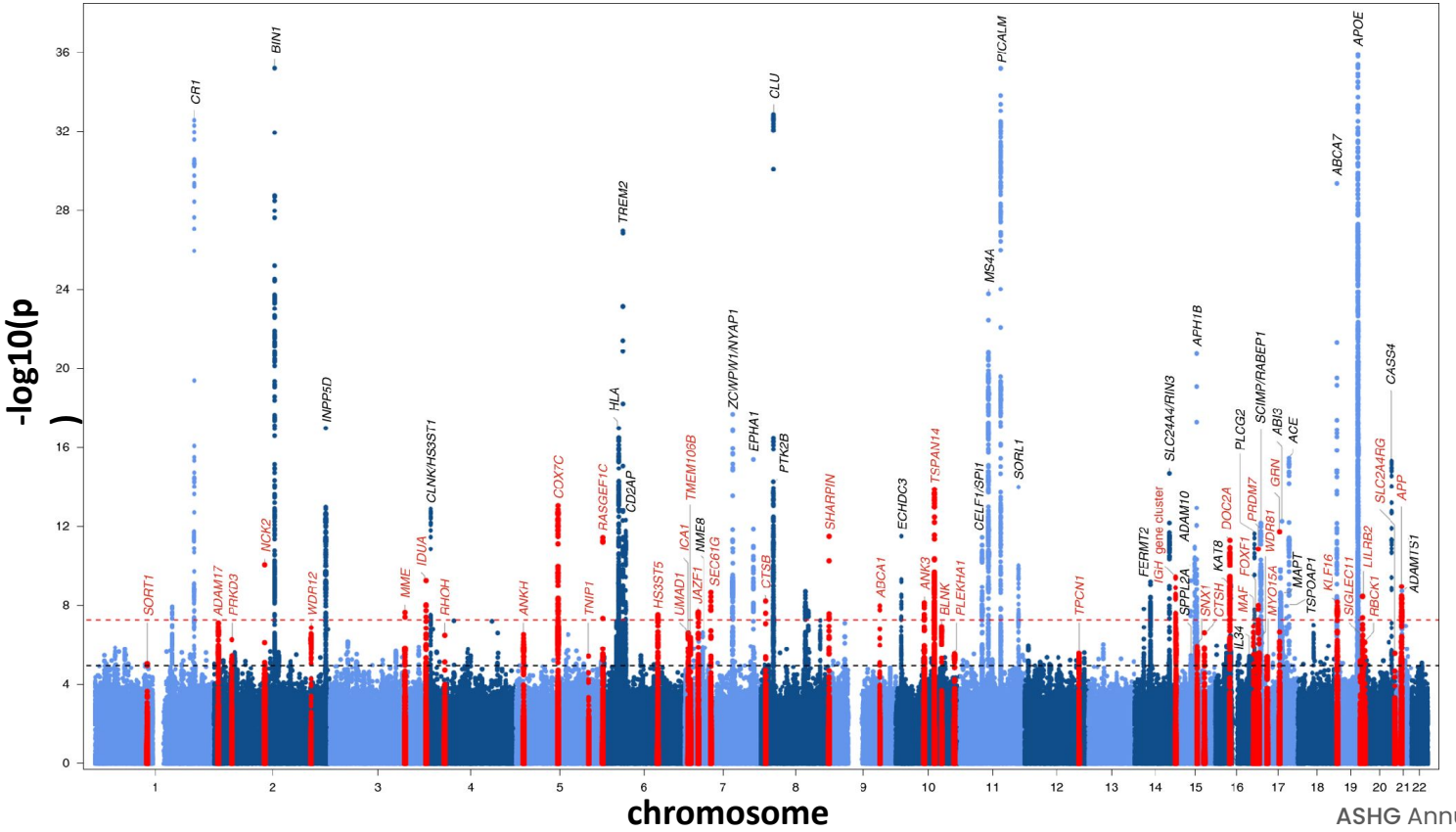


Alzheimer's disease is leading cause of dementia



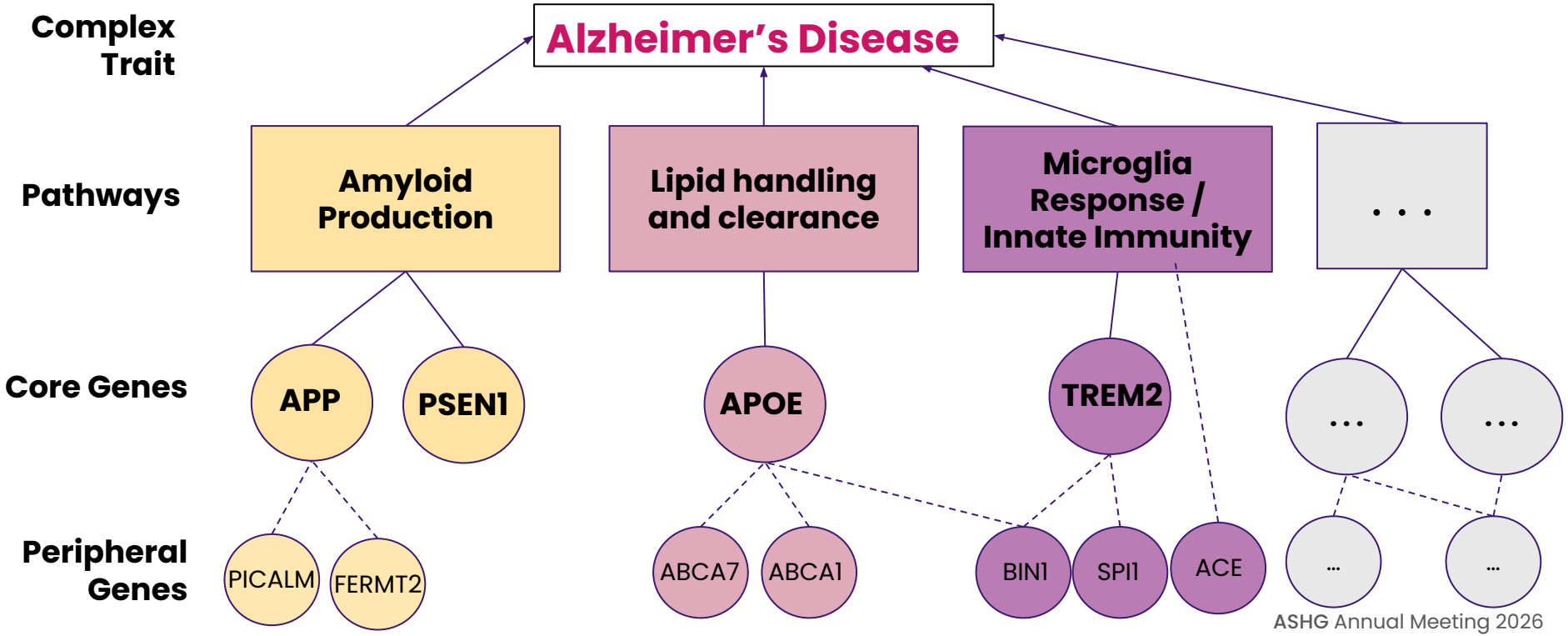
GWAS useful tool to detect genetic risk factors

Bellenguez et al. 2022 (N=789K, 111K AD/proxy cases 678K controls)

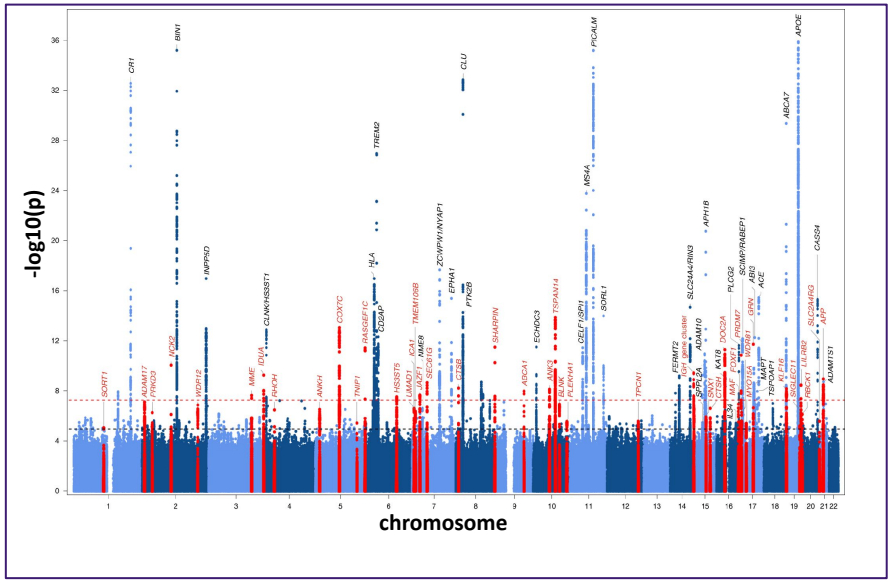


GWAS helps us identify essential pathways and genes underlying complex traits

Illustrative omnigenic model

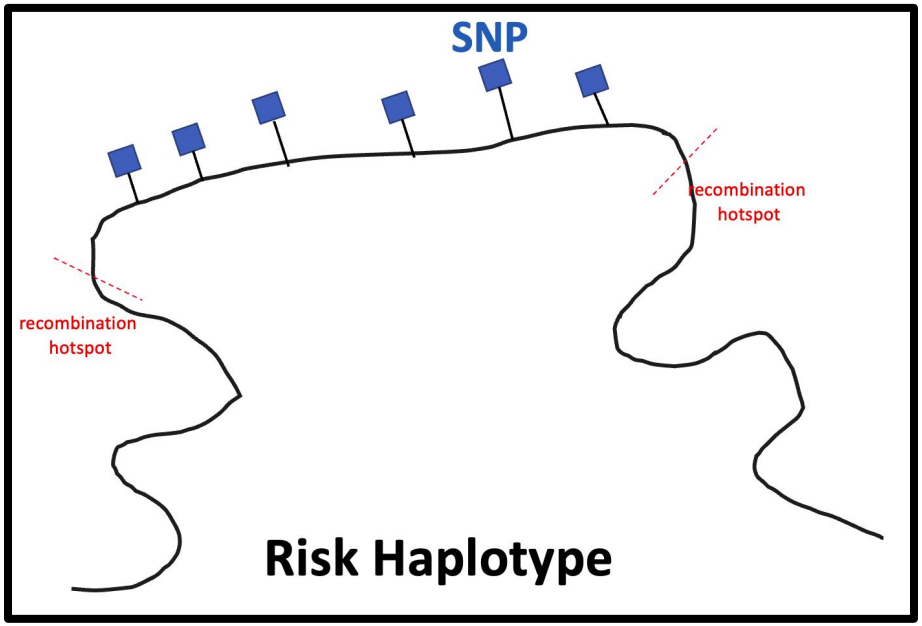


Despite successes, GWAS has several limitations

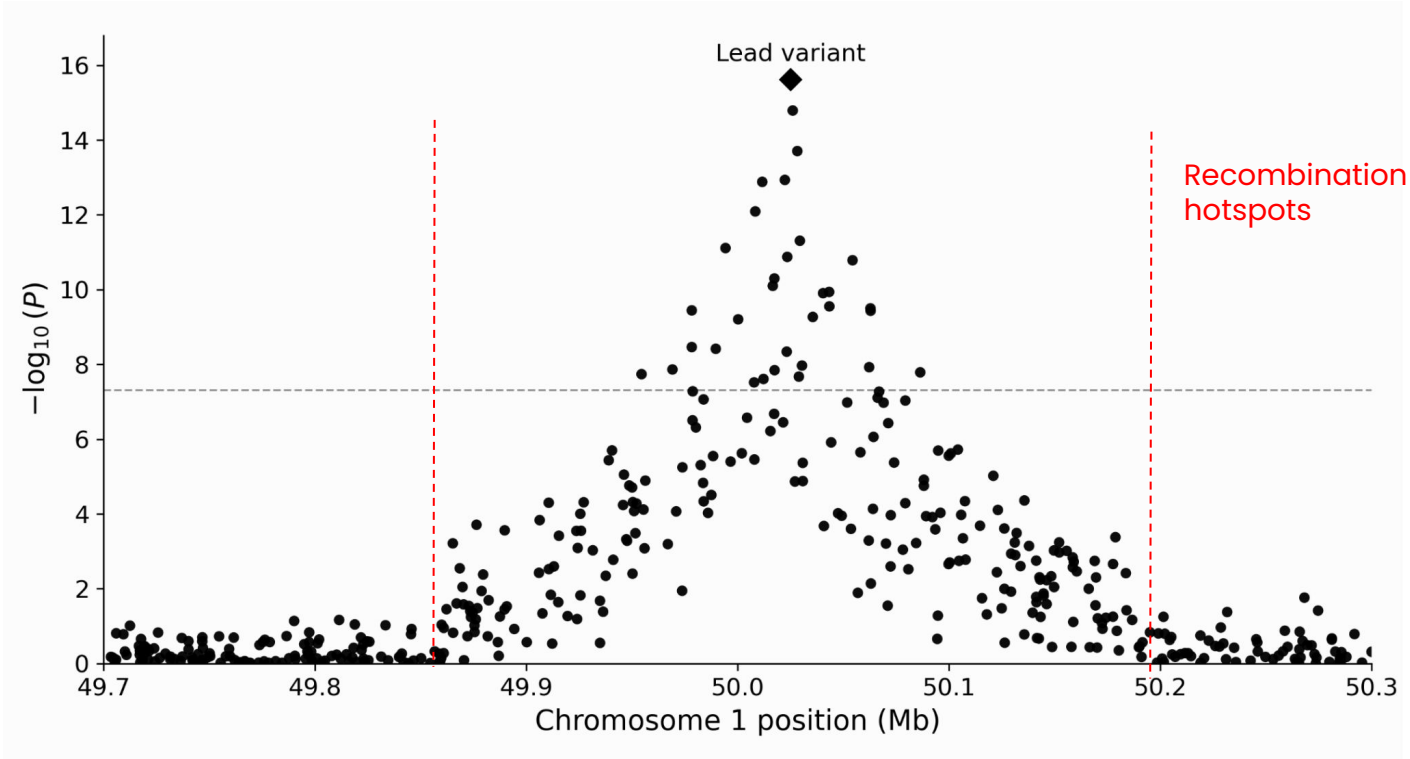


1. **Linkage-disequilibrium makes precise resolution of causal variants difficult**

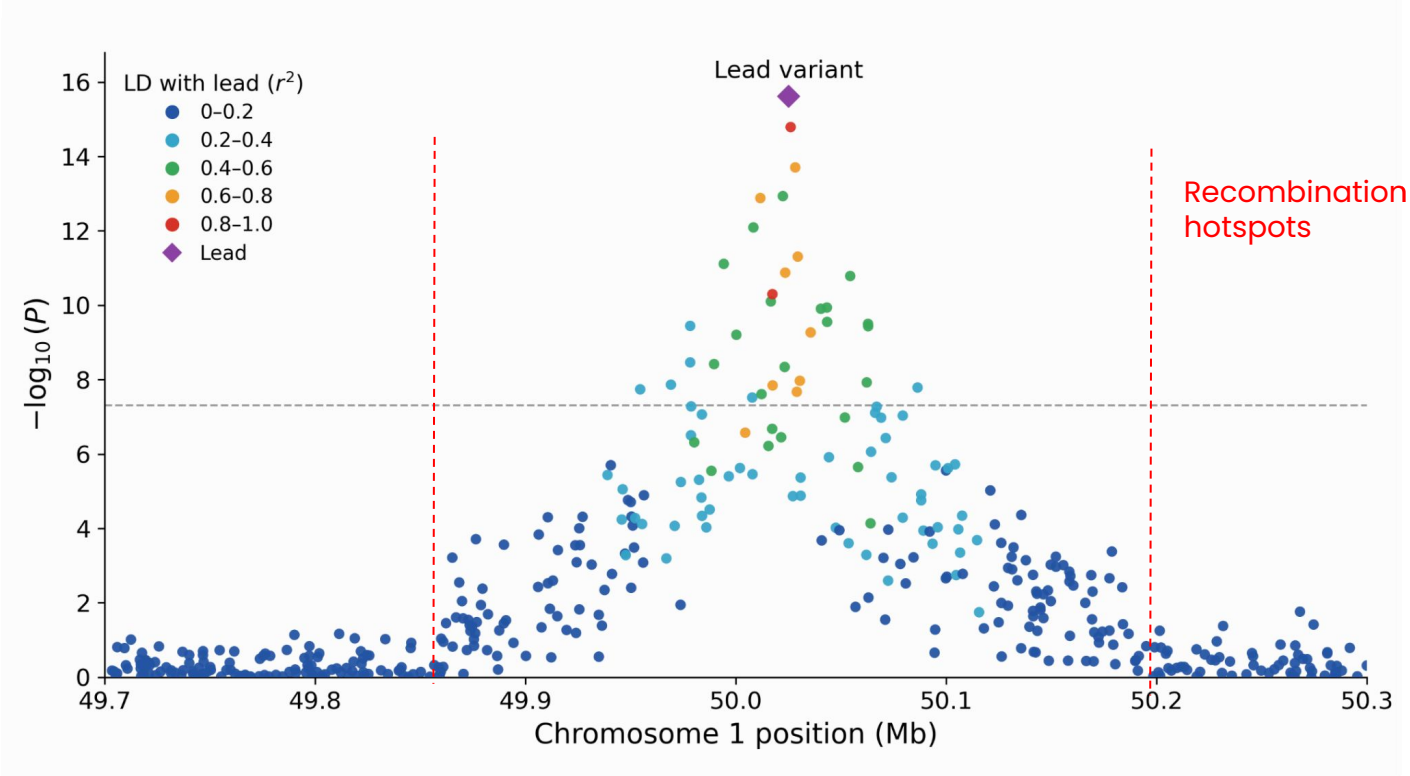
Linkage Disequilibrium complicates precise resolution of causal variants



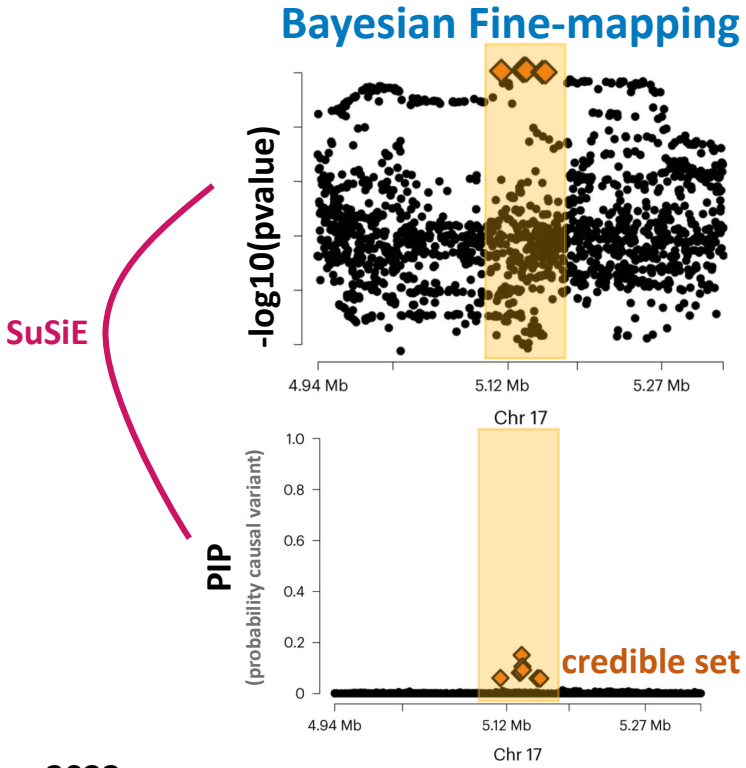
Typical GWAS loci contains potentially hundreds of linked variants that reach significance



Typical GWAS loci contains potentially hundreds of linked variants that reach significance



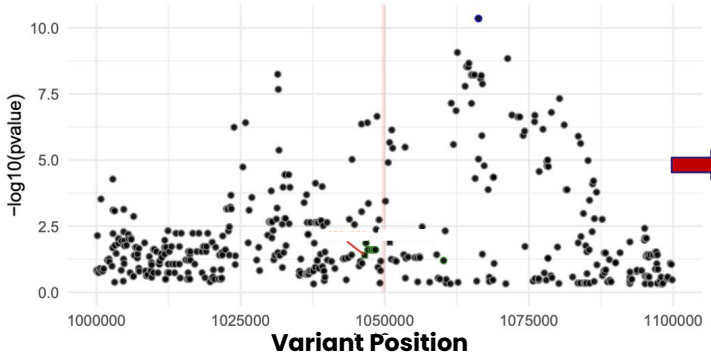
Genetic Fine-mapping can help disentangle LD and nominate potential causal variants



Yang *et al* Nature Genetics, 2023

SuSiE – Sum of Singular Effects

input:

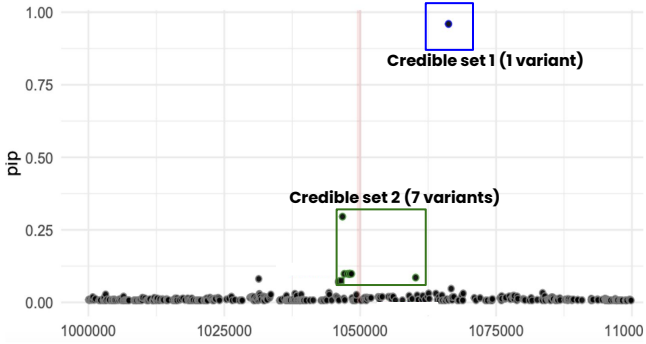


Effect Size
P-values
LD (correlation) between varaints

SuSiE

models the association at a locus as a sum of individual variant effects, accounting for LD to identify distinct signals and credible sets of potential causal variants

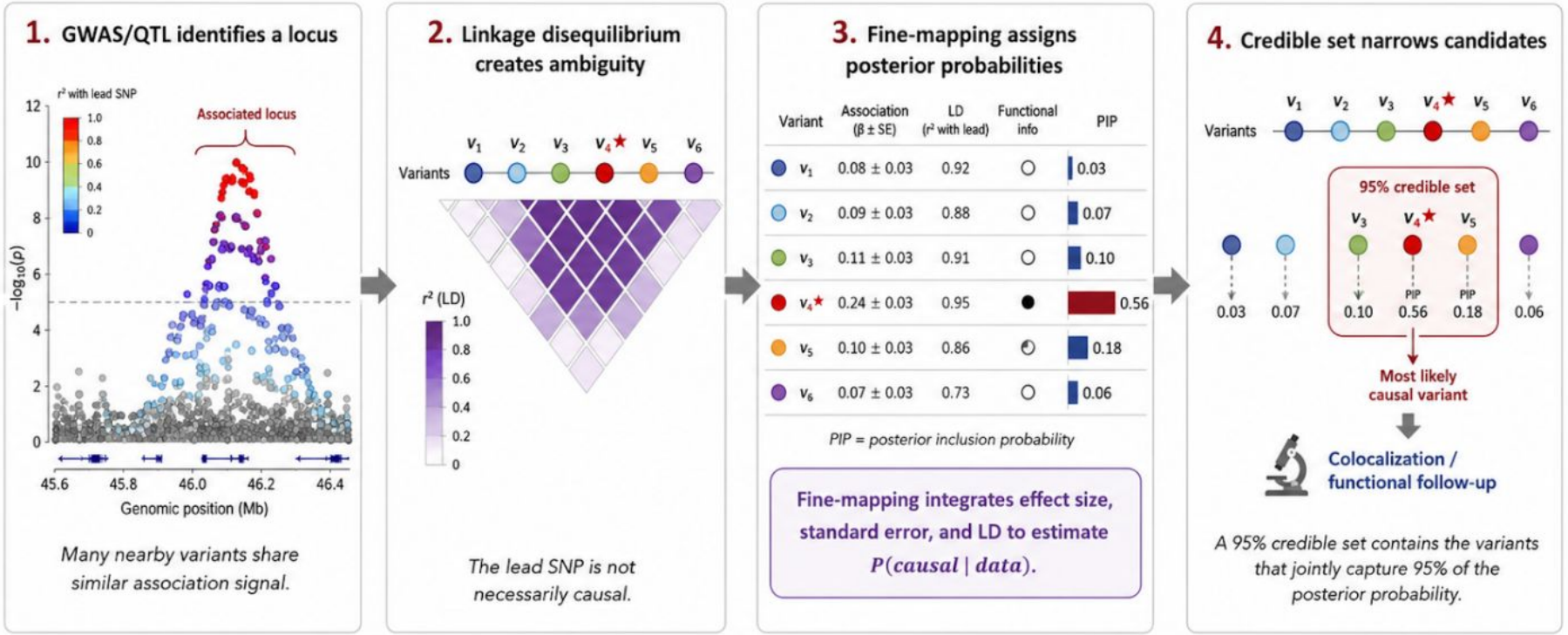
output:



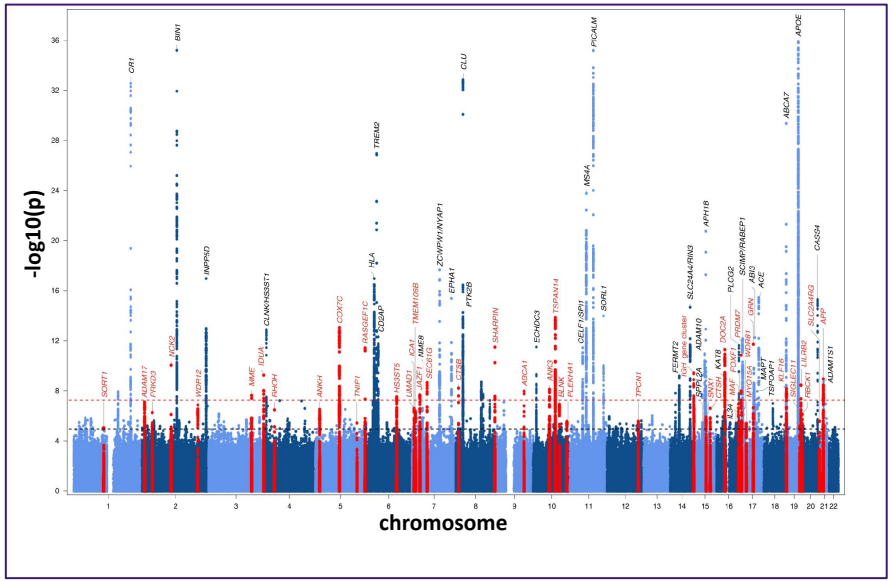
Posterior Inclusion Probability (PIP):
Posterior probability any given variant is causal for the association

95% Credible Set:
A set of variants with a 95% posterior probability of containing the causal variant for a given association signal.

Finemapping: From association at a locus to likely causal variant



Major limitations of GWAS

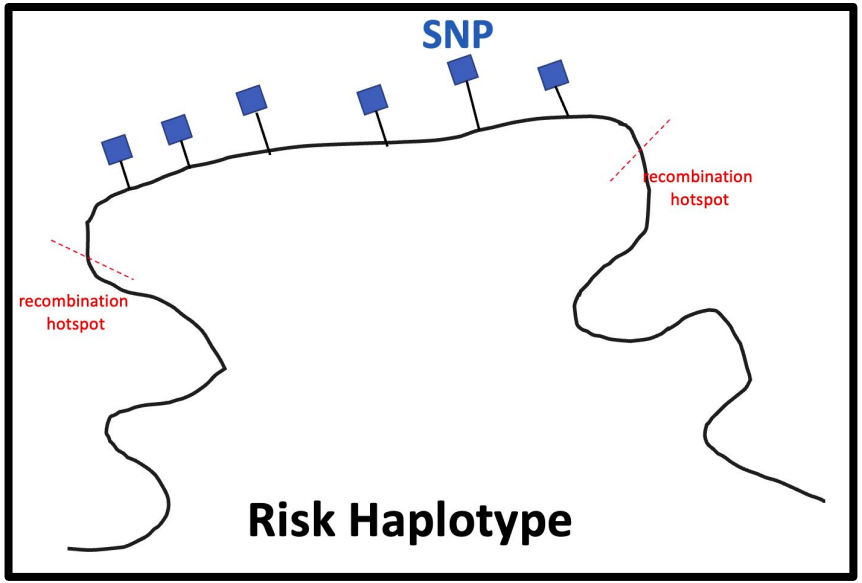


1. Linkage-disequilibrium makes precise resolution of causal variants difficult

2. **SNP-focused analyses may miss substantial genetic variation**

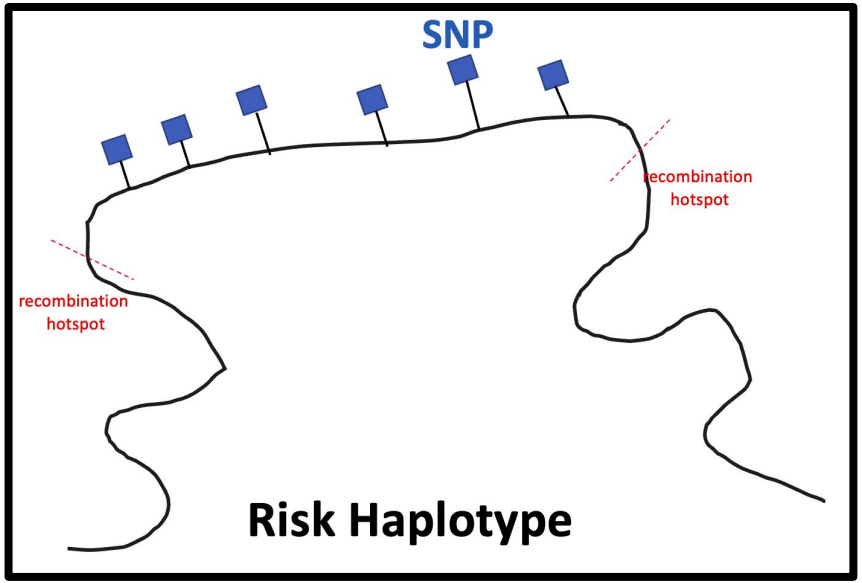
SNP-Centric approach of GWAS may bias detection of causal SVs

GWAS scope:

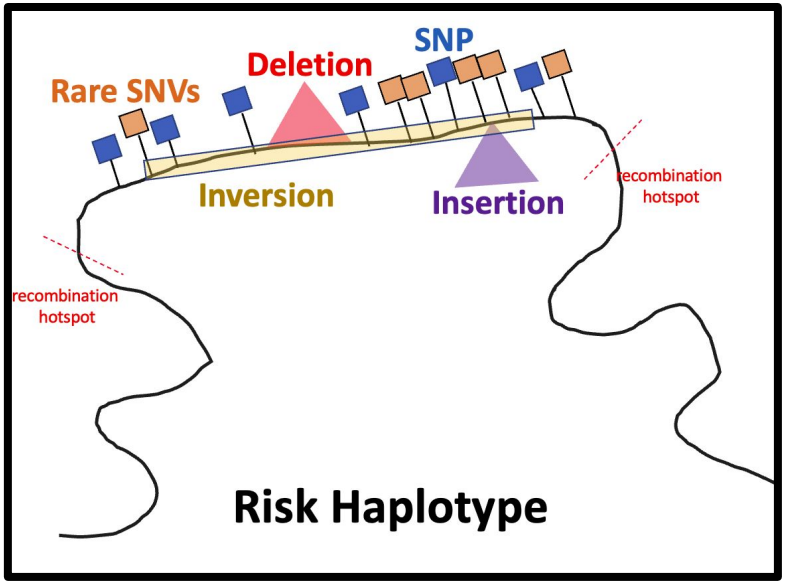


SNP-Centric approach of GWAS may bias detection of causal SVs

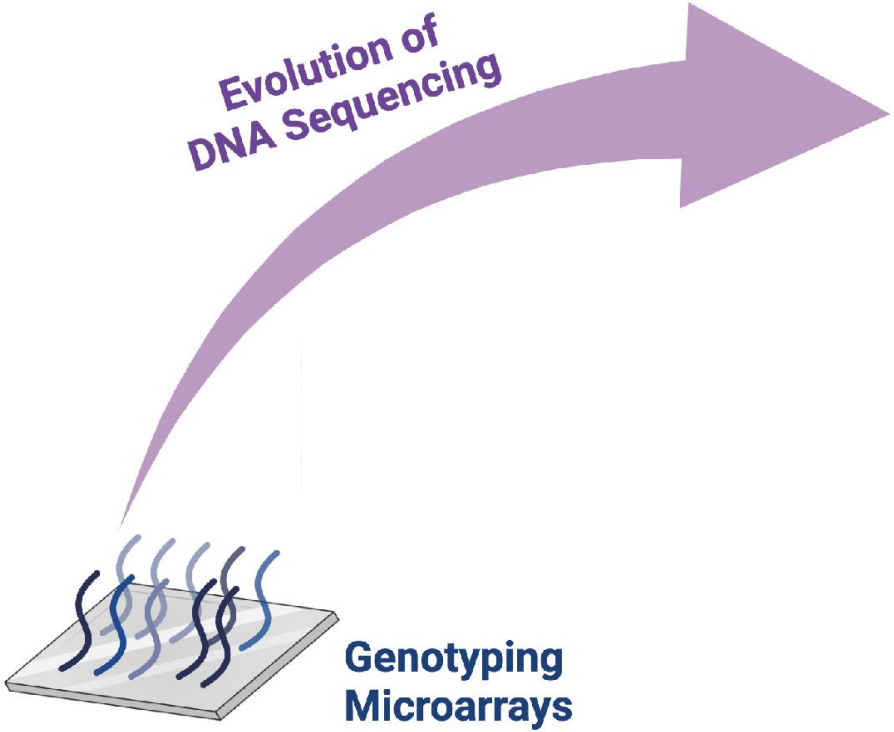
GWAS scope:



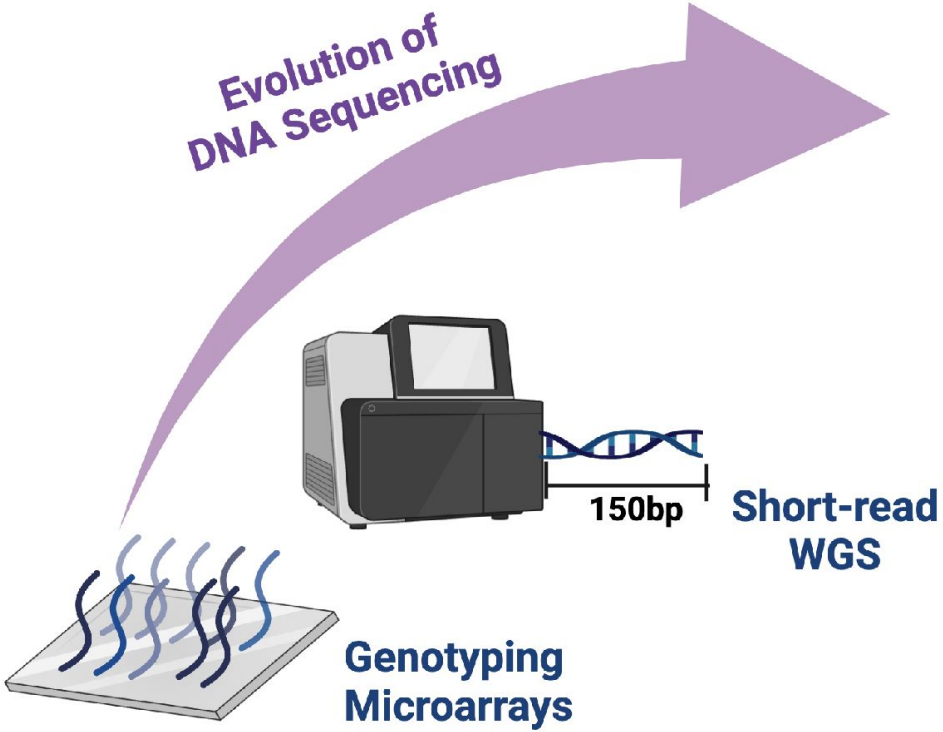
Reality:



Short-read vs long-read Sequencing

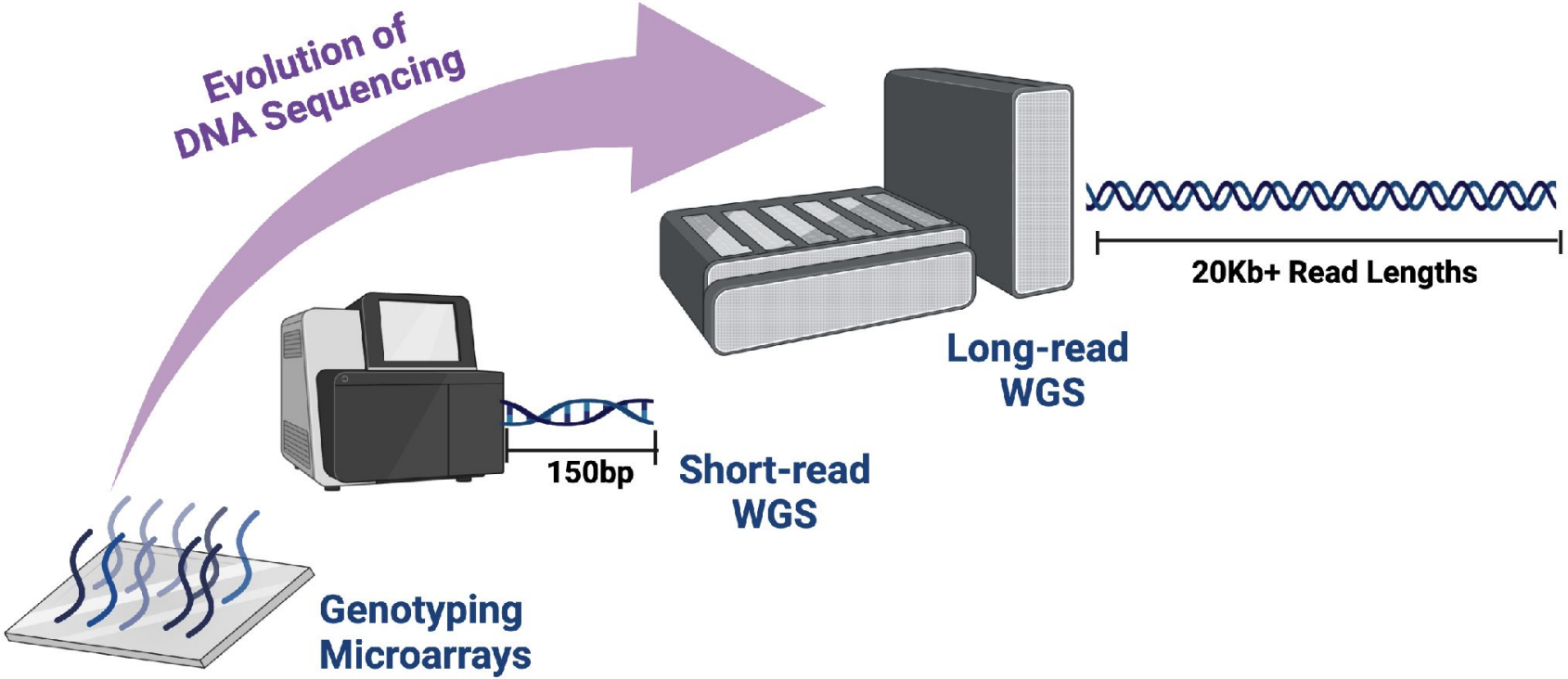


Short-read vs long-read Sequencing

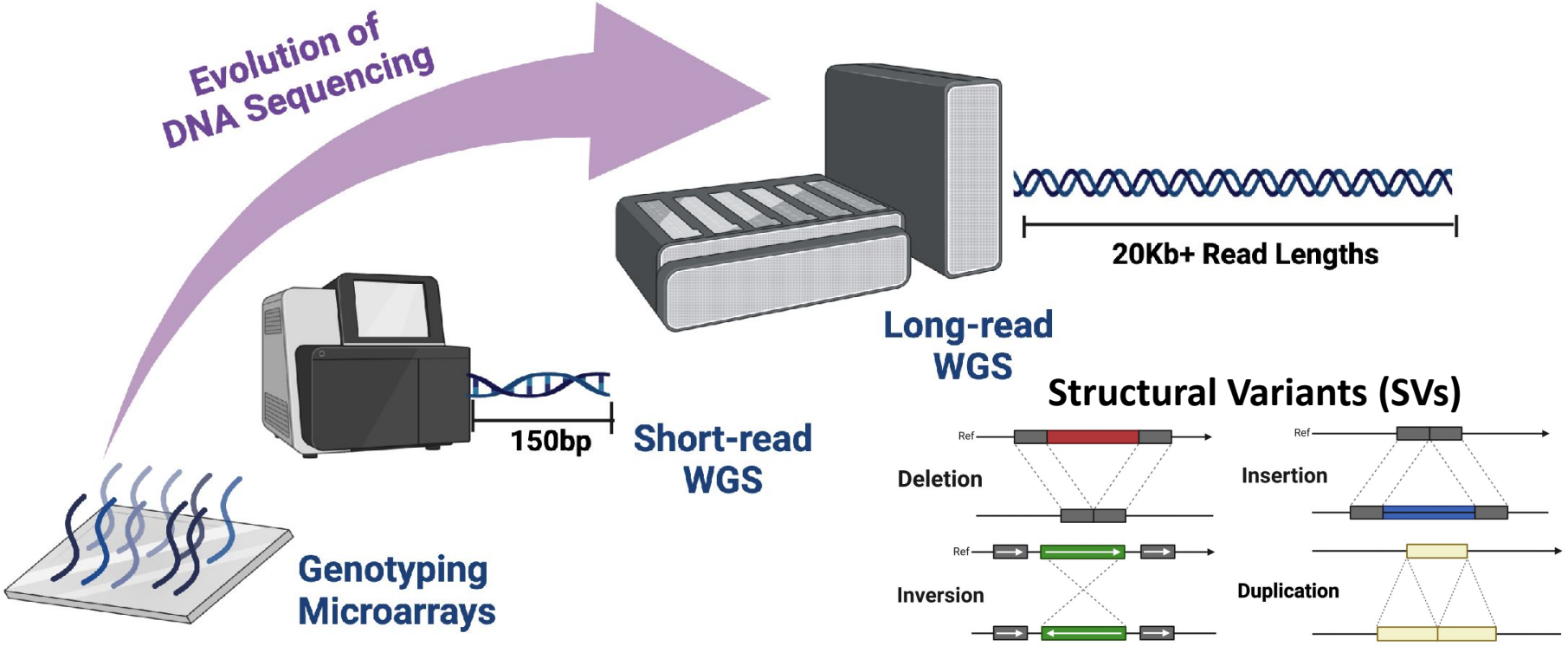


More comprehensive SNV coverage including rare variants

Short-read vs long-read Sequencing



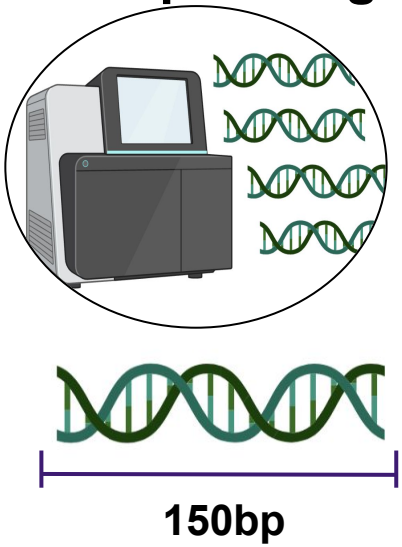
Short-read vs long-read Sequencing



Short reads are difficult to map to a repetitive genome

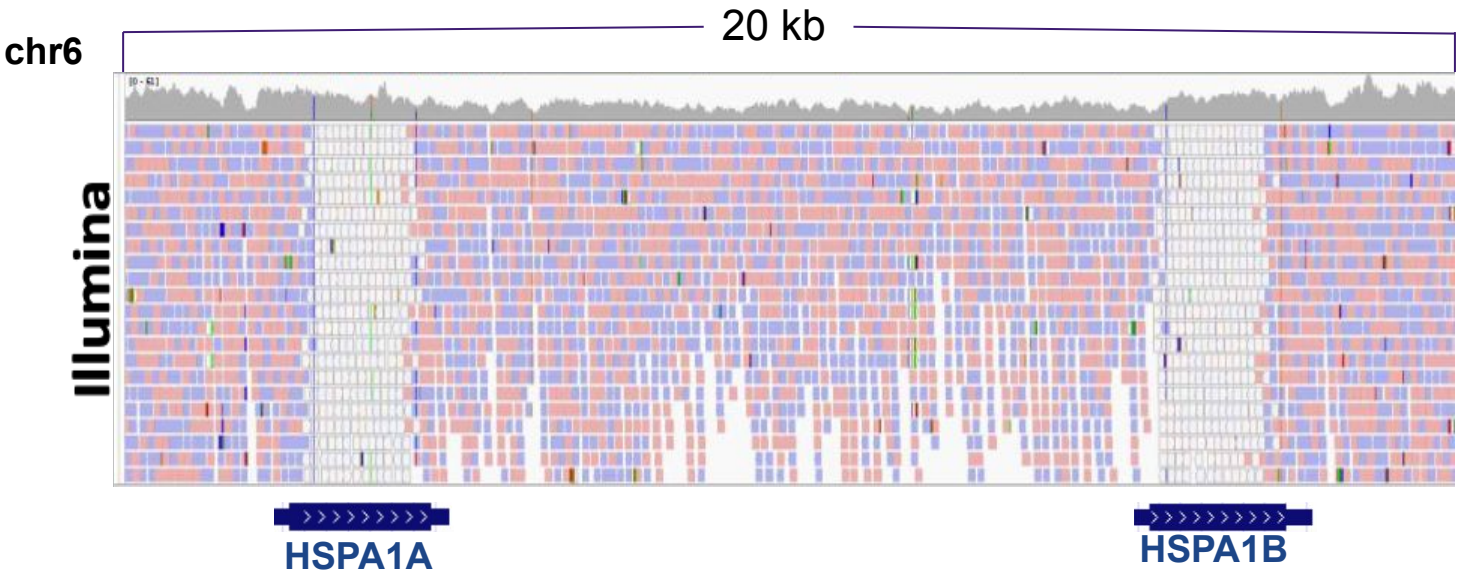


Illumina Short-read Sequencing

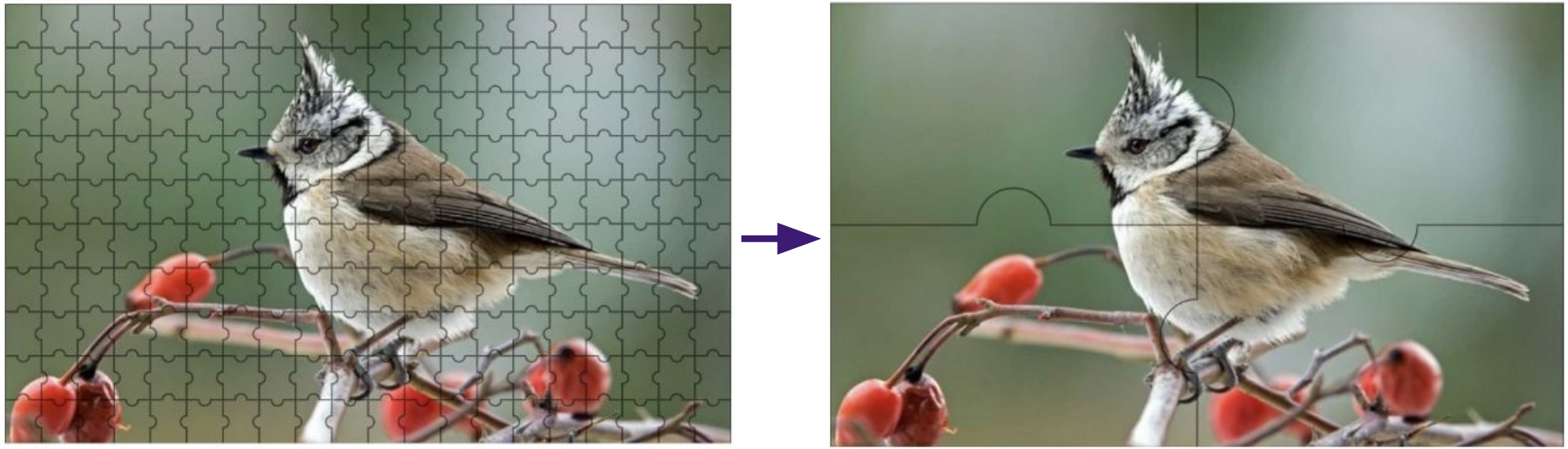


https://www.seqme.eu/en/next-gen-sequencing/article/long-read-sequencing-technologies

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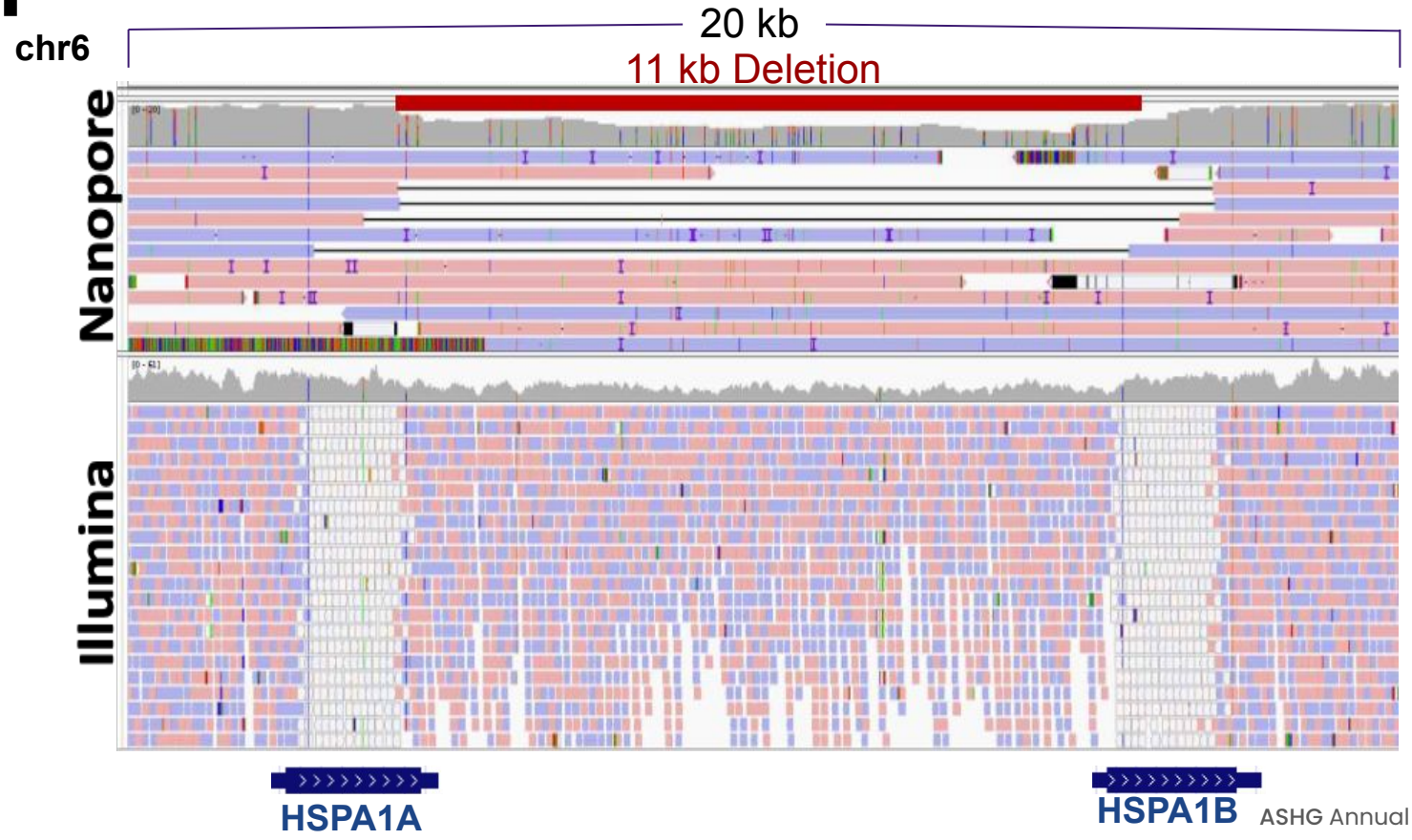


Longer reads can enable better mapping

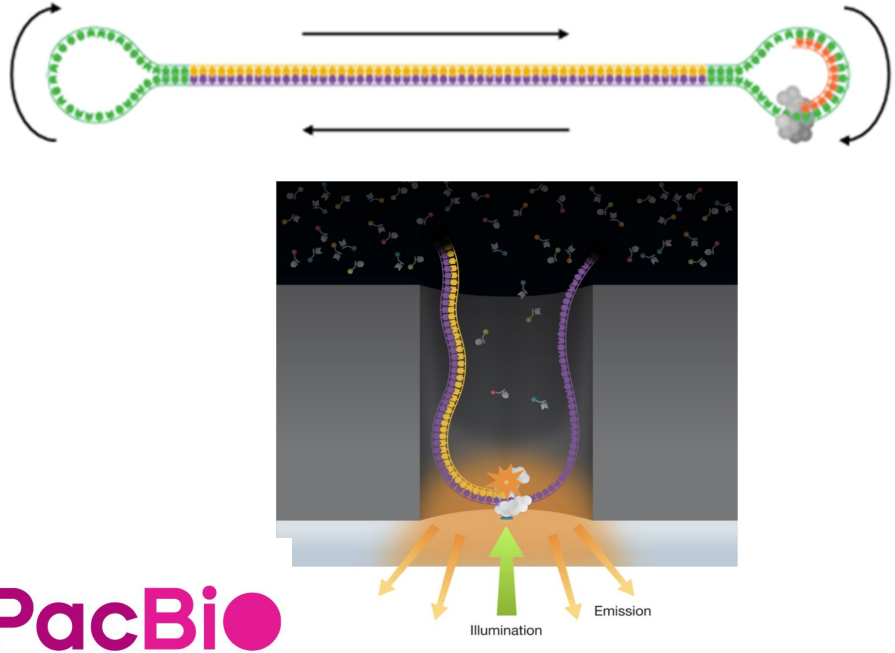


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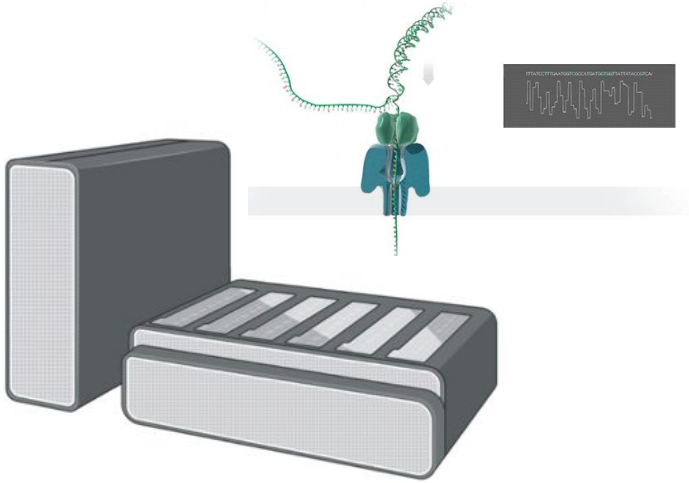
Long reads can accurately map and resolve complex SVs



Third-generation sequencing provides long reads but has struggled with throughput and accuracy



PacBio



Oxford NANOPORE Technologies

Pacbio offers reliable, high accuracy reads at a higher cost per sample



PacBio
Revio

Higher accuracy
(Q30)

Lower throughput
(90Gb/flow cell)

Less scalable
(4 independent flow cells / machine)

Shorter read lengths
(15-20kb)

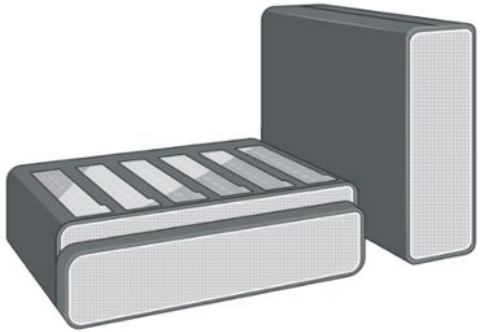
Nanopore maximizes scalability and read lengths at the cost of lower accuracy

Lower accuracy
(Q20)

High throughput
(120-180Gb/flow cell)

High scalability
(48 independent flow cells / machine)

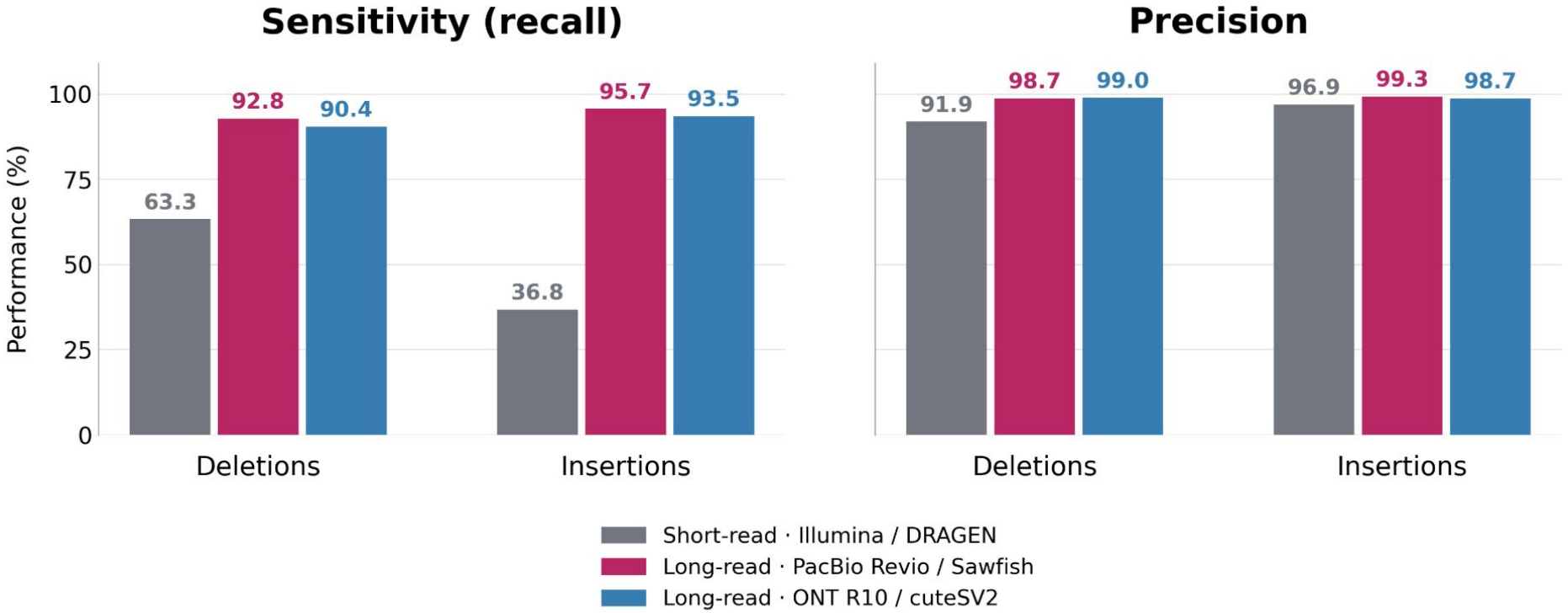
Long read lengths
(20-100+kb)



Oxford NANOPORE Technologies
Promethion 48

Long reads improve structural variant discovery

HG002 • T2T-Q100 v1.1 benchmark • ~38–43× coverage

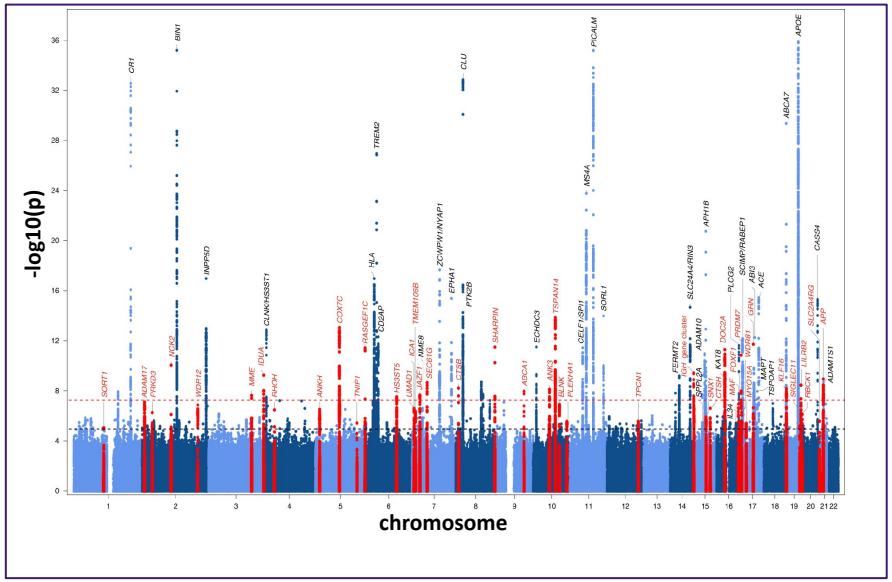


Eveleigh et al., Genome Biology (2026), Table S6 | DOI: 10.1186/s13059-026-04048-4

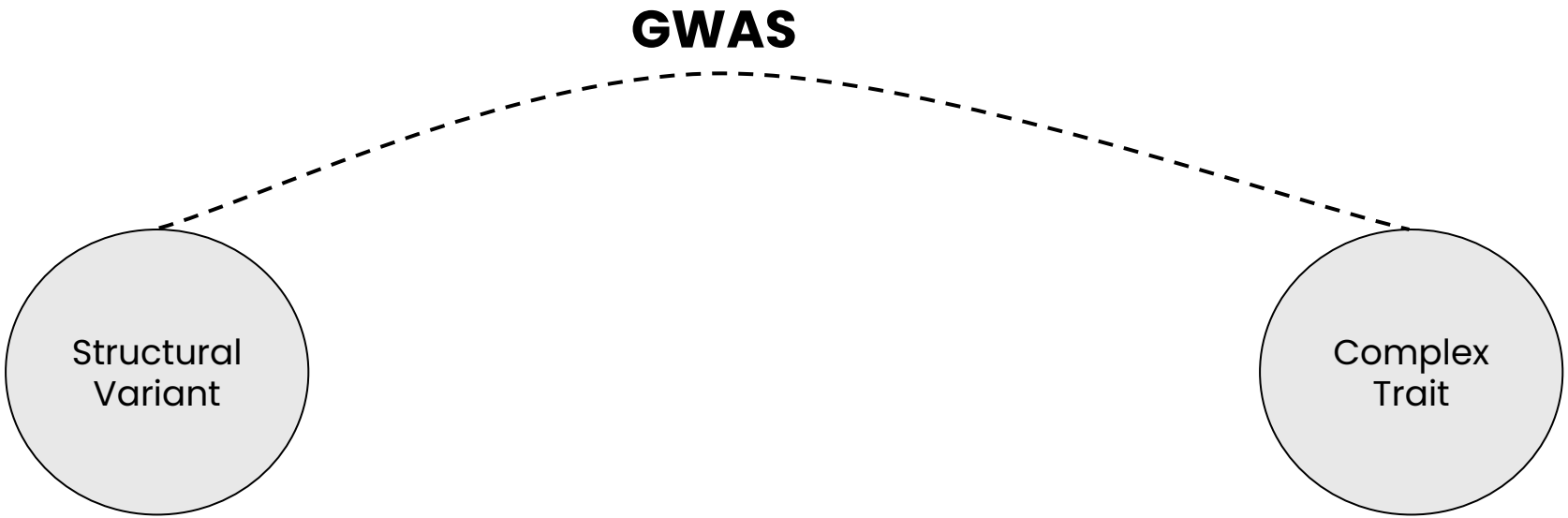
Population long-read sequencing is expanding catalogs of human structural variation

Resource	Population / focus	Long-read sample size	Platform	Main contribution
All of Us	Diverse US population	~14,000	PacBio HiFi	SV discovery linked to health records
1000 Genomes — ONT	26 global populations (Vienna, UWONT)	~1,500	Nanopore	Diverse population SV reference
1000 Chinese Pangenome	Chinese population	1,116	PacBio HiFi	Population-specific assemblies and SV catalog
deCODE	Icelandic population	3,622	Nanopore	Population-scale SV discovery and trait associations
UDN, GREGoR, GA4K	Rare disease patients and families	>3,000	PacBio HiFi + Nanopore	Rare and complex SVs underlying disease

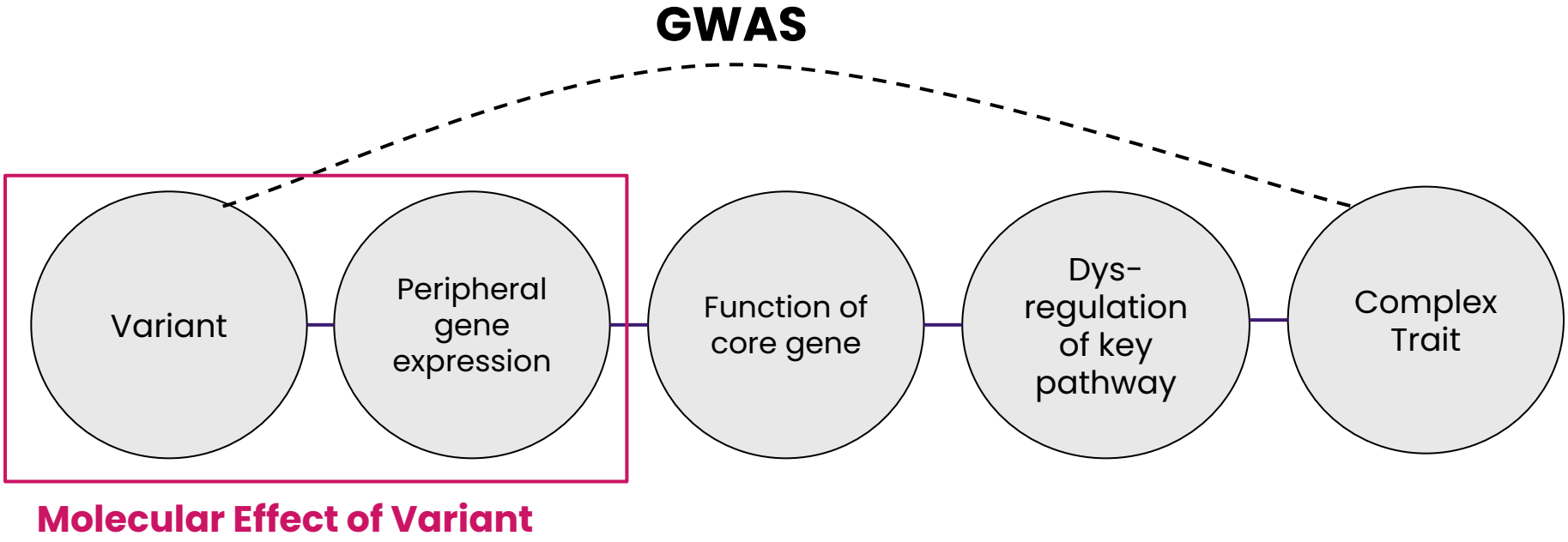
Major limitations of GWAS


-log10(p)
chromosome
BIN1
CR1
MS4A
PICALM
CLU
TREM2
APOE
ABCA7
CASS4
HLA
EPHA1
SORL1
APH1B
PTK2B
SHARPIN
ECHDC3
TSPAN14
CELF1/SPI1
ADAM10
SLC24A4/RIN3
FERMT2
ACE
ABI3
PLCG2
KAT8
DOC2A
MAF
FOXF1
PRDM7
MAF21A8
MGAT1
GRN
MAPT
CTSH
TSPOAP1
SCIMP/RABEP1
SLC39A13
ADAM17
PRNCD
WDR12
MMP
IDUA
CLNK/HSD17
ANKH
CD2AP
TMEM106B
IQCK
MAB9
SCIN
HSBP1
ADGA1
MMP2
BLNK
FCHSD1
TPCN1
ZCWPW1/NYAP1
PSEN1
SORT1
PRNCD
WDR12
CD33
TNIP1
HS3ST5
IQCJ-SCHIP1
SCIN
ADGA1
MMP2
BLNK
FCHSD1
TPCN1
FERMT2
SLC24A4/RIN3
ADAM10
SORL1
KAT8
DOC2A
MAF
FOXF1
PRDM7
MAF21A8
MGAT1
GRN
MAPT
TSPOAP1
ACE
ABI3
SCIMP/RABEP1
SLC39A13
PSEN1
ADAM17
PRNCD
WDR12
MMP
IDUA
CLNK/HSD17
ANKH
CD33
TNIP1
HS3ST5
IQCJ-SCHIP1
SCIN
HSBP1
ADGA1
MMP2
BLNK
FCHSD1
TPCN1
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SORT1
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TMEM106B
IQCK
MAB9
SCIN
HSBP1

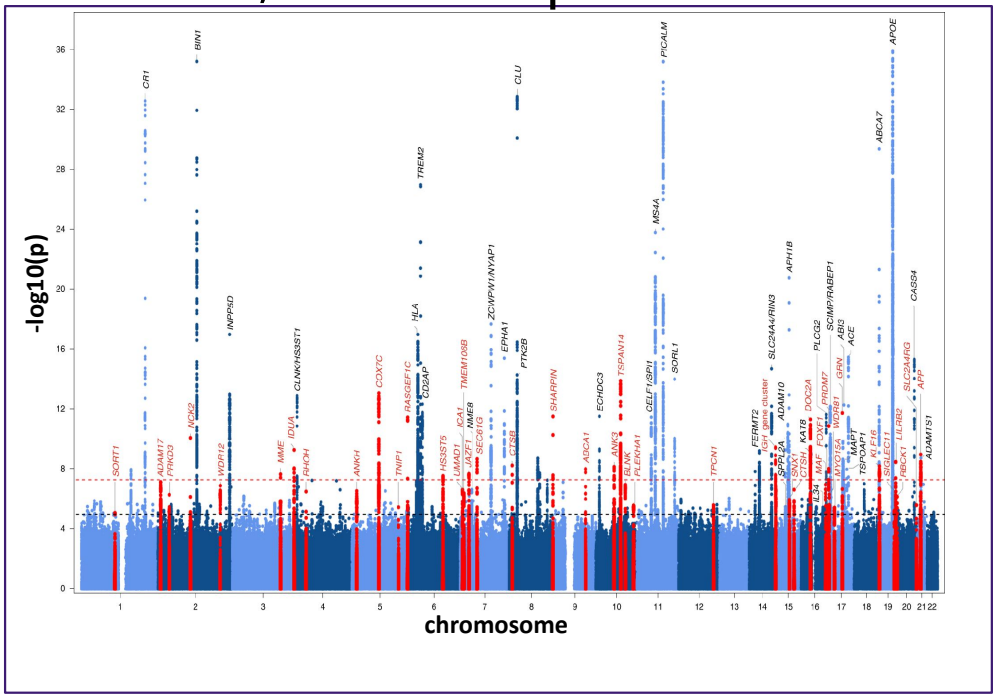
GWAS attempts to test a variant's effect directly on a complex trait



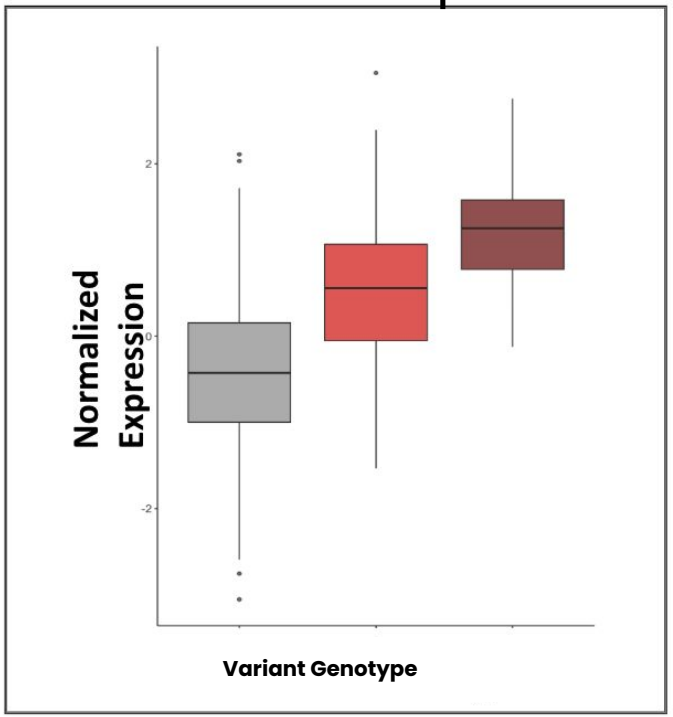
In reality a variant has an indirect effect often through a long causal network



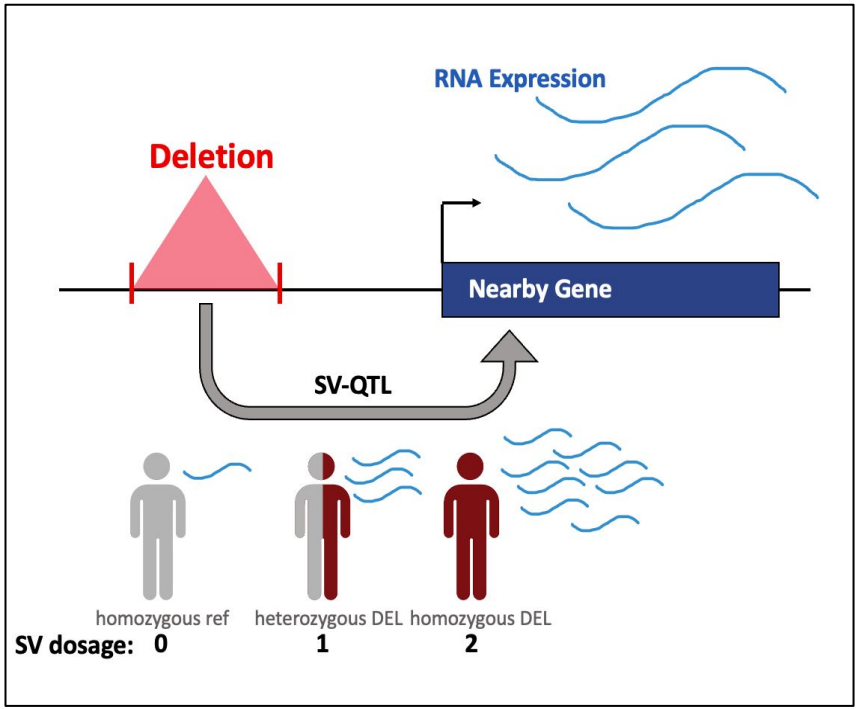
GWAS
~100,000s samples

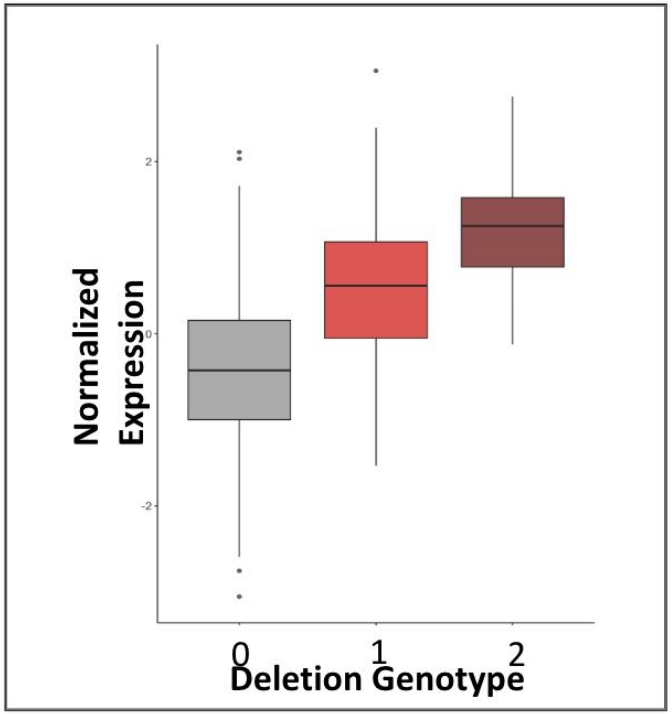


Molecular Trait Mapping (QTLs)
~100s of samples



Quantitative Trait Loci (QTL) analyses let us identify molecular effects of variants

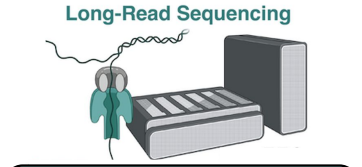




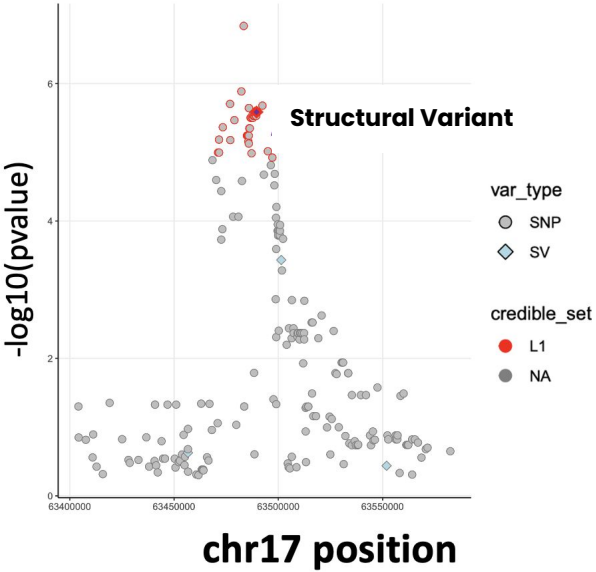
Publicly available Multi-omics cohorts have mapped molecular traits across multiple layers of regulation

Resource	QTL type	Tissue(s)	Molecular phenotype	Sample size
GTEx v8	**e**QTL, **s**QTL	49 tissues	Gene **expression**; alternative **splicing**	73–706 per tissue
deCODE	**p**QTL	Plasma	**Protein** abundance	35,559
UK Biobank PPP	**p**QTL	Plasma	**Protein** abundance	54,219
GoDMC	**m**QTL	Whole blood	DNA **methylation**	32,851
ROSMAP	**e**QTL, **m**QTL, **ha**QTL	Prefrontal cortex	Expression; methylation; **histone acetylation**	Up to 494

Extending QTL resources to long-read cohorts can determine molecular effects of SVs



Structural Variant QTLs

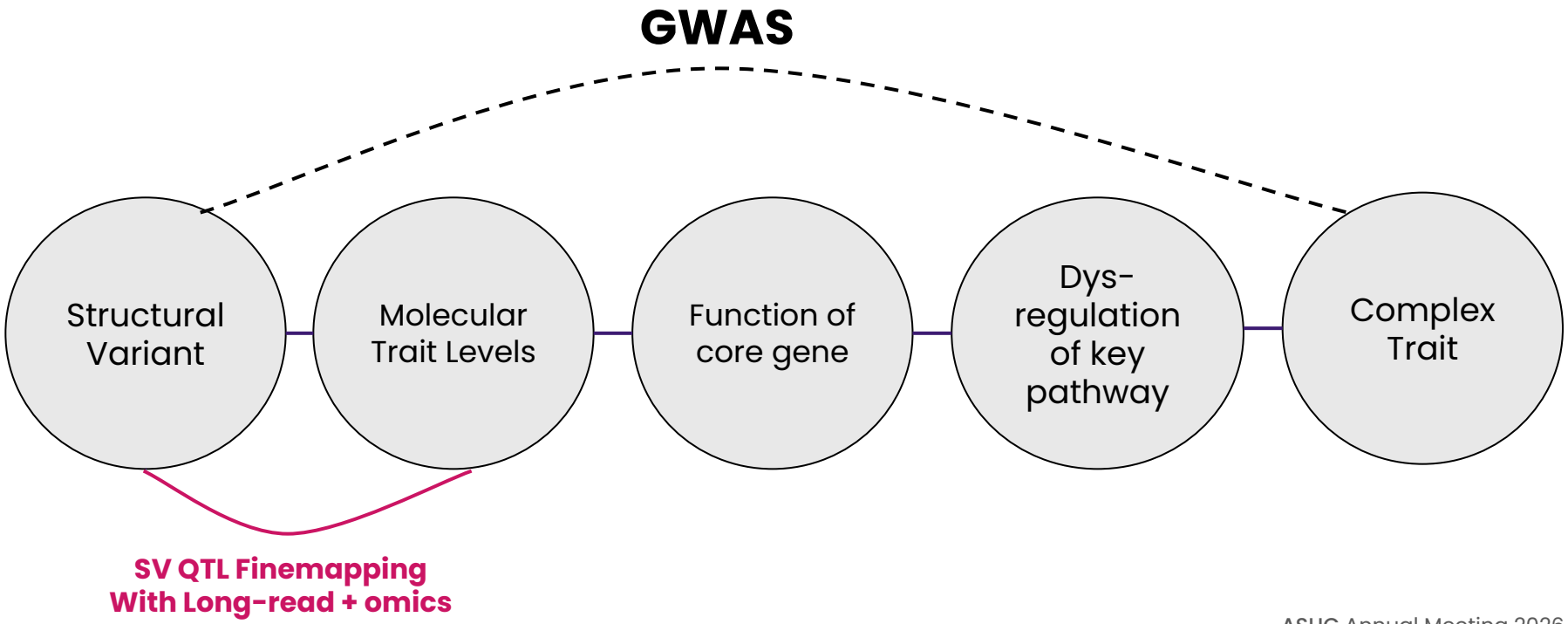


SuSiE Fine Mapping

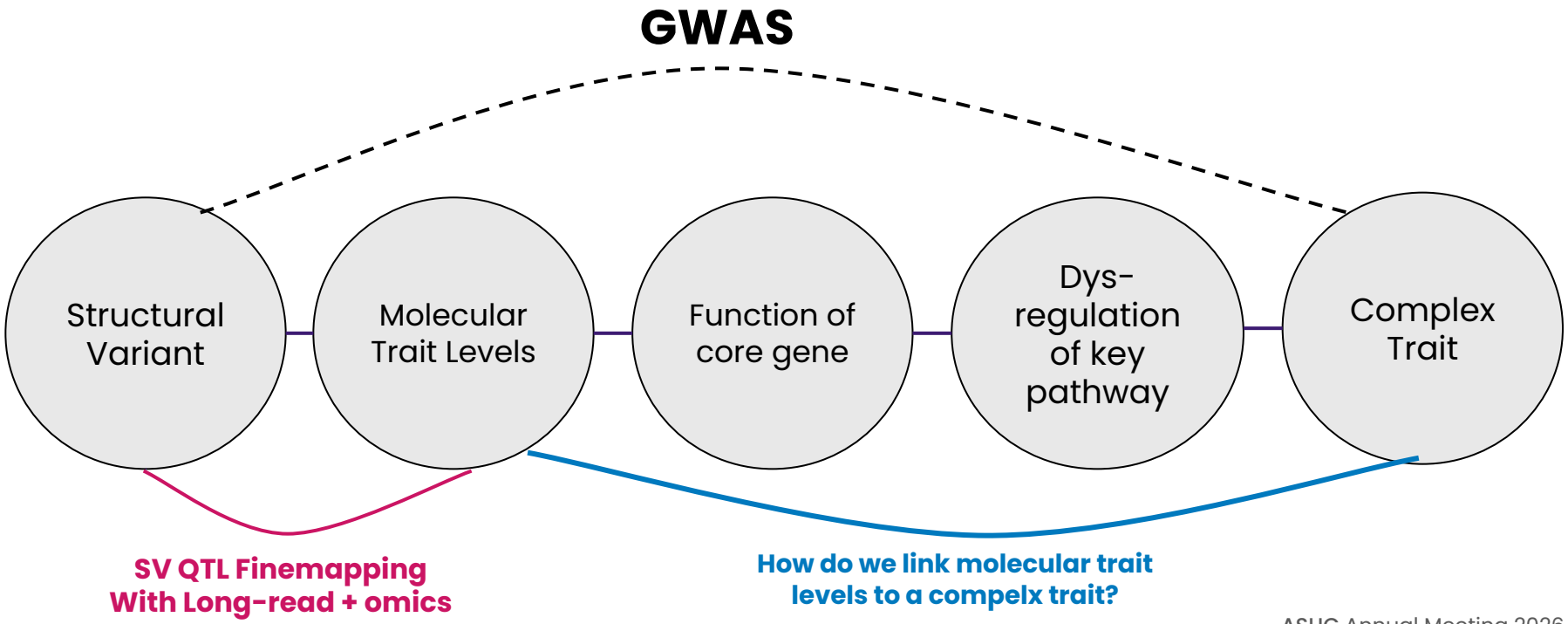
Credible Set

SV (pip=.70)
SNV1 (pip=.15)
SNV2 (pip=.10)

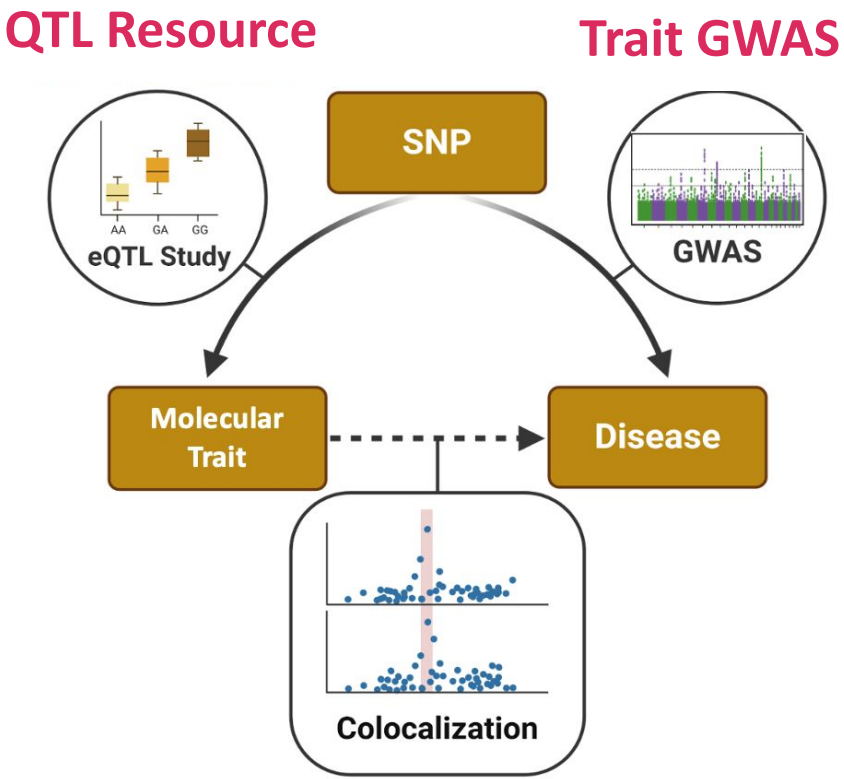
In reality a variant has an indirect effect often through a long causal network

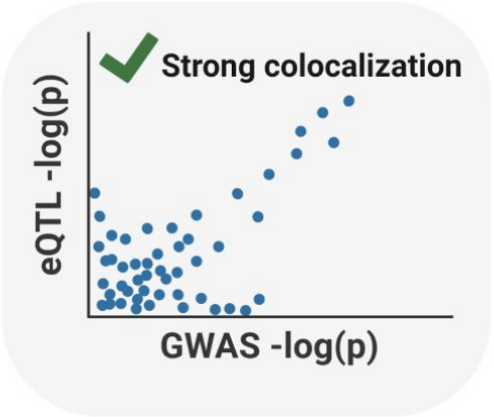


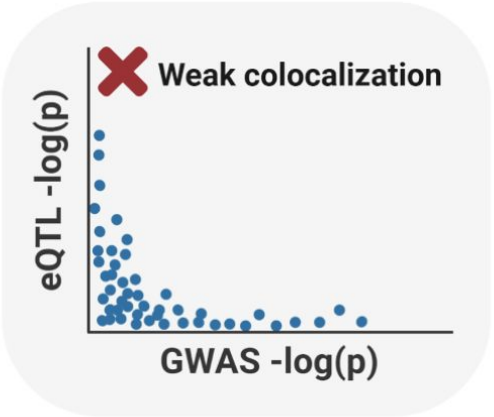
In reality a variant has an indirect effect often through a long causal network



Colocalization integrates GWAS can identify QTLs relevant to trait

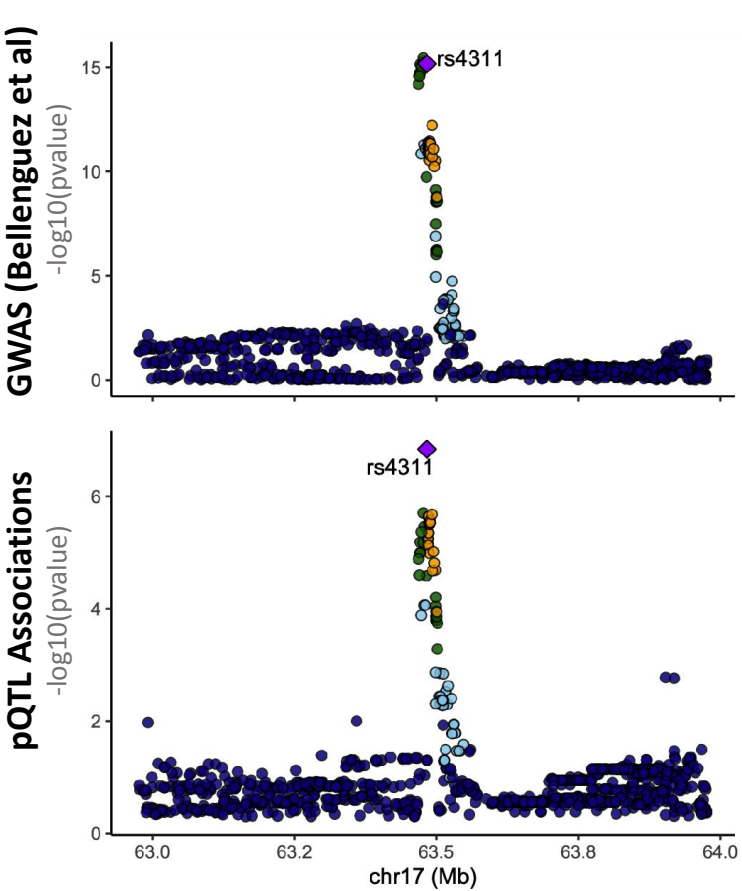


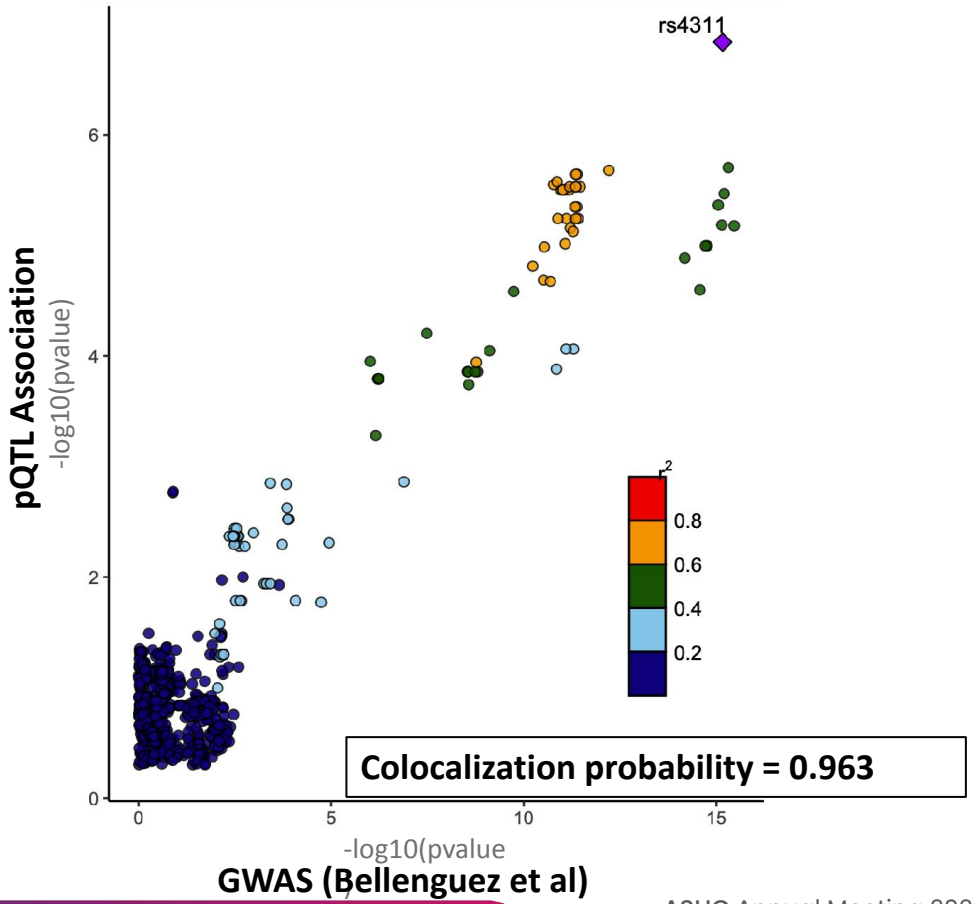




Slides adapted from Pagé Goddard

Coloc measures if variant is a shared signal between two traits

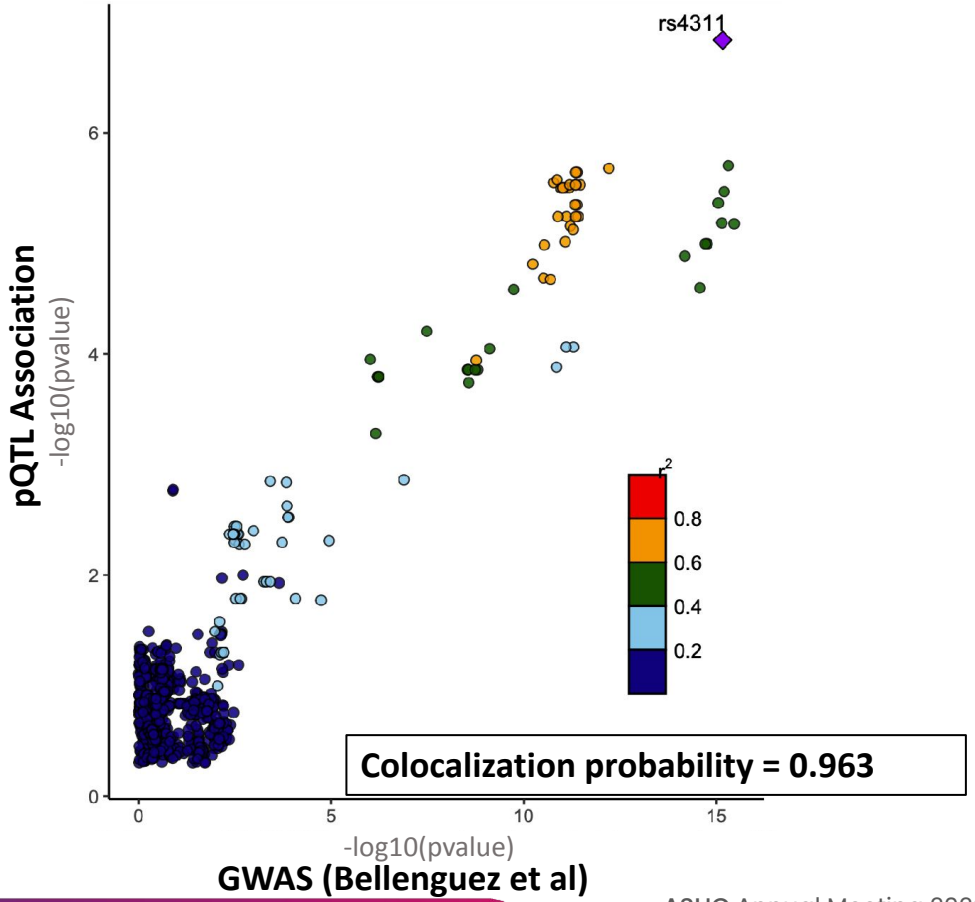




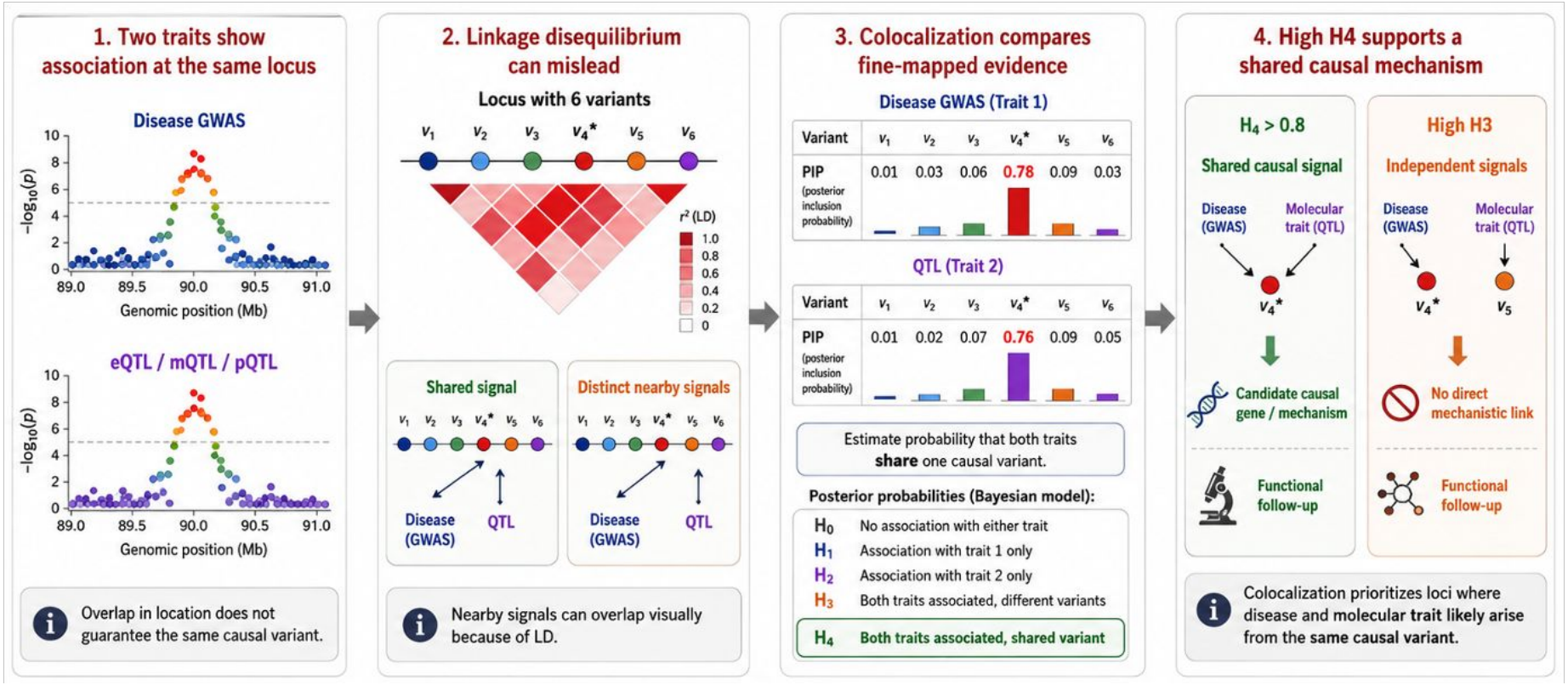
Coloc measures if variant is a shared signal between two traits

Posterior probabilities (Bayesian model):

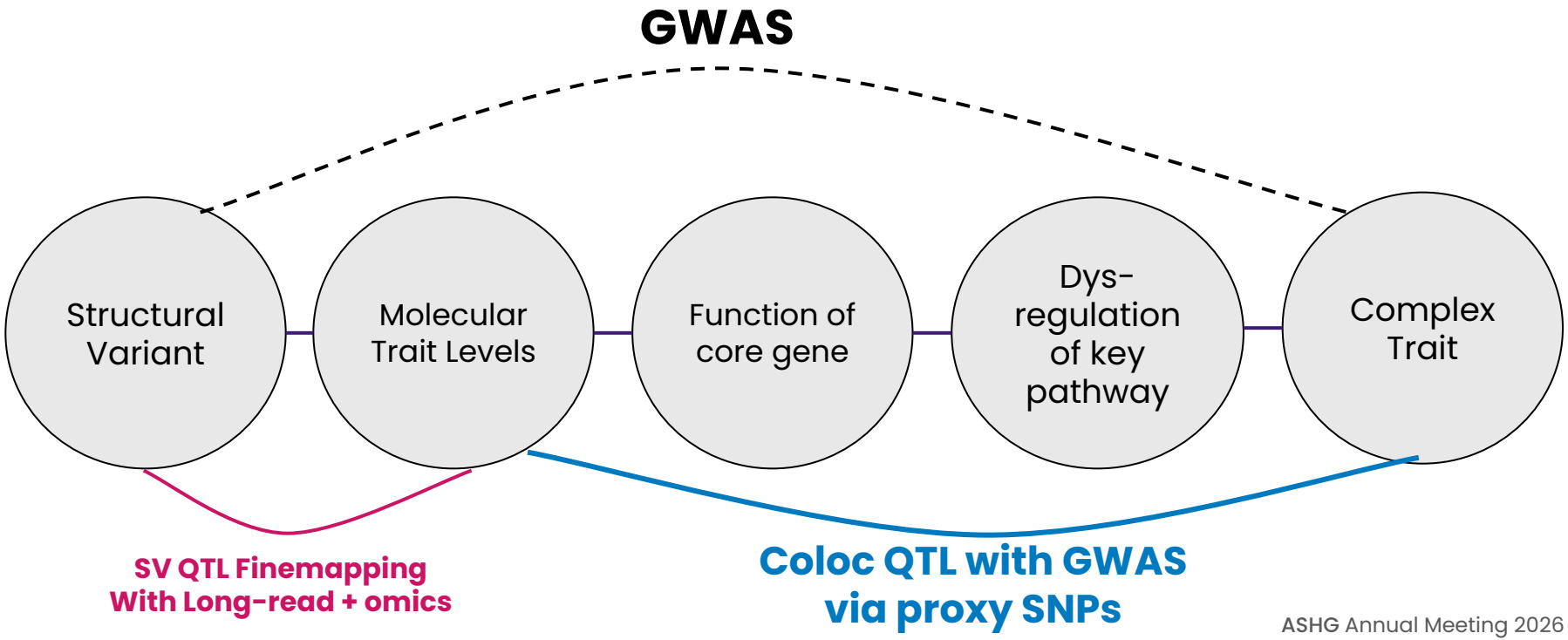
H_0	No association with either trait
H_1	Association with trait 1 only
H_2	Association with trait 2 only
H_3	Both traits associated, different variants
H_4	**Both traits associated, shared variant**



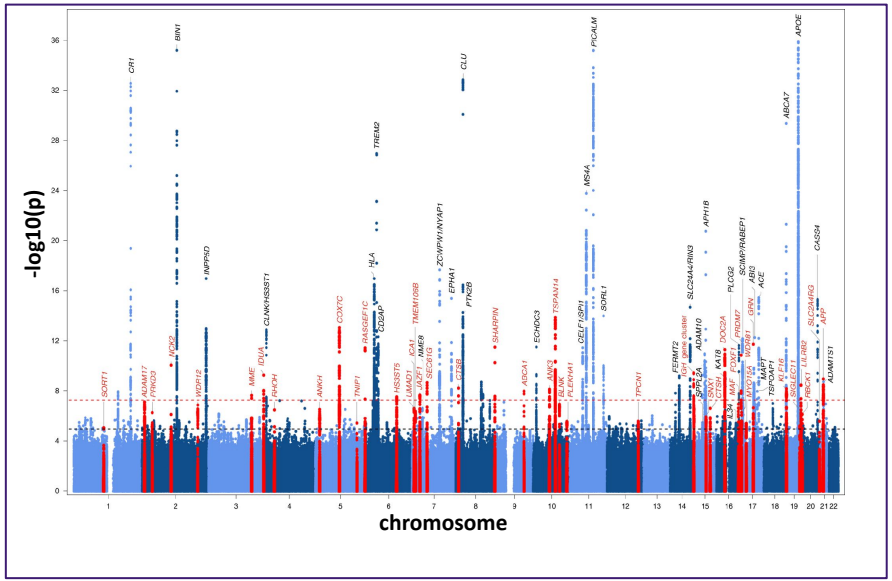
Colocalization: From shared locus to shared causal signal



In reality a variant has an indirect effect often through a long causal network



Major limitations of GWAS



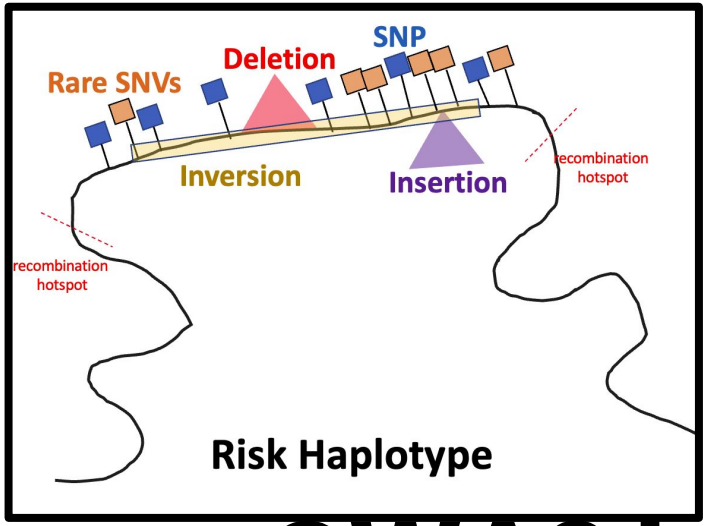
1. **Linkage-disequilibrium makes precise resolution of causal variants difficult**

2. **SNP-focused analyses may miss substantial genetic variation**

3. **Limited interpretation of noncoding variants hinders identification of causal genes and mechanisms**

Putting it all together

Problem	Solution
1. Linkage-disequilibrium makes precise resolution of causal variants difficult	**Genetic Fine-mapping tools** can disentangle LD
2. SNP-focused GWAS may miss substantial genetic variation 3. Abundance of non-coding variants impedes detection of causal genes and molecular mechanisms	**Long-read sequencing** allows more comprehensive detection of Structural and repeat variation
	Measuring an intermediate molecular phenotype through multi-omics **QTL** and **colocalization** can help interpret risk loci

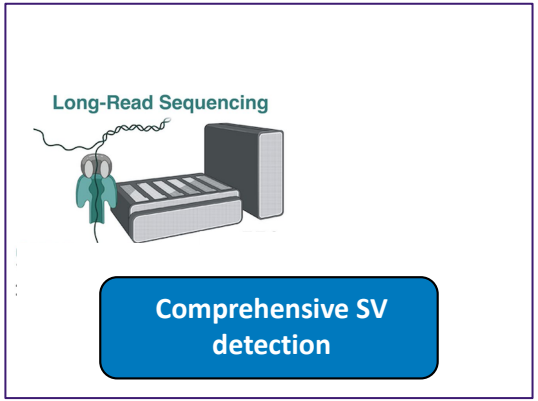


How many GWAS hits for complex traits or disease could be driven by Structural Variants?

Conceptual workflow to find causal SVs from long-read sequencing in GWAS

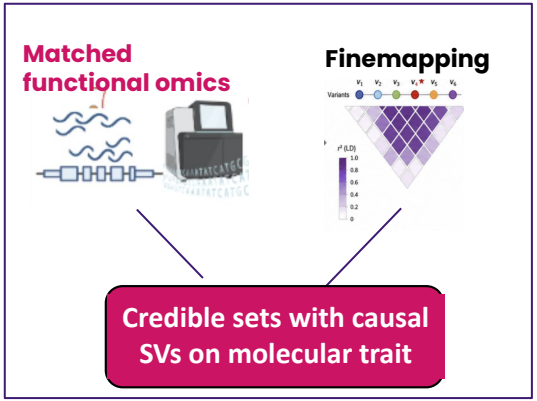
01

Long-read Cohort with Matched multi-omics



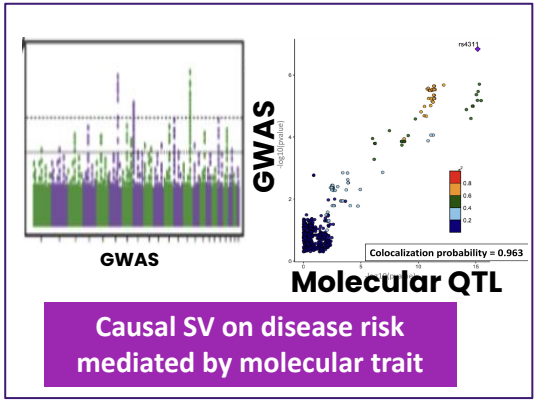
02

Joint statistical finemapping of SNPs + SVs with SuSiE



03

Coloc with GWAS over Proxy SNPs



Alternative approaches: imputation

Imputation can be used to directly measure effect of SV in large, well-powered GWAS



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Genome-wide associations of structural variants with human traits through imputation from long-read assemblies

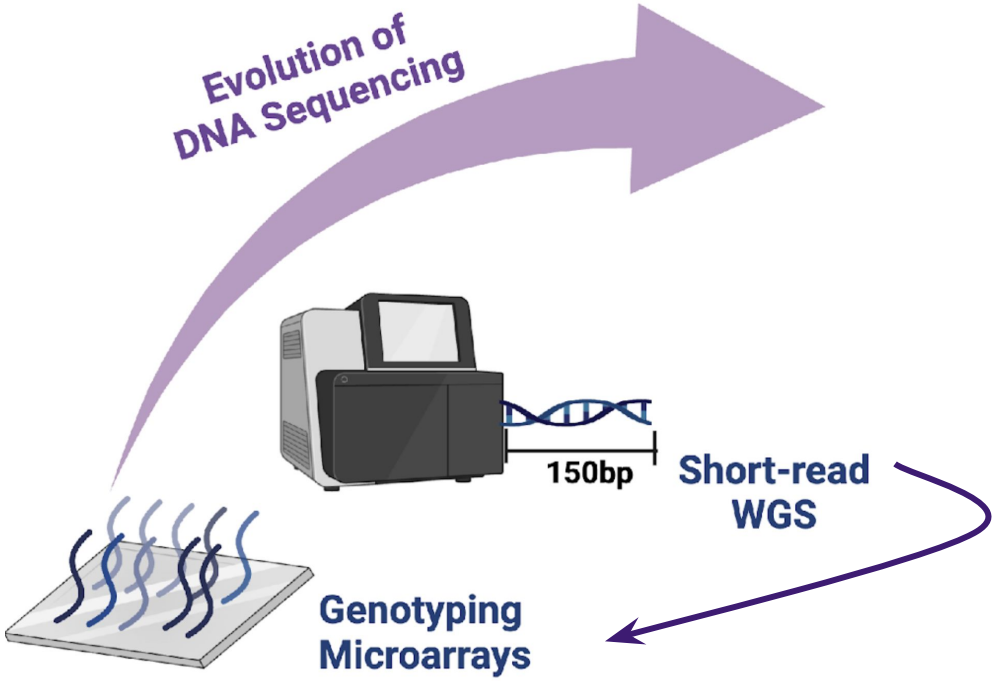
Wei-Yang Bai, Shuli Liu, Zhongqu Duan, Ji-Jian Yang, Jie Chen, Junren Hou, Lianfeng Wu, Nan Li, Ting Qi & Jian Yang

Nature Genetics **58**, 1258–1267 (2026) | Cite this article

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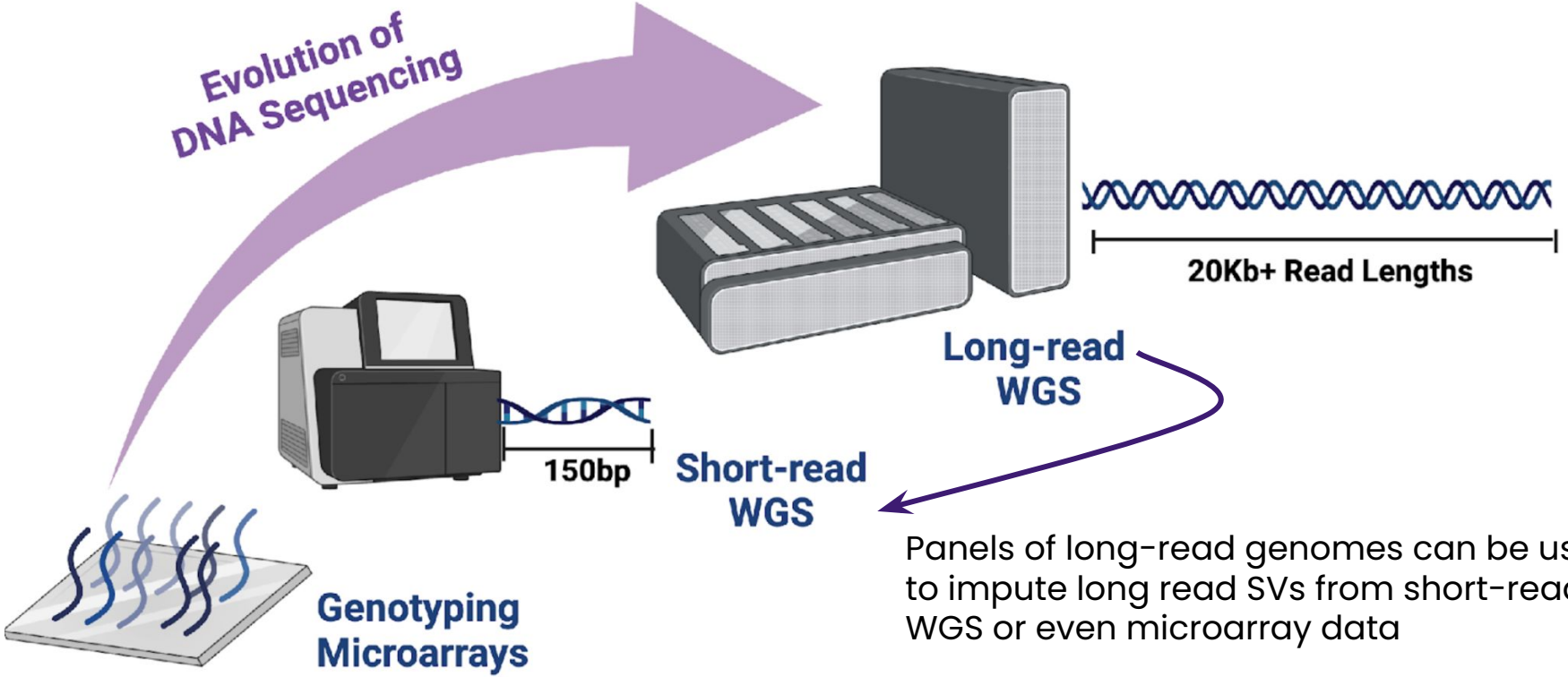
31k Accesses | **13** Citations | **46** Altmetric | Metrics

Imputation panels can fill in missing variants based on LD patterns from typed markers



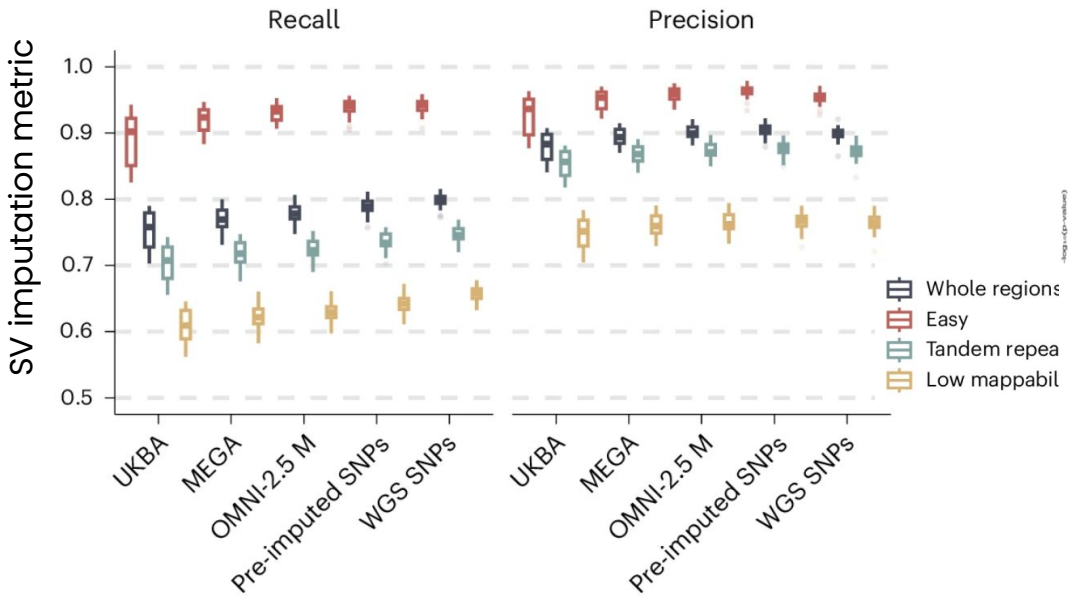
Large panels of short-read WGS have been used to impute whole-genome SNVs from array data

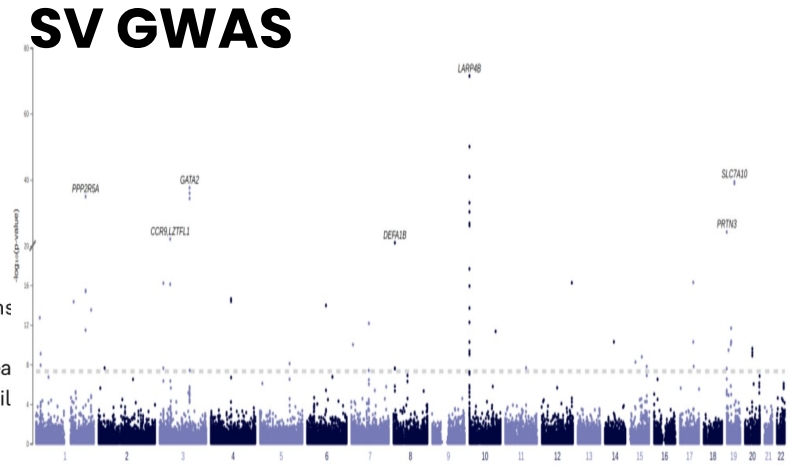
Analogously, SVs from long-read sequencing may be imputed from genome wide SNVs



Panels of long-read genomes can be used to impute long read SVs from short-read WGS or even microarray data

Yang et al 2026 show SVs can be imputed and tested directly in GWAS





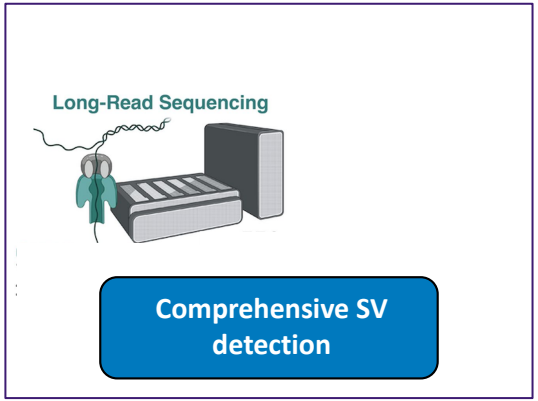
Downsides of imputation

- **Incomplete coverage:** rare and complex SVs may be impossible to impute, so GWAS will still be missing variants
- **Reduced association power:** imputation uncertainty weakens associations; increasing GWAS sample size helps, but poorly imputed variants can remain effectively untestable.
- **Limited mechanistic information:** an SV–disease association doesn't identify the affected gene, tissue, or molecular phenotype.

Conceptual workflow to find causal SVs from long-read sequencing in GWAS

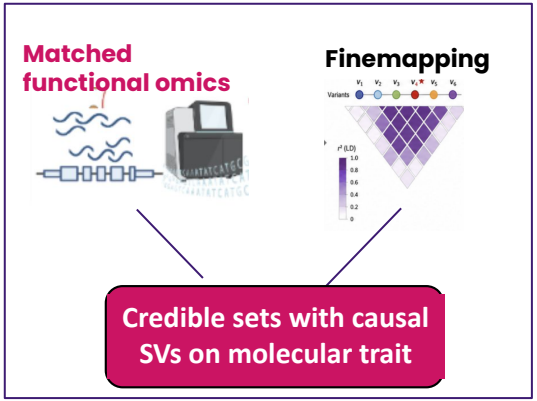
01

Long-read Cohort with Matched multi-omics



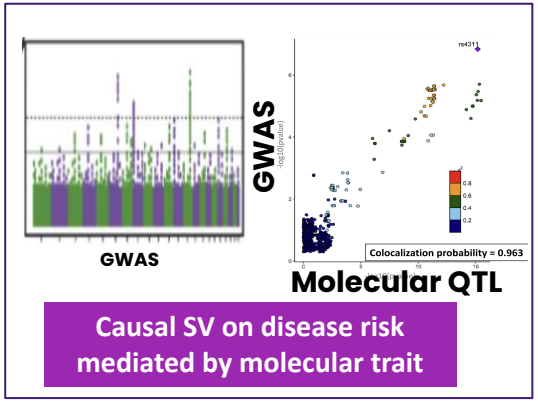
02

Joint statistical finemapping of SNPs + SVs with SuSiE



03

Coloc with GWAS over Proxy SNPs



02

ADRC/SAMS long-read and multi-omics resource

Long-read QTL resource for SV finemapping

› medRxiv [Preprint]. 2025 Oct 29:2025.10.10.25337775. doi: 10.1101/2025.10.10.25337775.

Long-read genome sequencing and multi-omics in aging and neurodegeneration

Tanner D Jensen [1], Yann Le Guen [2], Lia Talozzi [3 4], Sherry Yang [5], John Gorzynski [1 6], Andrés Peña-Tauber [3], Ilaria Stewart [3], Alexis Ferrasse [6], Daniel Nachun [7], Maggie T Arriaga [1], Justin Lee [2], Rafael Catoia Pulgrossi [2], Junyoung Park [3], Jingyu Zhang [3], Anthony D Wagner [8], Elizabeth C Mormino [3], Kathleen L Poston [3], Victor W Henderson [3 9], Zihuai He [2 3], Tony Wyss-Coray [3 10], Stephen B Montgomery [7], Euan A Ashley [6], Michael D Greicius [3]

Affiliations + expand

PMID: 41282933 PMCID: PMC12633103 DOI: 10.1101/2025.10.10.25337775

Paperpile PDF

To be published in *Nature Genetics* this year

We sequenced a Long-read, multi-omics neurodegeneration Cohort

Alzheimer's **D**isease **R**esearch **C**enter
\+ **S**tanford **A**ging and **M**emory **S**tudy

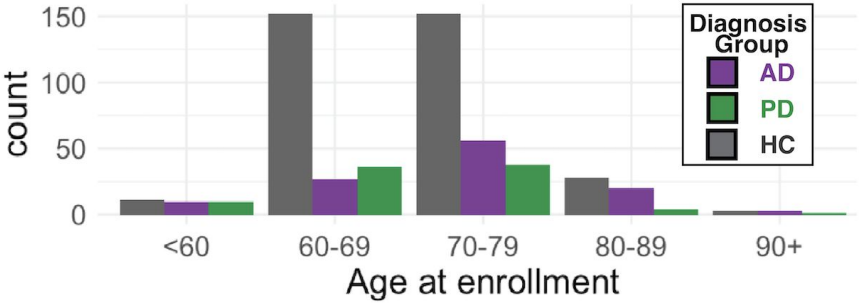


551 participants

Alzheimer's (n = **116**)

Parkinson's (n = **89**)

Healthy Control (n = **346**)



Deep Phenotyping



Longitudinal
Clinical Screening

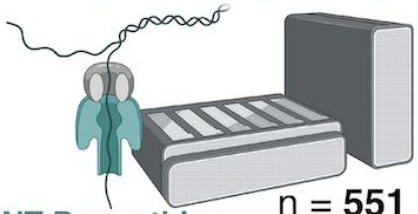


Brain MRI,
Amyloid PET
Tau PET

We sequenced a Long-read, multi-omics neurodegeneration Cohort

Multi-Platform Genome Sequencing

Long-Read Sequencing



ONT Promethion
18x Coverage
19Kb Read Length

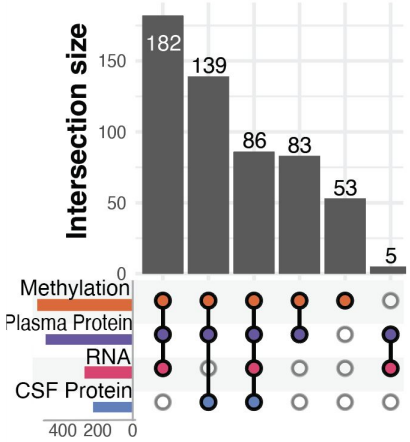
n = **551**
(366 with SRS)

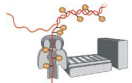
Short-Read Sequencing



n = **477**
(366 with LRS) **40x** Coverage

Matched Multi-omics Data

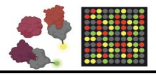




LRS Methylation
PBMCs n = **544**

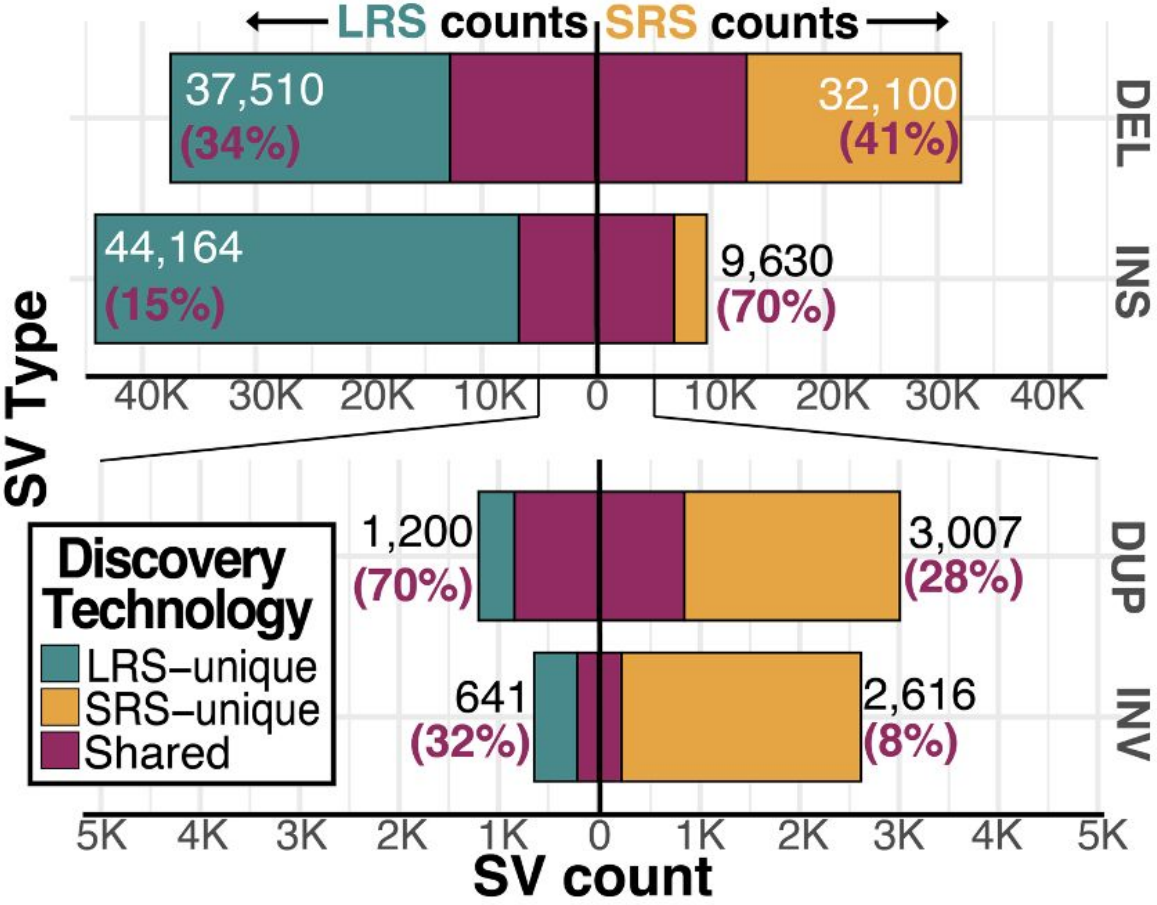


Transcriptomics
PBMCs n = **275**

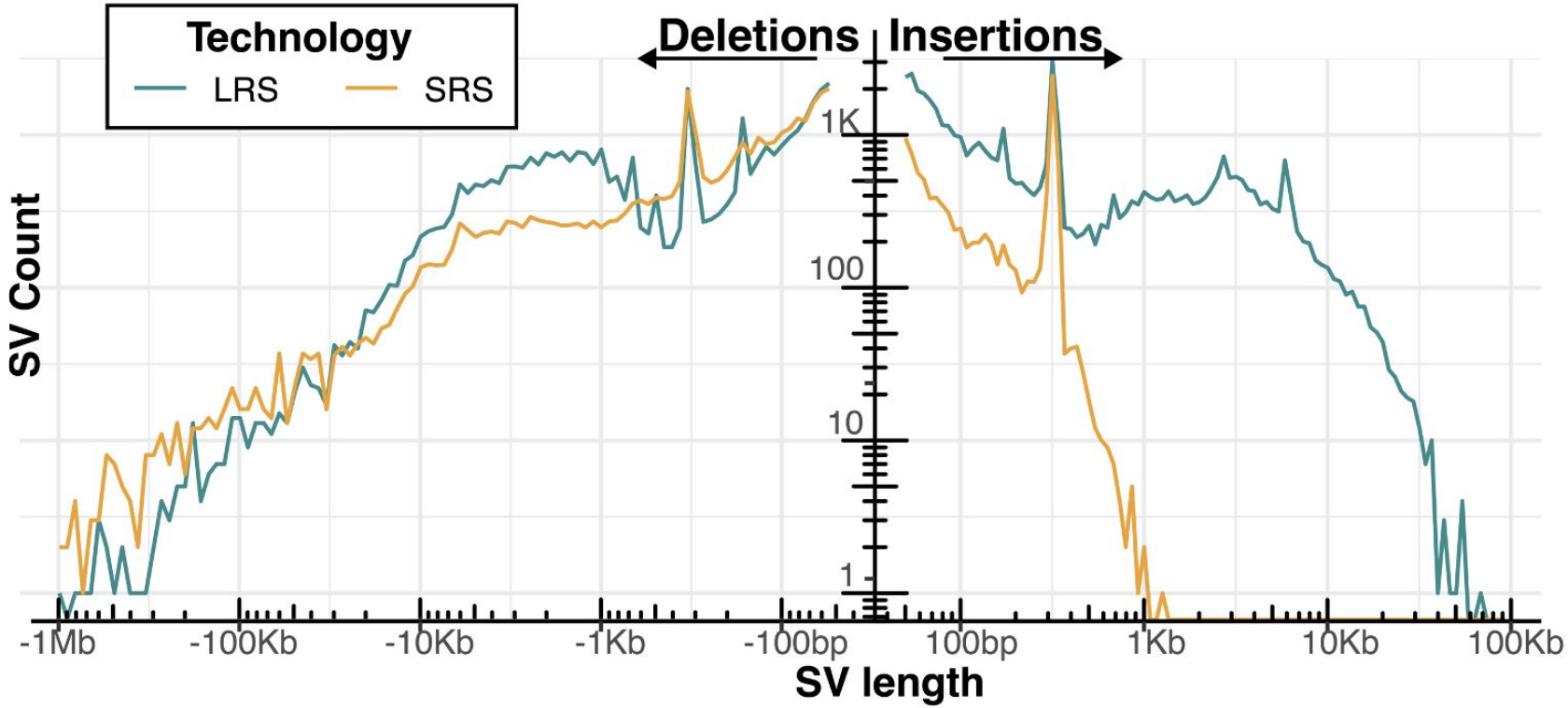


Proteomics
Plasma n = **495**,
CSF n = **226**

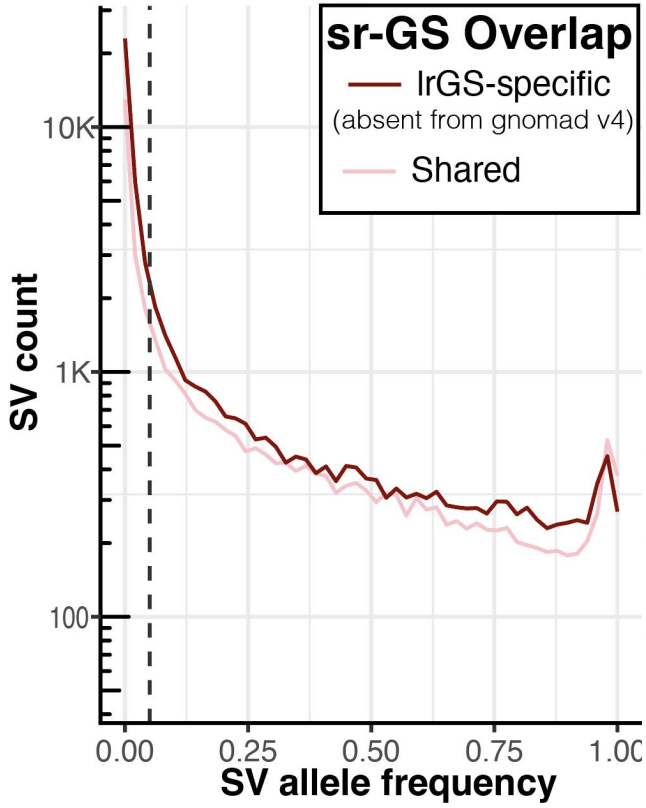
Long-read sequencing identifies more SVs than short reads



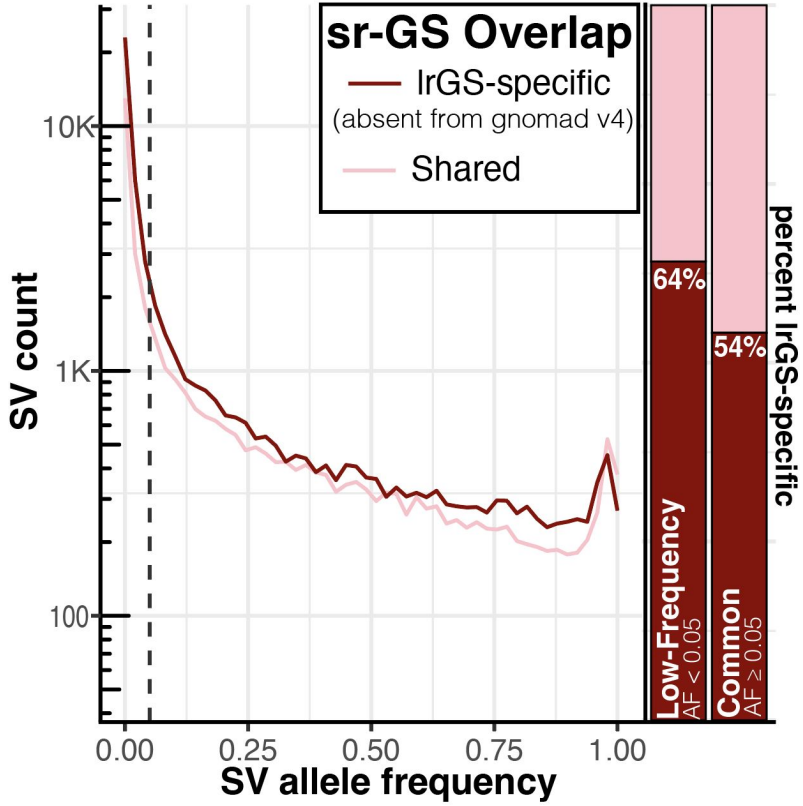
Long reads can esolve longer insertions than short reads



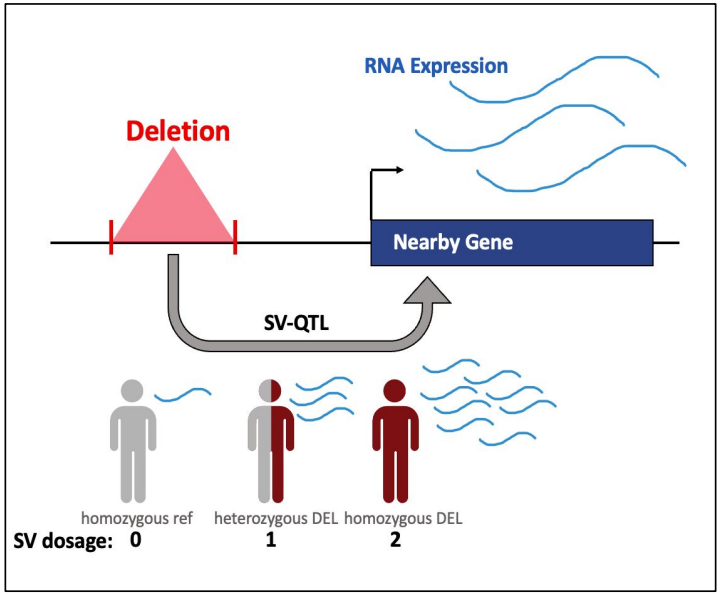
Population-scale sequencing allowed ascertainment of SV allele frequency

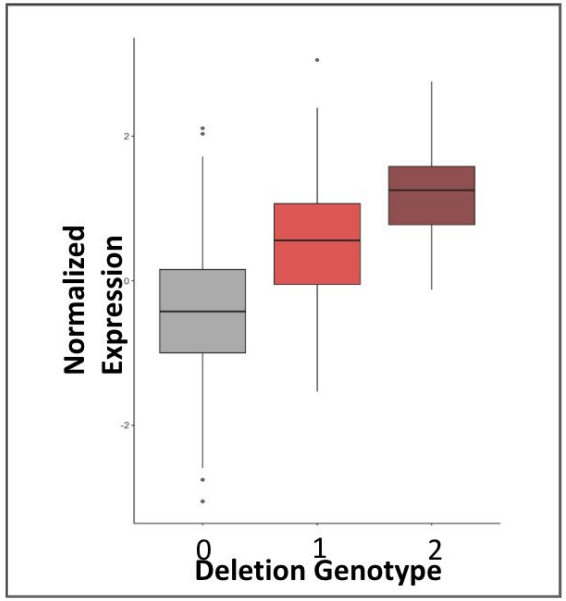


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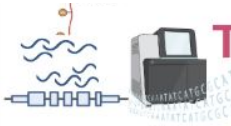
Using multi-omics data from ADRC/SAMS we mapped SV-QTLs across four molecular traits



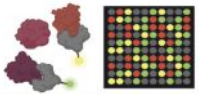




LRS Methylation
PBMCs n = **544**

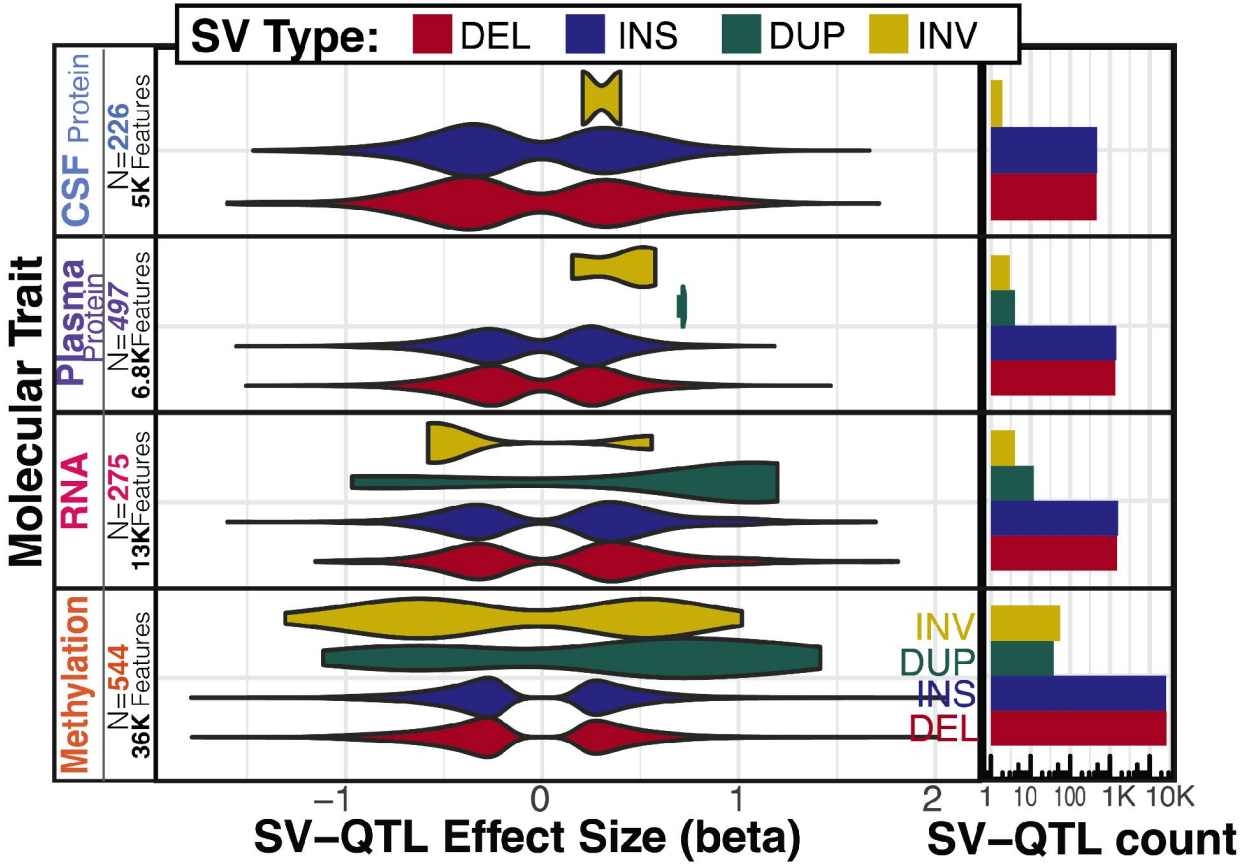


Transcriptomics
PBMCs n = **275**

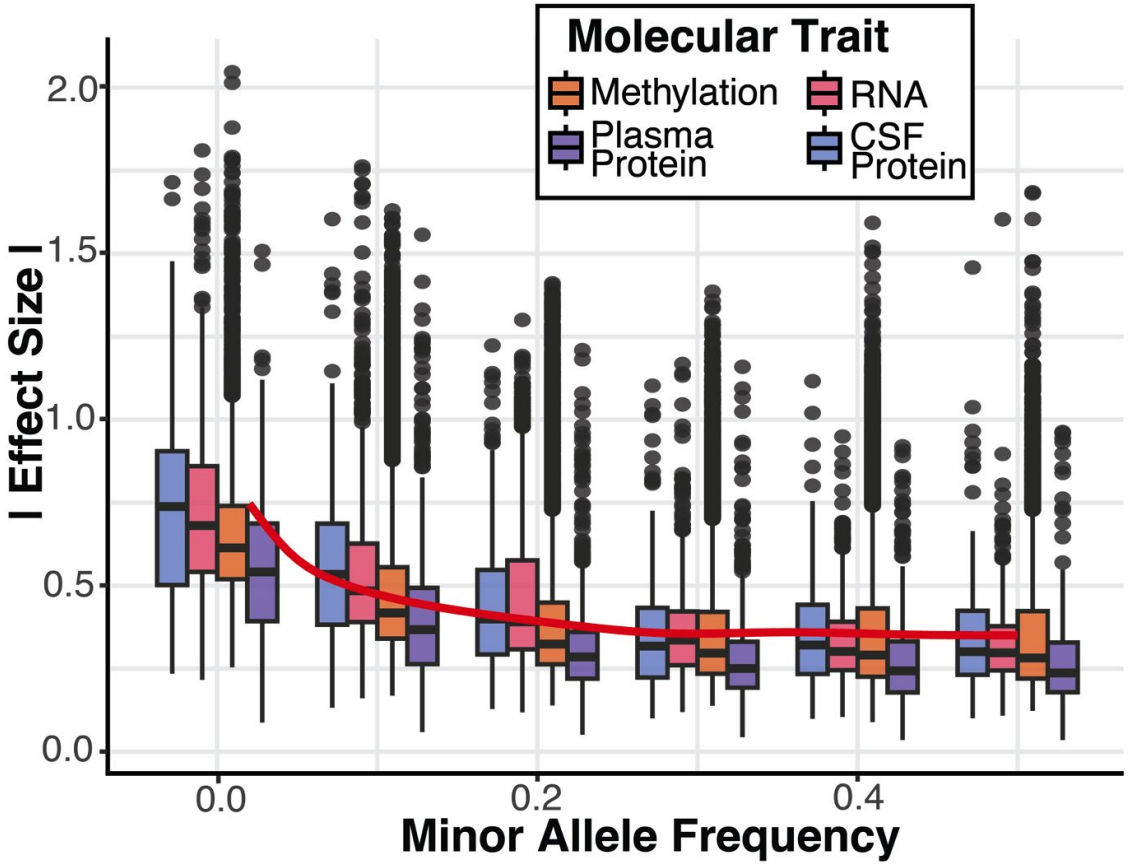


Proteomics
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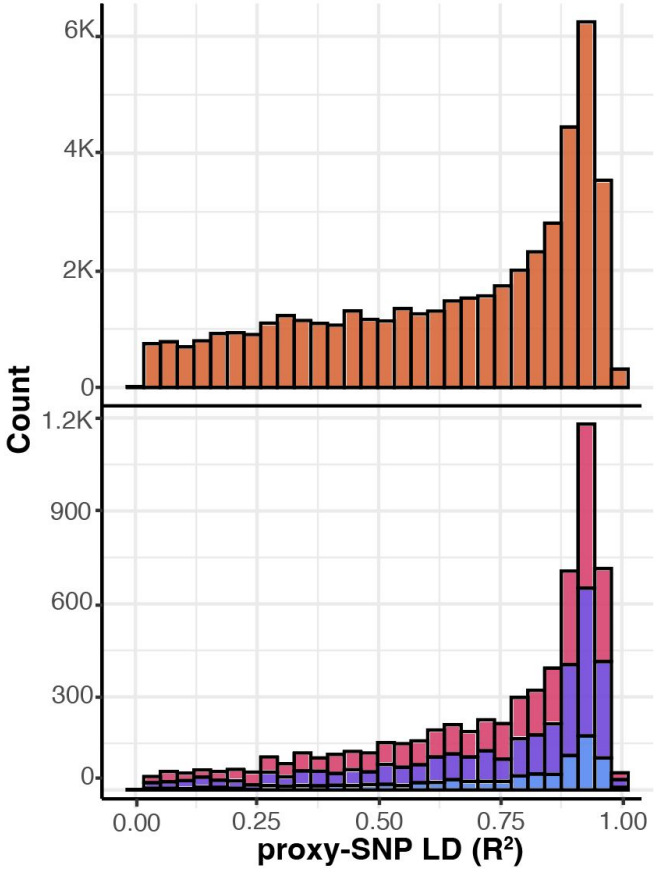
We detect thousands of SV-QTLs across different SV Types and omes



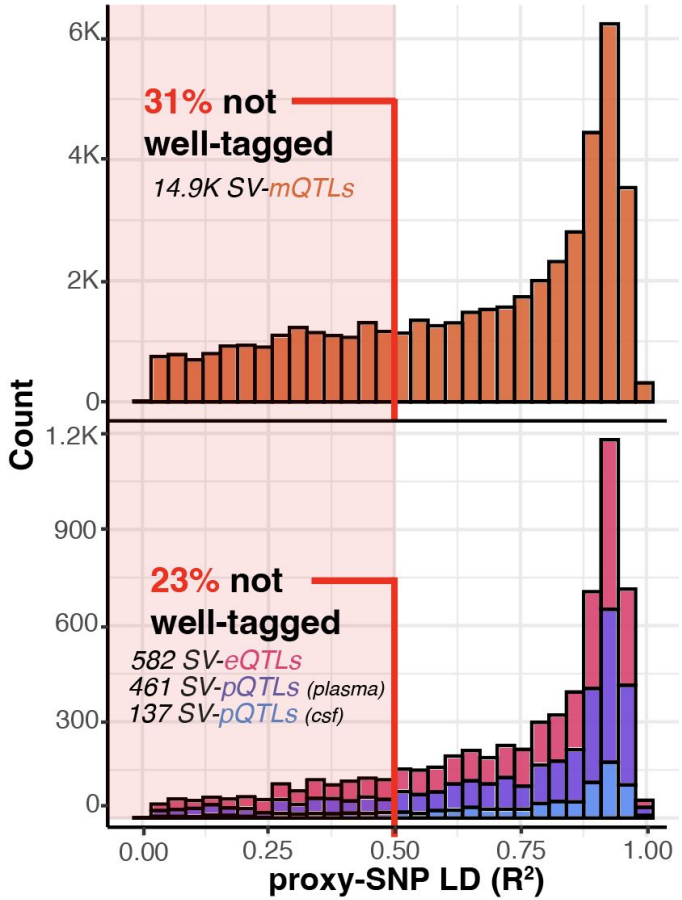
Low-frequency structural variants had larger effect sizes



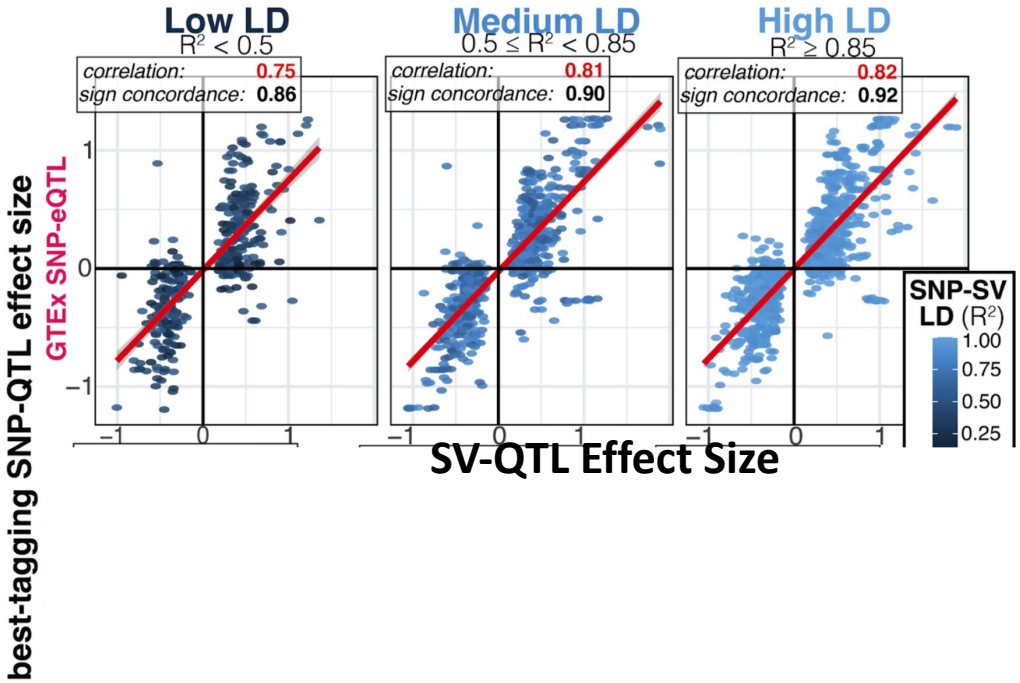
Most SV-QTLs are well-tagged by a nearby proxy-SNP



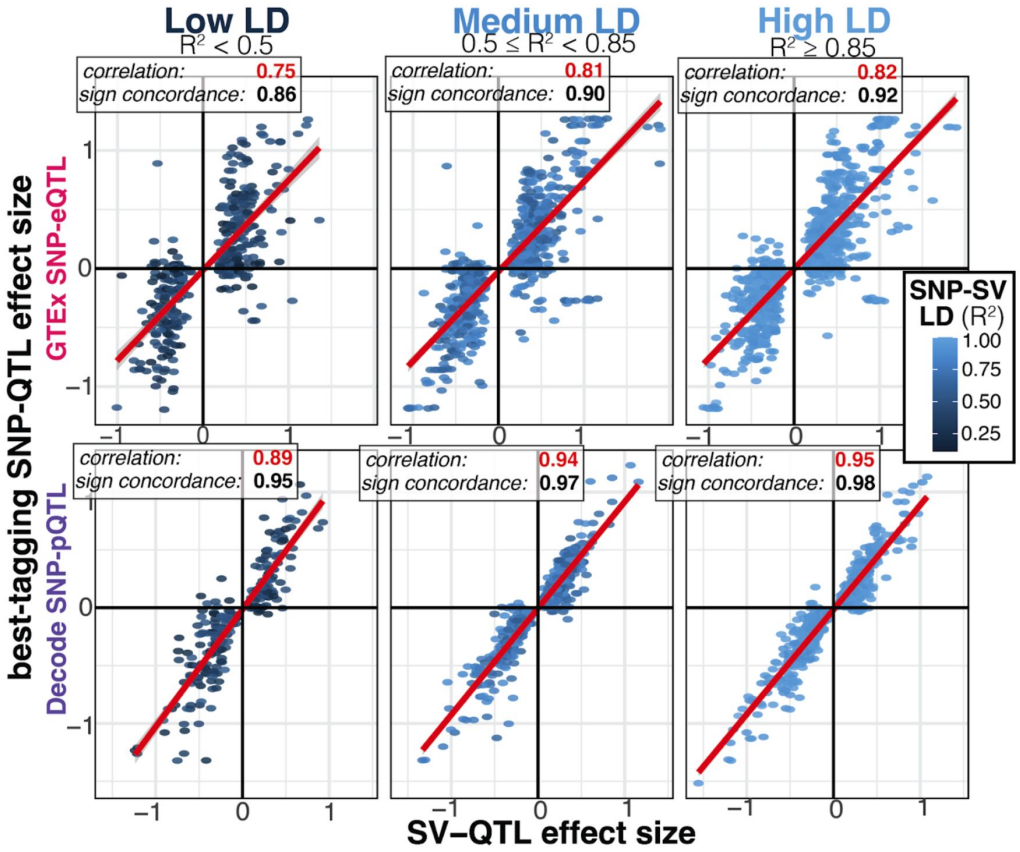
Up to 31% of SV-QTLs aren't well-tagged and are novel associations



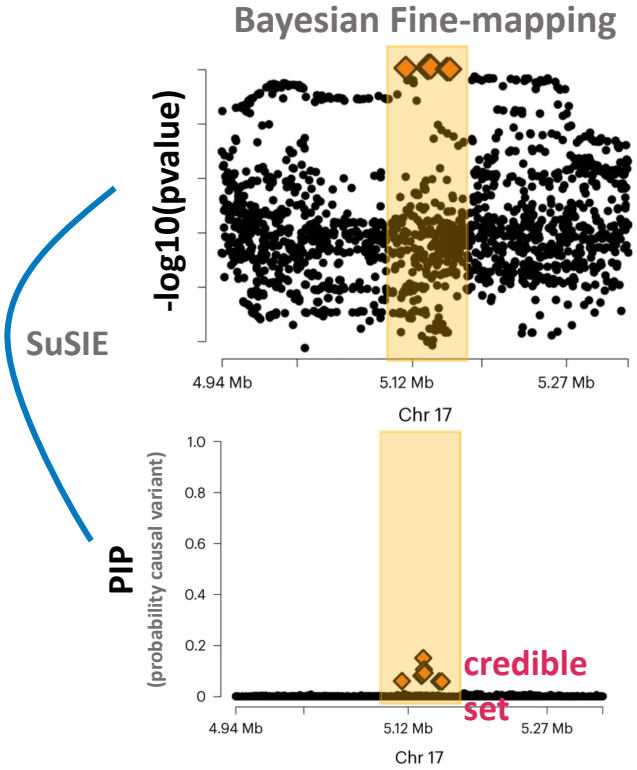
SV-QTL betas are highly correlated with SNP-betas from deCODE pQTLs



SV-QTL betas are highly correlated with SNP-betas from deCODE pQTLs

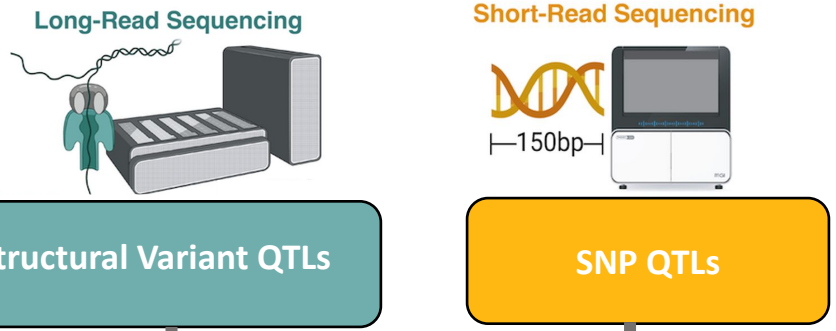


Joint statistical fine-mapping can identify associations that may be driven by SV



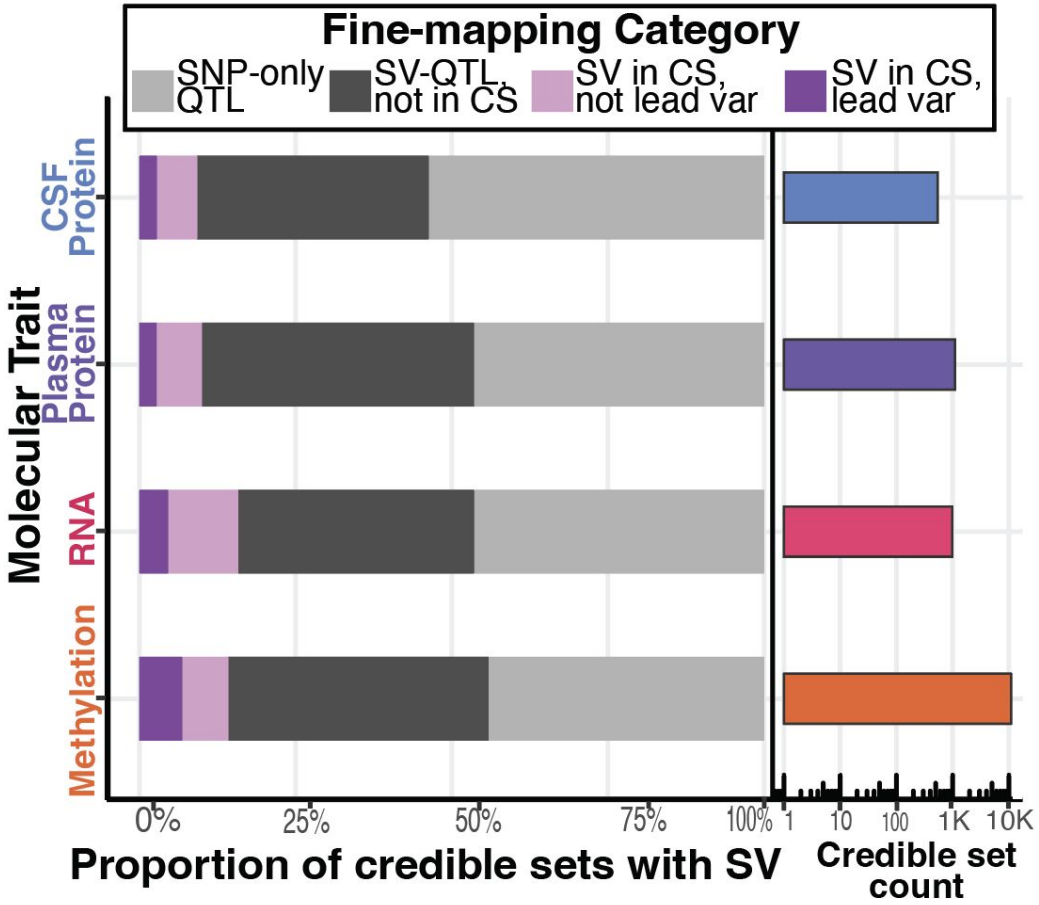
PIP (Posterior Inclusion Probability**)**: certainty that a specific variant is causal

Credible Set: subset of variants likely to contain true causal variant

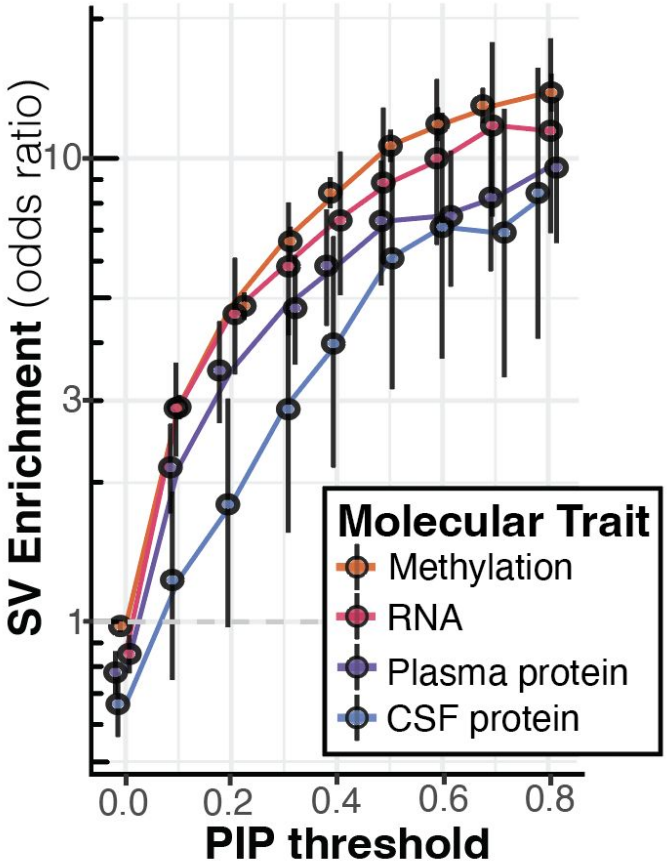


Does Credible Set Contain SV?

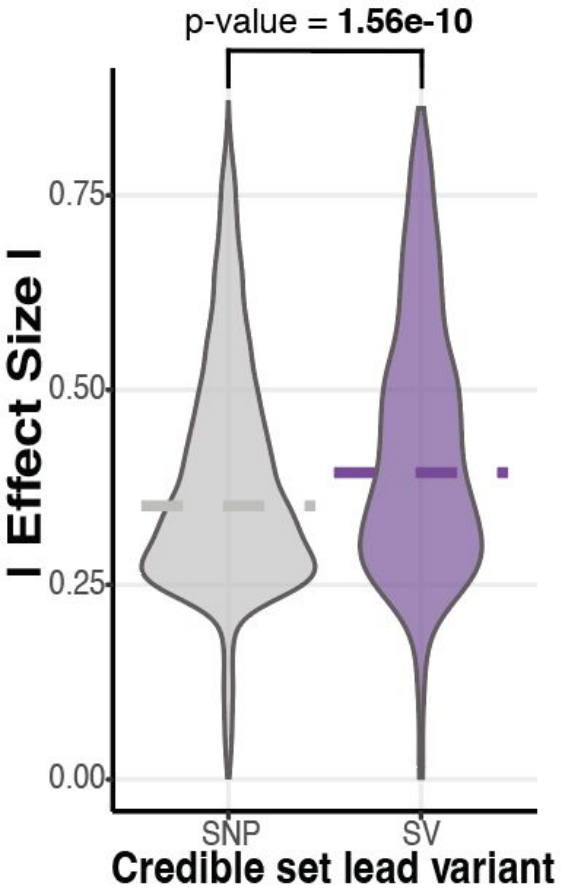
12% of QTL credible sets contained a structural variant



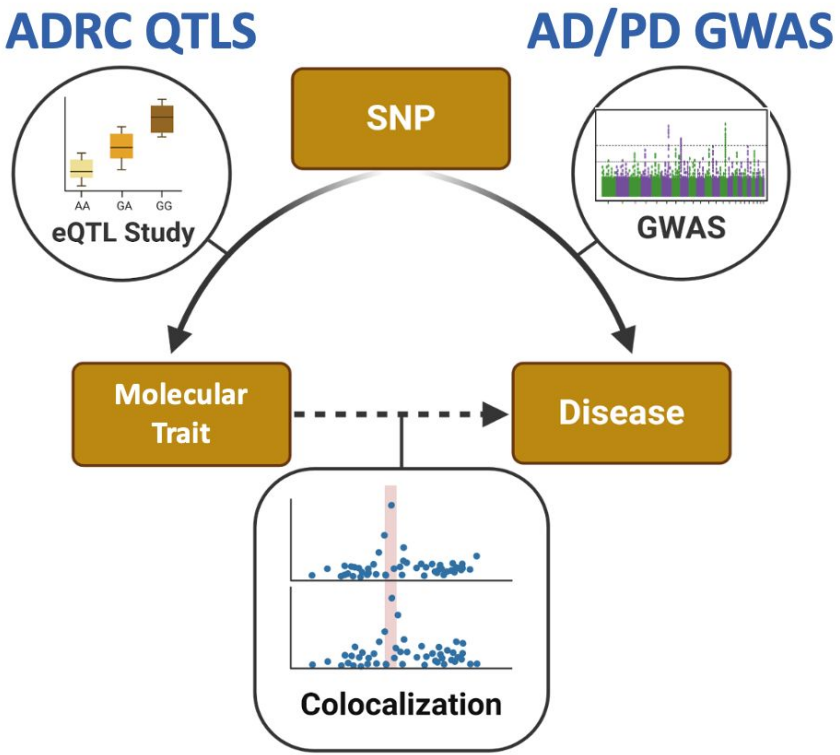
SVs 10 times more likely than SNPs to be fine-mapped with high PIP

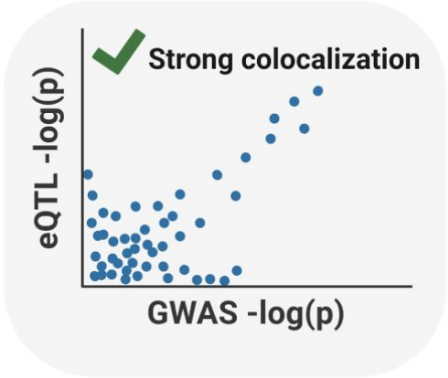


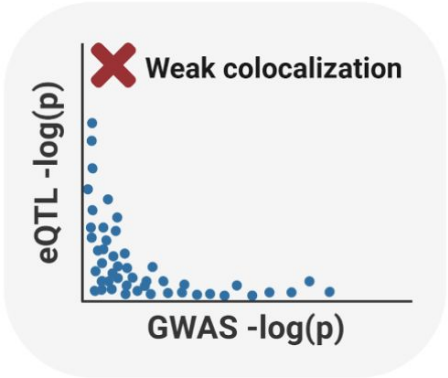
Credible sets with a lead SVs had stronger effect sizes



GWAS integration and colocalization can identify QTLs relevant to disease risk

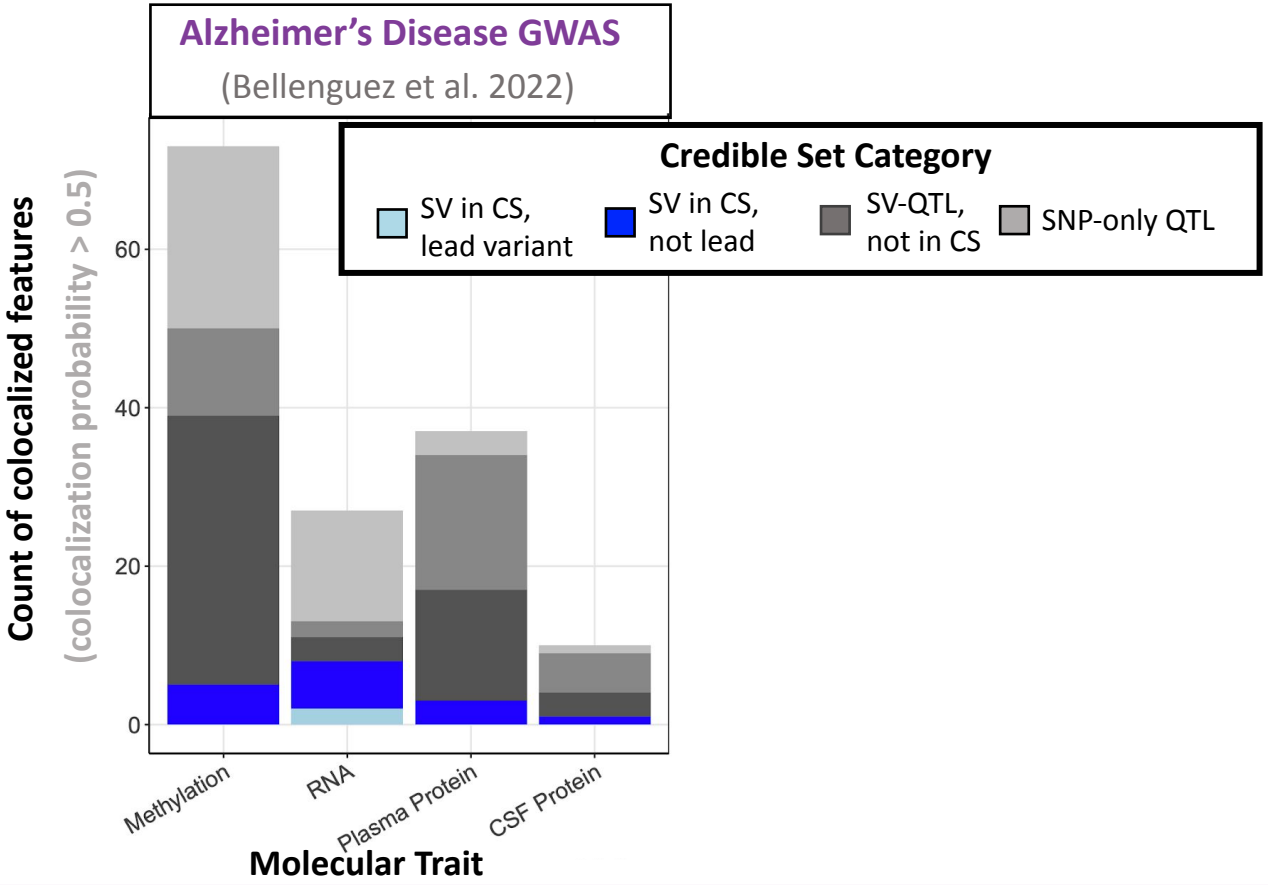




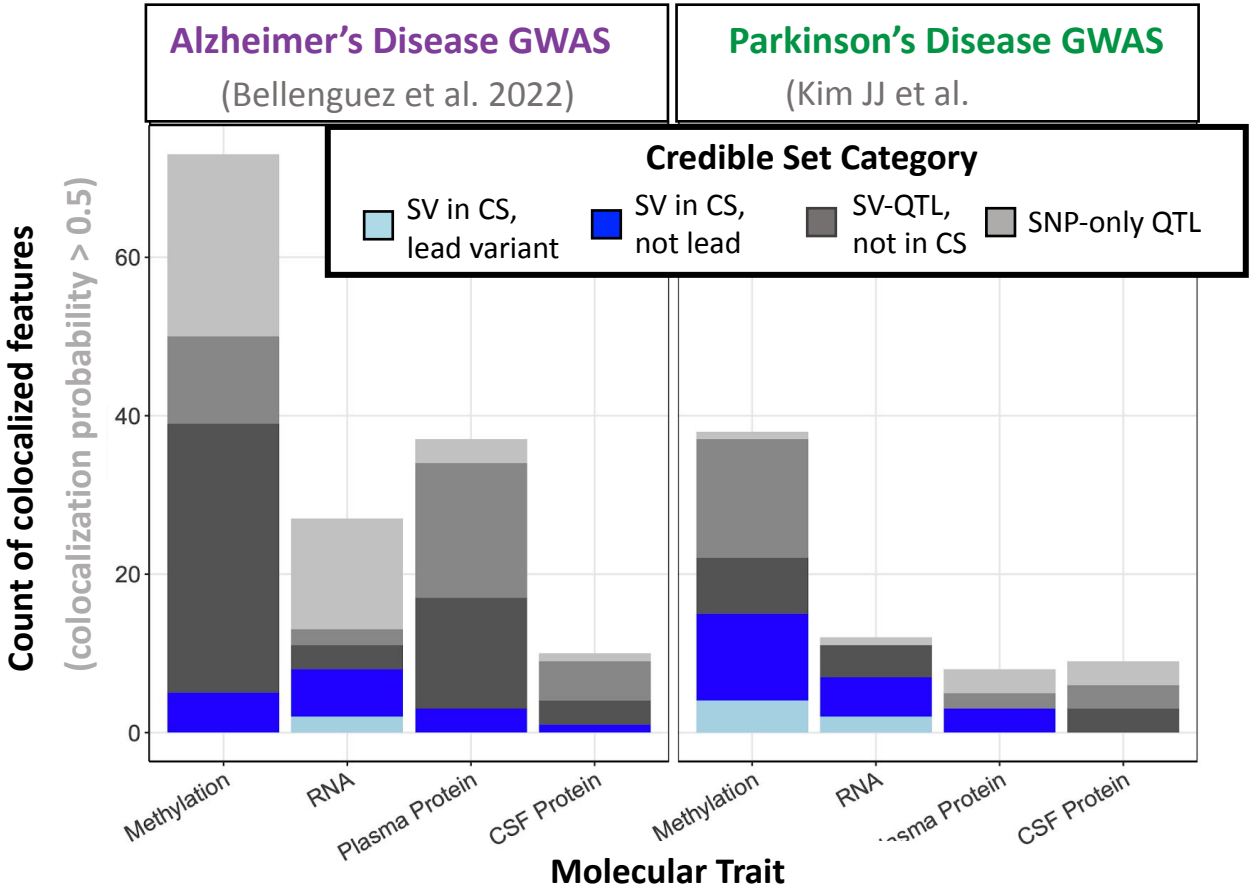


Slides adapted from Pagé Goddard

40 QTL associations that colocalized with AD or PD GWAS may contain causal SV



40 QTL associations that colocalized with AD or PD GWAS may contain causal SV



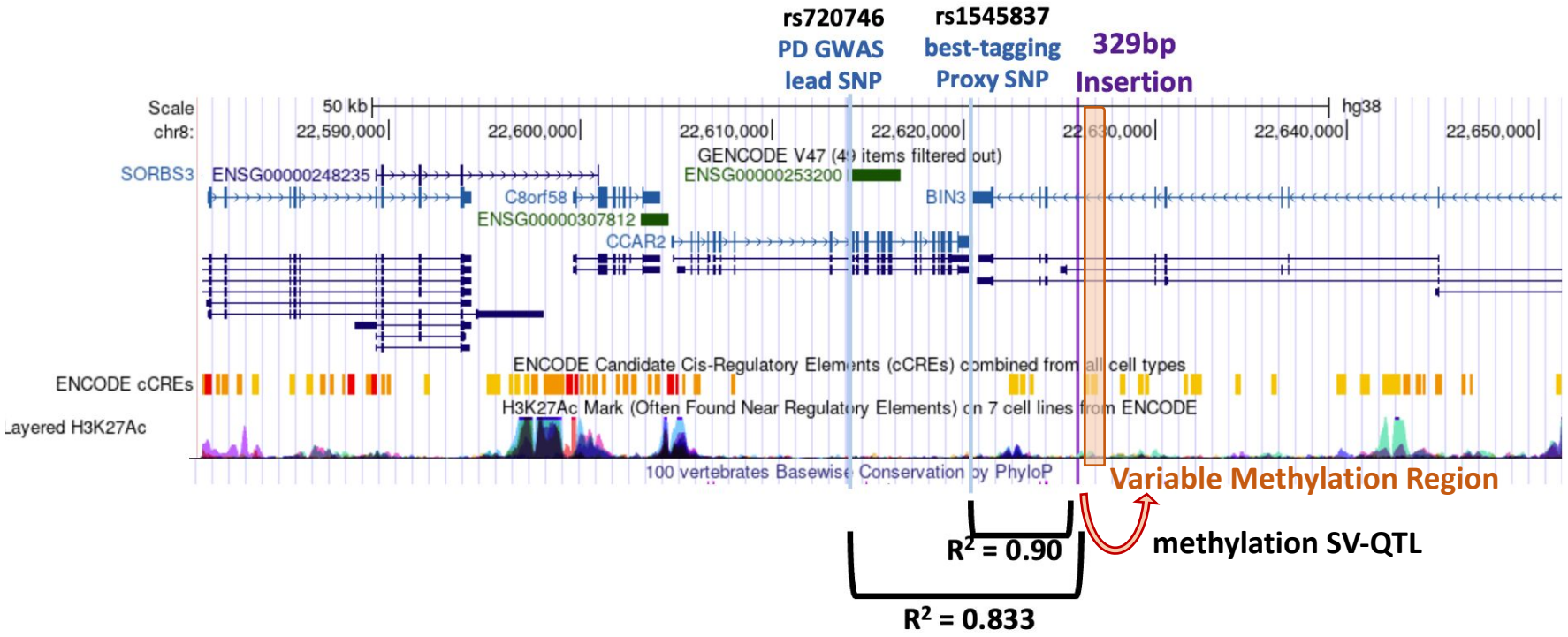
8

Alzheimer's Loci colocalized with an ADRC QTL with potentially causal SV

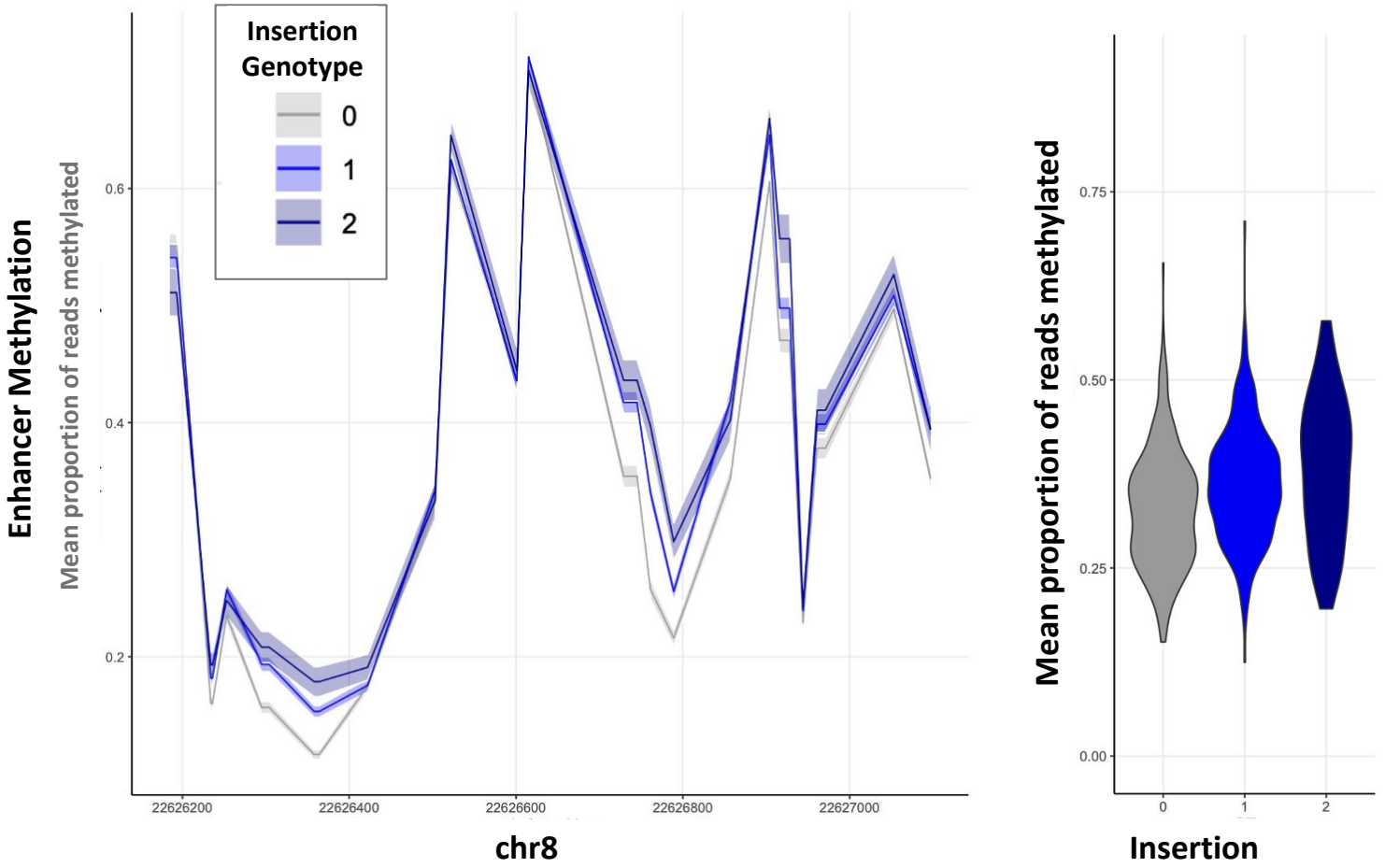
5

Parkinson's Loci colocalized with an ADRC QTL with potentially causal SV

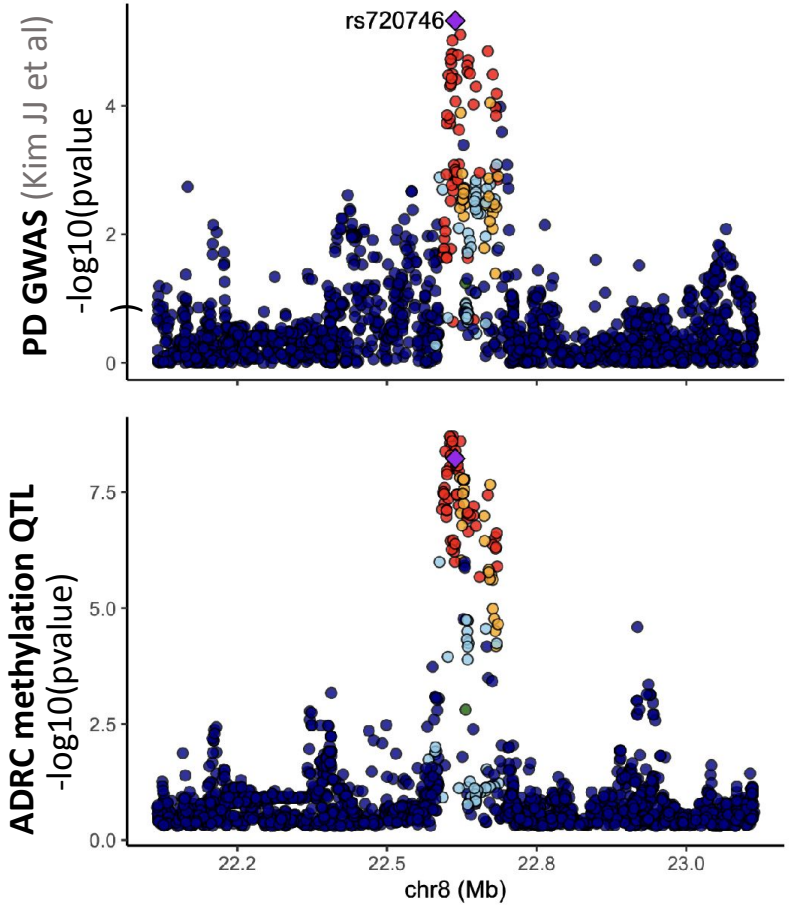
Insertion in *BIN3* associated with variable methylation levels of nearby enhancer

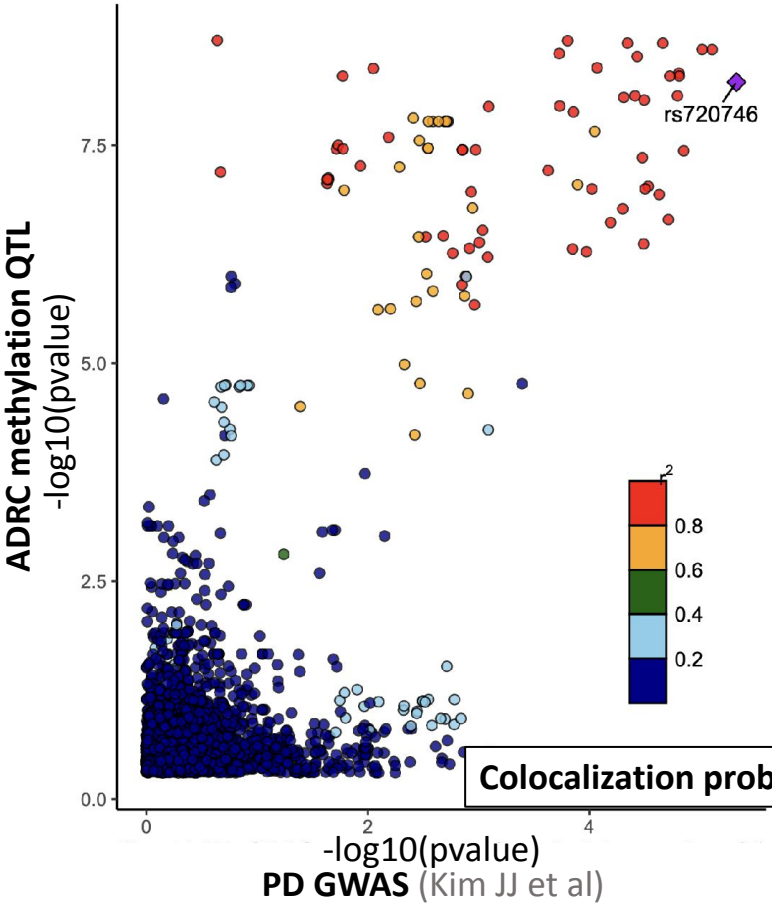


Insertion in *BIN3* associated with variable methylation levels of nearby enhancer



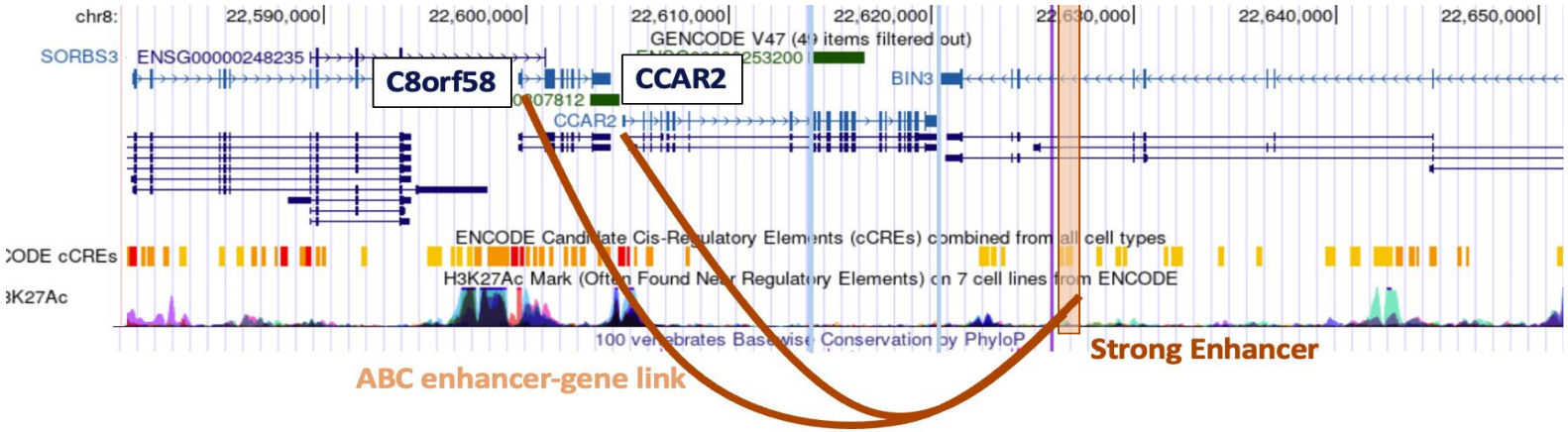
Methylation for enhancer in *BIN3* is associated with Parkinson's risk

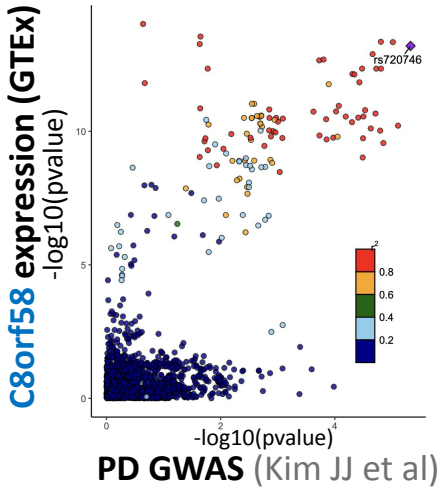


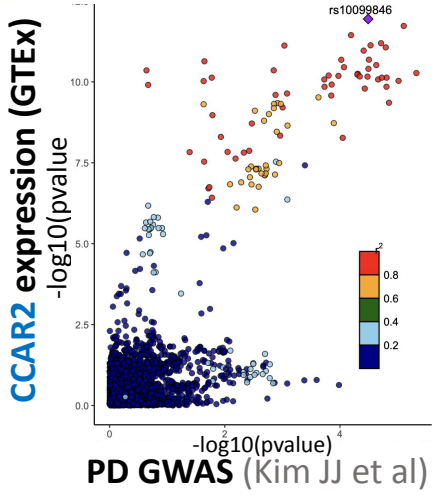


Colocalization probability = 0.894

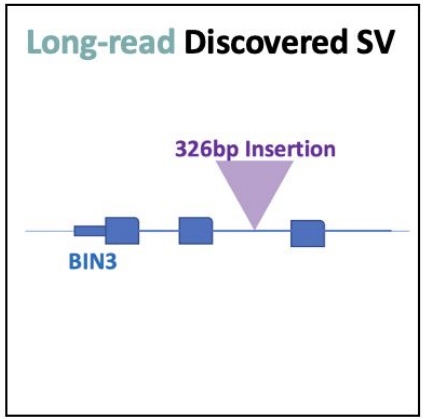
Enhancer-gene linking maps this enhancer to nearby genes whose expression also colocalizes with Parkinson's

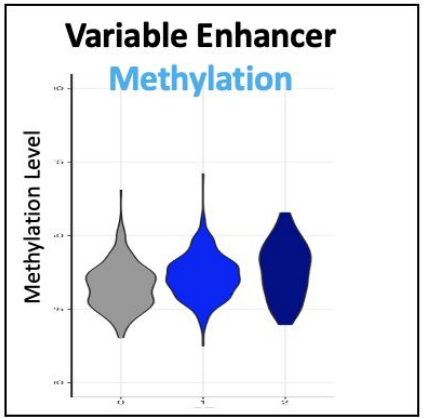


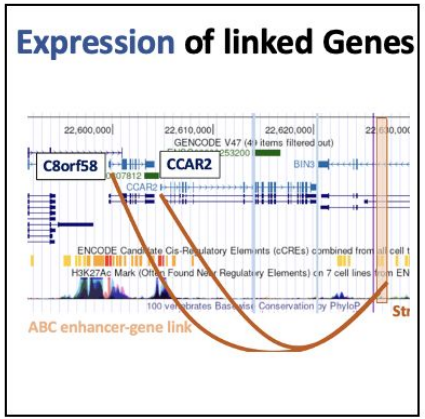


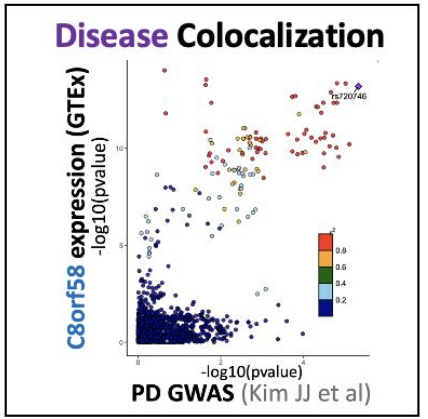


Integrating Long-Read Sequencing and Multi-Omics data of a neurodegenerative disease cohort allow us to pinpoint functional SVs that may drive disease risk

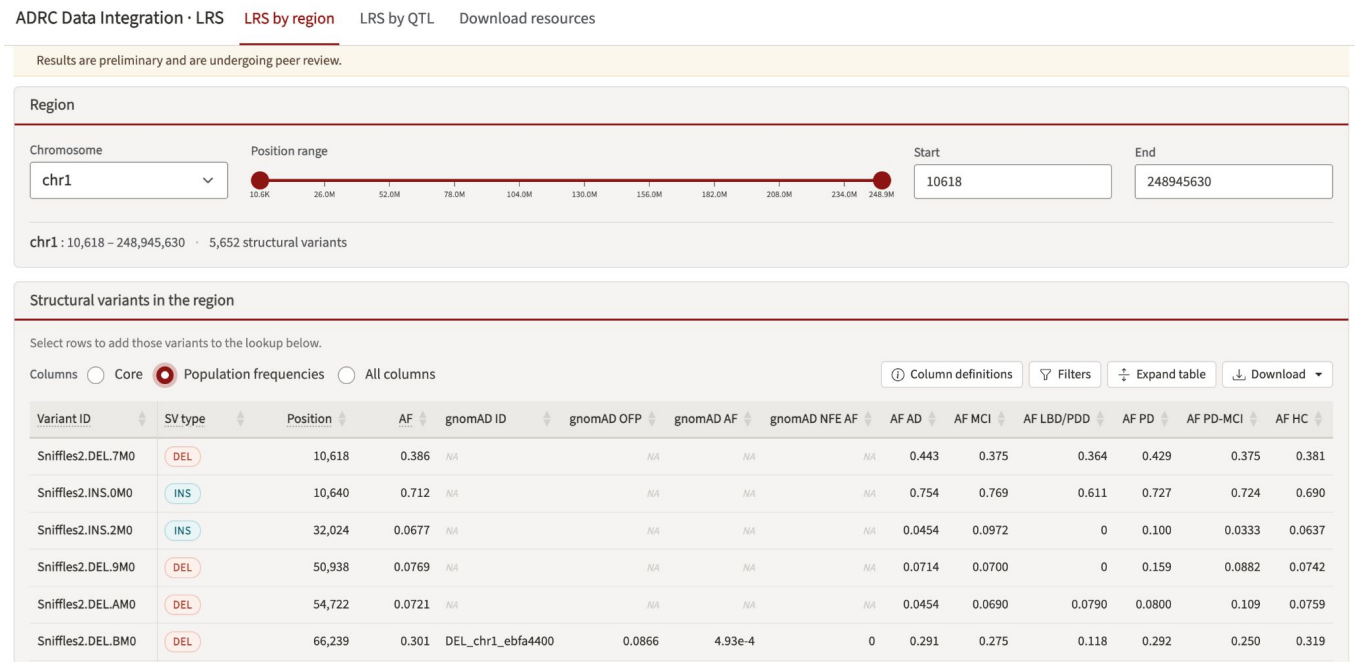








Query variants and QTL results in Shiny App



Variant ID	SV type	Position	AF	gnomAD ID	gnomAD OFP	gnomAD AF	gnomAD NFE AF	AF AD	AF MCI	AF LBD/PDD	AF PD	AF PD-MCI	AF HC
Sniffles2.DEL.7M0	DEL	10,618	0.386	NA	NA	NA	NA	0.443	0.375	0.364	0.429	0.375	0.381
Sniffles2.INS.0M0	INS	10,640	0.712	NA	NA	NA	NA	0.754	0.769	0.611	0.727	0.724	0.690
Sniffles2.INS.2M0	INS	32,024	0.0677	NA	NA	NA	NA	0.0454	0.0972	0	0.100	0.0333	0.0637
Sniffles2.DEL.9M0	DEL	50,938	0.0769	NA	NA	NA	NA	0.0714	0.0700	0	0.159	0.0882	0.0742
Sniffles2.DEL.AM0	DEL	54,722	0.0721	NA	NA	NA	NA	0.0454	0.0690	0.0790	0.0800	0.109	0.0759
Sniffles2.DEL.BM0	DEL	66,239	0.301	DEL_chr1_ebfa4400	0.0866	4.93e-4	0	0.291	0.275	0.118	0.292	0.250	0.319

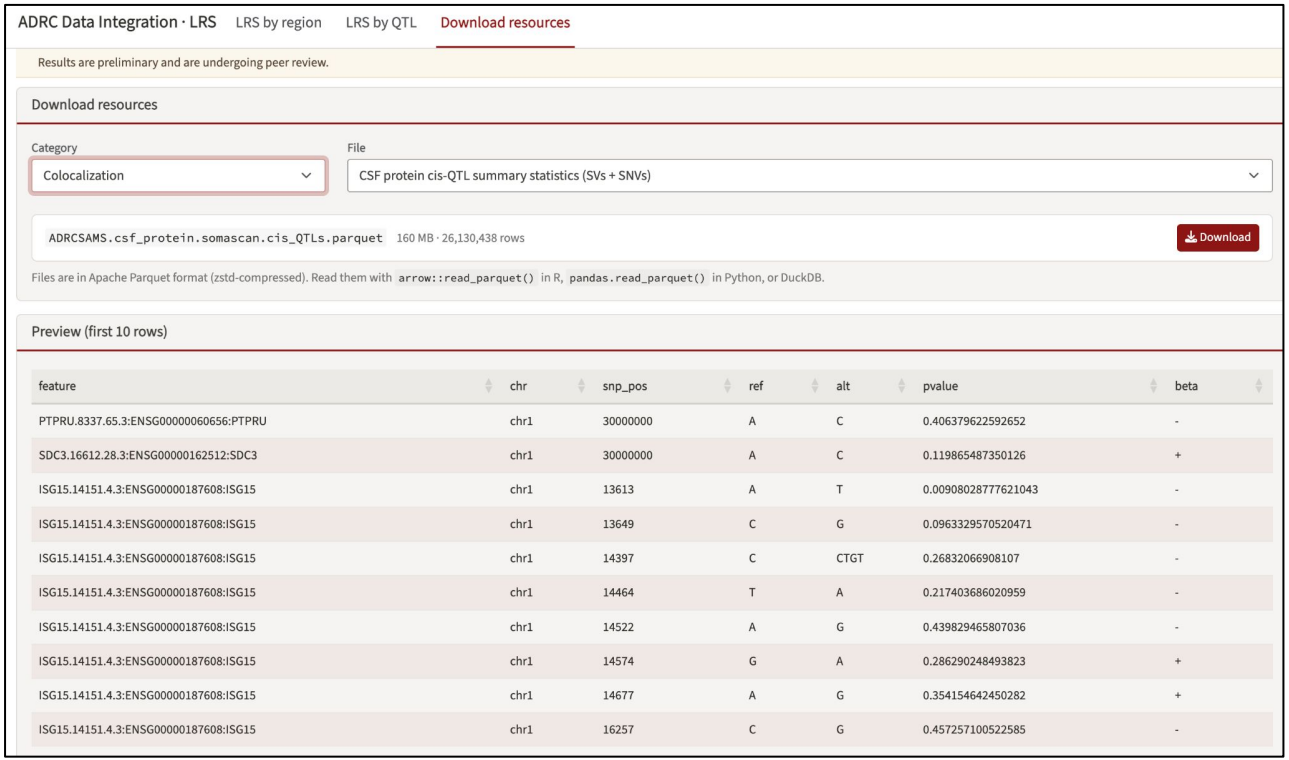
https://qsushiny.shinyapps.io/adrc_lrs/

Shiny app

Existing QTL SV finemapping summary for ADRC/SAMS molecular traits

1	Pheno	Ome	has_QTL	lowest_SV_pval	lowest_SNV_pva	number_credible_se	highest_pip_va	SV_in_cs	Finemapping_category	Finemapped_svs
36284	ENSG00000002933:TMEM176A	RNA	TRUE	3.84E-07	1.13E-21	1	SNP	FALSE	SV-QTL_noncausal	
36287	ENSG00000003249:DBNDD1	RNA	TRUE	2.68E-08	3.06E-11	1	SNP	FALSE	SV-QTL_noncausal	
36300	ENSG00000005020:SKAP2	RNA	TRUE	0.001032891	1.54E-09	1	SNP	FALSE	SNP-only_QTL	
36321	ENSG00000006282:SPATA20	RNA	TRUE	2.04E-05	8.58E-24	1	SNP	FALSE	SV-QTL_noncausal	
36325	ENSG00000006432:MAP3K9	RNA	TRUE	3.16E-05	6.55E-09	1	SNP	FALSE	SV-QTL_noncausal	
36333	ENSG00000006652:IFRD1	RNA	TRUE	0.003588094	4.60E-07	1	SNP	FALSE	SNP-only_QTL	
36340	ENSG00000007129:CEACAM21	RNA	TRUE	0.000421358	8.93E-11	1	SNP	FALSE	SNP-only_QTL	
36363	ENSG00000008311:AASS	RNA	TRUE	0.036391797	1.24E-07	1	SNP	FALSE	SNP-only_QTL	
36376	ENSG00000010030:ETV7	RNA	TRUE	0.000638799	5.83E-14	1	SNP	FALSE	SNP-only_QTL	
36405	ENSG00000013573:DDX11	RNA	TRUE	2.04E-07	2.84E-36	1	SNP	FALSE	SV-QTL_noncausal	
36423	ENSG00000019169:MARCO	RNA	TRUE	0.000448768	1.98E-08	1	SNP	FALSE	SNP-only_QTL	
36437	ENSG00000021762:OSBPL5	RNA	TRUE	0.006736311	6.07E-09	1	SNP	FALSE	SNP-only_QTL	
36440	ENSG00000022556:NLRP2	RNA	TRUE	2.07E-14	3.49E-13	1	SV	TRUE	SV-QTL_lead_var	Sniffles2.DEL.17B9M12
36463	ENSG00000033122:LRRC7	RNA	TRUE	2.19E-05	2.22E-14	1	SNP	FALSE	SV-QTL_noncausal	
36472	ENSG00000036448:MYOM2	RNA	TRUE	9.08E-20	7.28E-27	2	SNP	TRUE	SV-QTL_not_lead	Sniffles2.INS.327M7
36494	ENSG00000041515:MYO16	RNA	TRUE	0.014533309	4.16E-23	1	SNP	FALSE	SNP-only_QTL	
36497	ENSG00000042493:CAPG	RNA	TRUE	0.001625246	9.29E-12	1	SNP	FALSE	SNP-only_QTL	
36504	ENSG00000043591:ADRB1	RNA	TRUE	0.004098744	1.19E-09	1	SNP	FALSE	SNP-only_QTL	
36505	ENSG00000044115:CTNNA1	RNA	TRUE	0.094233423	2.90E-07	1	SNP	FALSE	SNP-only_QTL	
36512	ENSG00000047662:FAM184B	RNA	TRUE	1.15E-06	1.18E-09	1	SNP	FALSE	SV-QTL_noncausal	
36523	ENSG00000049192:ADAMTS6	RNA	TRUE	0.003954342	3.52E-09	1	SNP	FALSE	SNP-only_QTL	
36524	ENSG00000049247:UTS2	RNA	TRUE	0.000227751	1.06E-12	1	SNP	FALSE	SNP-only_QTL	
36526	ENSG00000049283:EPN3	RNA	TRUE	0.0542711	7.52E-09	1	SNP	FALSE	SNP-only_QTL	
36533	ENSG00000050555:LAMC3	RNA	TRUE	0.010348555	9.41E-10	2	SNP	FALSE	SNP-only_QTL	
36542	ENSG00000051523:CYBA	RNA	TRUE	0.000307181	5.32E-12	1	SNP	FALSE	SNP-only_QTL	
36543	ENSG00000051620:HEBP2	RNA	TRUE	2.02E-10	1.84E-19	1	SNP	FALSE	SV-QTL_noncausal	
36549	ENSG00000053524:MCF2L2	RNA	TRUE	0.0114333	1.18E-07	1	SNP	FALSE	SNP-only_QTL	
36560	ENSG00000054793:ATP9A	RNA	TRUE	0.005706909	9.56E-15	1	SNP	FALSE	SNP-only_QTL	
36565	ENSG00000055483:USP36	RNA	TRUE	2.08E-06	3.42E-08	1	SNP	FALSE	SV-QTL_noncausal	
36583	ENSG00000059122:FLYWCH1	RNA	TRUE	1.88E-22	2.54E-23	1	SNP	TRUE	SV-QTL_not_lead	Sniffles2.INS.2AEMF

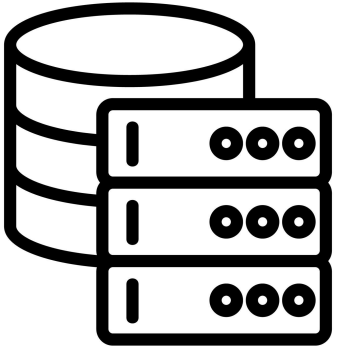
Full QTL summary stats downloadable allowing colocalization with any complex trait GWAS



feature	chr	snp_pos	ref	alt	pvalue	beta
PTPRU.8337.65.3:ENSG00000060656:PTPRU	chr1	30000000	A	C	0.406379622592652	-
SDC3.16612.28.3:ENSG00000162512:SDC3	chr1	30000000	A	C	0.119865487350126	+
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	13613	A	T	0.00908028777621043	-
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	13649	C	G	0.0963329570520471	-
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	14397	C	CTGT	0.26832066908107	-
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	14464	T	A	0.217403686020959	-
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	14522	A	G	0.439829465807036	-
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	14574	G	A	0.286290248493823	+
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	14677	A	G	0.354154642450282	+
ISG15.14151.4.3:ENSG00000187608:ISG15	chr1	16257	C	G	0.457257100522585	-

Full lr-GS BAMs available through NIAGADs

- Individual-level data available through approved Data Access Request



03

Interactive workbook: Google Co-lab code to finemap and colocalize AD and PD risk SVs

Practical advice on running finemapping and coloc experiments

https://github.com/Junkkkk/sv-finemapping-coloc-tutorial/tree/main

Q&A – Notebook 1

1. At which leads is an SV high-tagging ($r^2 \geq 0.6$)? Does high tagging by itself tell you the SV is causal?
2. The HP deletion is only moderately tagged by rs217181, yet it explains HP protein much better. What would an SNV-only fine-mapping of HP protein report?
3. One GWAS lead was not in our SNV panel. How could a missing lead change your list of candidate SVs?
4. Most SVs in these tables are common in this cohort. How would you use SV allele frequencies from a healthy ageing cohort when you filter candidate SVs in a patient with a rare disease?

At which leads is a SV “high-tagging” ($r^2 \geq 0.6$)? Does “high tagging” by itself tell you the SV is causal?

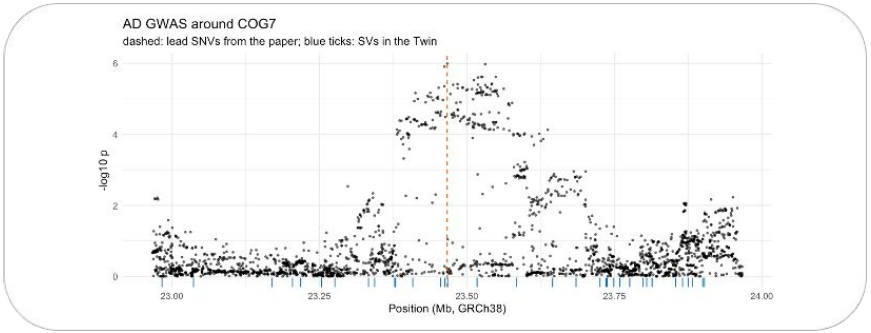
ANSWER

High-tagging SVs were found at **AD lead 16:23466676:C:CCA** and **PD lead 17:46276519:A:G**.

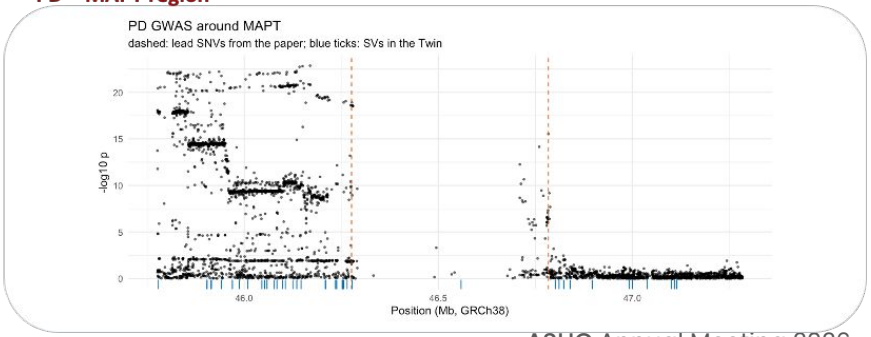
High LD means the lead and SV carry similar genotype information in this sample. It nominates the SV as a candidate, but it does not identify the causal allele.

Take-home: Tagging is evidence of correlation—not causality.

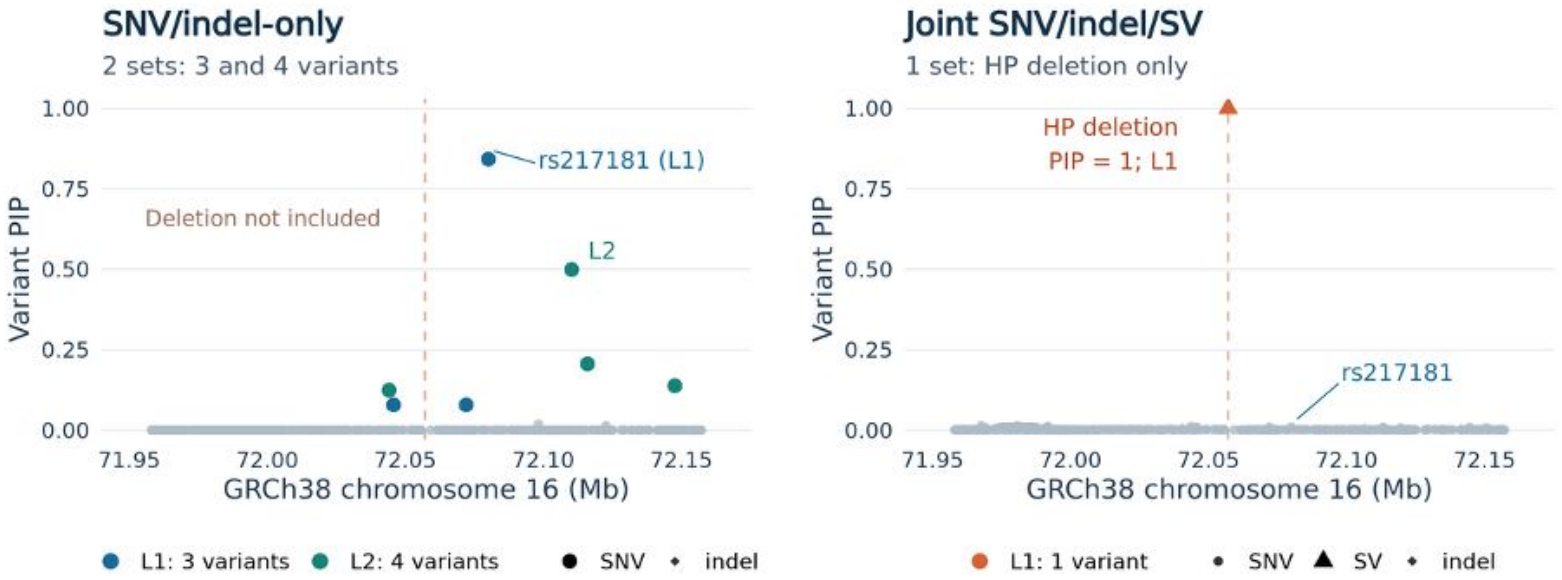
AD • COG7 region



PD • MAPT region

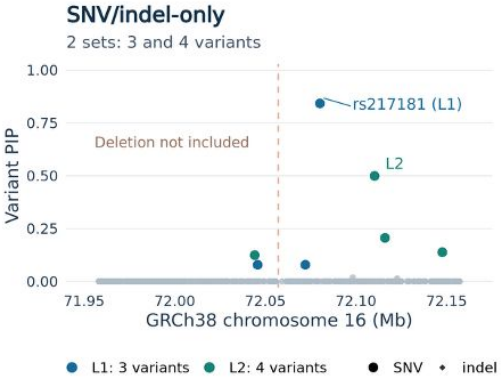


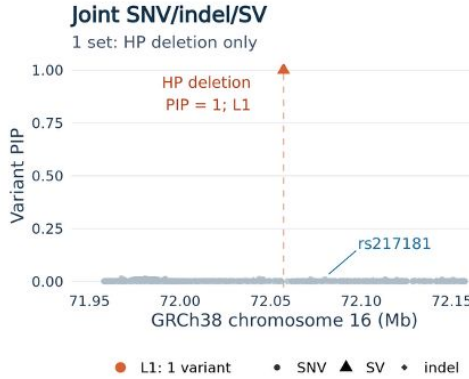
The HP deletion is only moderately tagged by rs217181, yet it explains HP protein much better. What would an SNV-only fine-mapping of HP protein report?



The HP deletion is only moderately tagged by rs217181, yet it explains HP protein much better. What would an SNV-only fine-mapping of HP protein report?

Only variants supplied to the model can receive PIP. **An SNV-only fit could nominate rs217181 or another proxy, but it could never nominate the missing deletion.** Incomplete tagging can also weaken or spread the association signal.





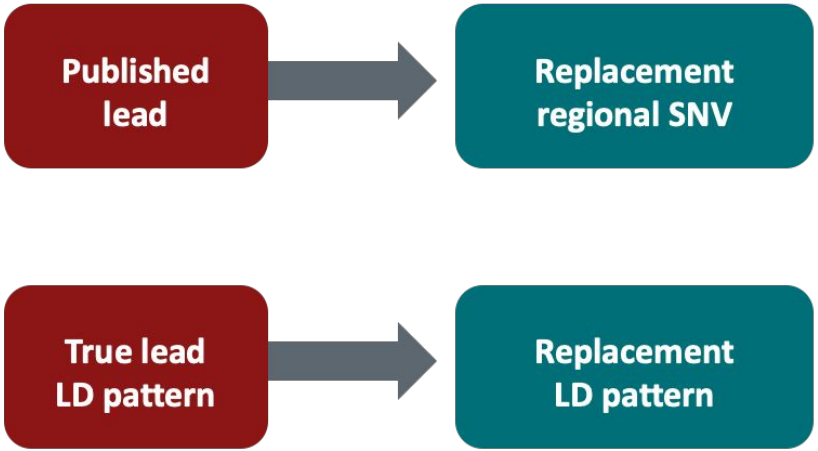
Take-home: A missing candidate cannot be recovered by the model.

One GWAS lead was not in our SNV panel. How could a missing lead change your list of candidate SVs?

ANSWER

The replacement SNV may have a different LD pattern from the missing lead. **It can miss SVs that tag the true lead or nominate SVs that tag only the replacement.** Report it as a regional replacement—not a validated proxy unless pairwise LD with the original lead is known.

Take-home: The anchor variant defines the LD question.



Most SVs in these tables are common in this cohort. How would you use SV allele frequencies from a healthy ageing cohort when you filter candidate SVs in a patient with a rare disease?

ANSWER

Deprioritize alleles whose frequency is incompatible with the disease model, considering **prevalence, penetrance, inheritance and ancestry**. Do not use one universal cutoff: an unaffected heterozygote does not exclude a recessive allele, and missing frequency is not zero.

Take-home: Frequency filters evidence; it does not diagnose pathogenicity.

Reference frequency is one piece of evidence



Missing / unmatched frequency: unknown, not AF = 0

Q&A – Notebook 2

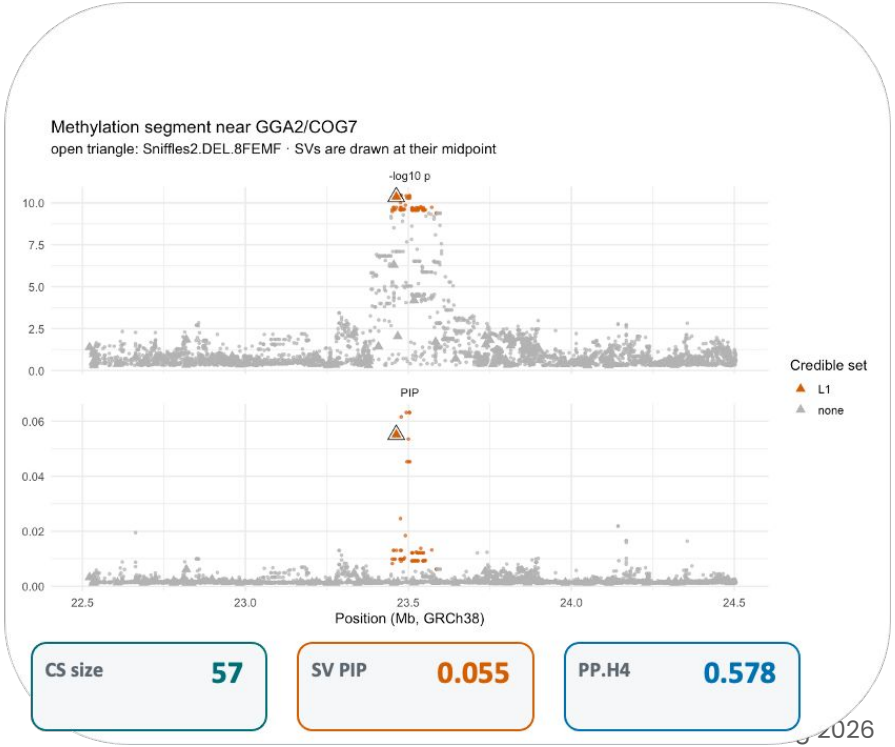
1. At COG7 the deletion is in the credible set but its PIP is small, and PP.H4 is only a little above 0.5. What does "in the credible set" mean, and what does it *not* mean? How confident are you in this locus?
2. Why is the MAPT credible set so much larger than the HP one? (Hint: compare the purity column and think about the inversion haplotype.)
3. Colocalization used SNVs only. How would you argue that the SV, and not a nearby SNV, is the shared causal variant?
4. Run the cell below, which fine-maps LRRC37A2 with `L = 1` (one causal signal only). Does the credible set change? Why or why not?

At COG7 the deletion is in the credible set but its PIP is small, and PP.H4 is only a little above 0.5. What does "in the credible set" mean, and what does it not mean? How confident are you in this locus?

ANSWER

Credible-set membership makes the deletion one plausible variant for the molecular-QTL signal—not a 95% causal variant. **PIP 0.055 gives weak variant-level support; PP.H4 0.578 gives only modest regional evidence for a shared AD–mQTL signal.** Prioritize it as plausible, not established.

Take-home: Credible set, PIP and PP.H4 answer different questions.



Why is the MAPT credible set so much larger than the HP one? (Hint: compare the purity column and think about the inversion haplotype.)

ANSWER

The inversion haplotype creates extensive LD, so thousands of variants carry nearly the same information and posterior support is spread across them. **High purity reflects strong correlation within the set—not high confidence in any one variant.** HP is more distinguishable.

Take-home: Strong association can coexist with poor variant resolution.

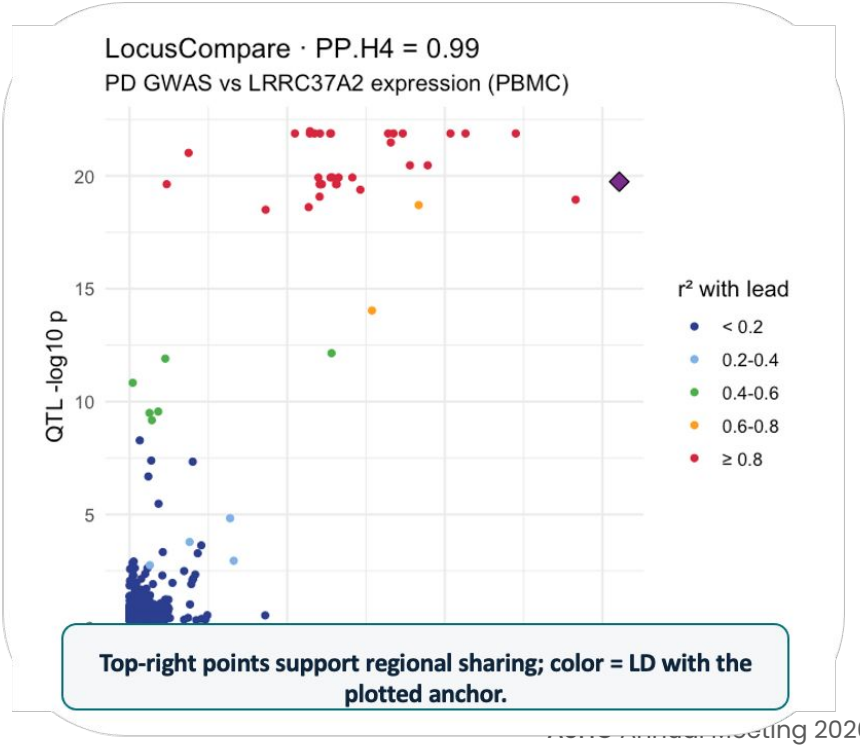


Colocalization used SNVs only. How would you argue that the SV, and not a nearby SNV, is the shared causal variant?

ANSWER

SNV-only coloc supports a shared regional GWAS–QTL signal; **it does not identify the SV as causal.** Distinguishing the SV may require direct or imputed disease association, joint fine-mapping or conditioning when LD permits, different LD patterns across populations, and functional evidence. If genotypes are nearly identical, the present data cannot resolve them.

Take-home: Regional sharing is not variant-level causality.



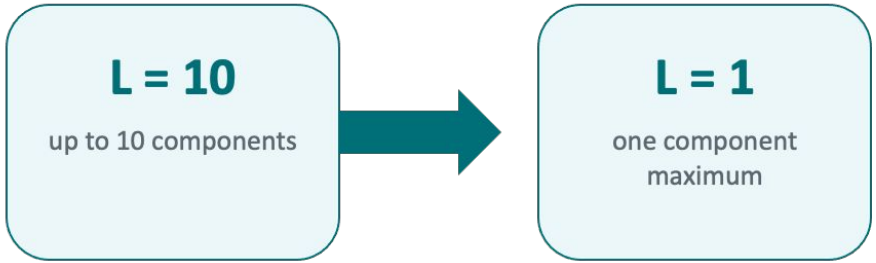
Run the cell below, which fine-maps LRRC37A2 with L = 1 (one causal signal only). Does the credible set change? Why?

ANSWER

The default L = 10 fit produced only one reportable signal, so limiting the model to one component leaves the main result unchanged. **L is an upper bound, not the number of true causal variants.** Stability does not prove that no additional effects exist.

Take-home: Sensitivity to L tests model stability—not biological signal count.

Same reported result



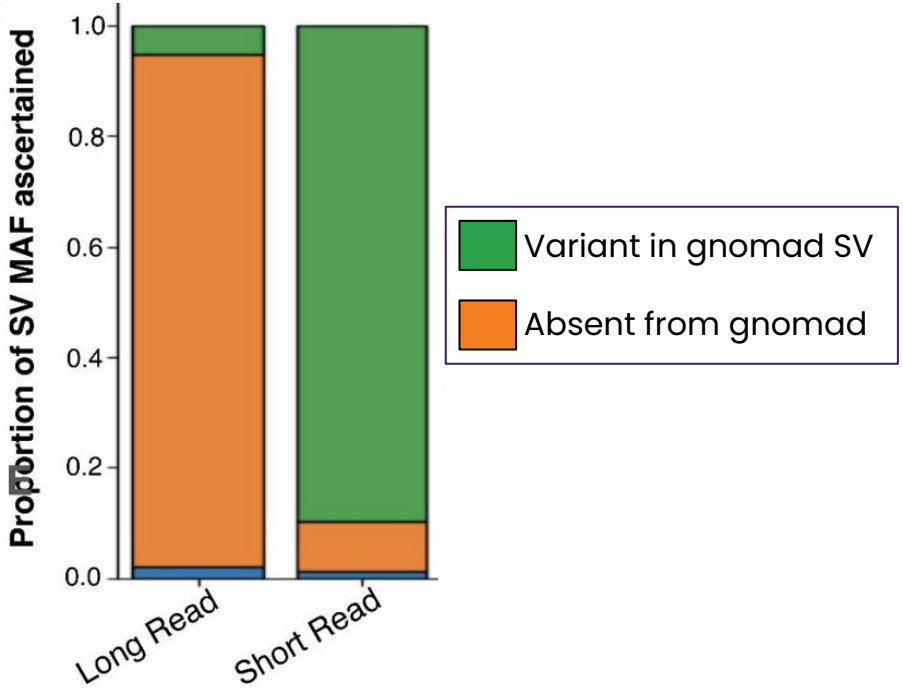
1 credible set • 2,656 variants
purity 0.885 • lead PIP 0.017

04

Rare structural variant prioritization

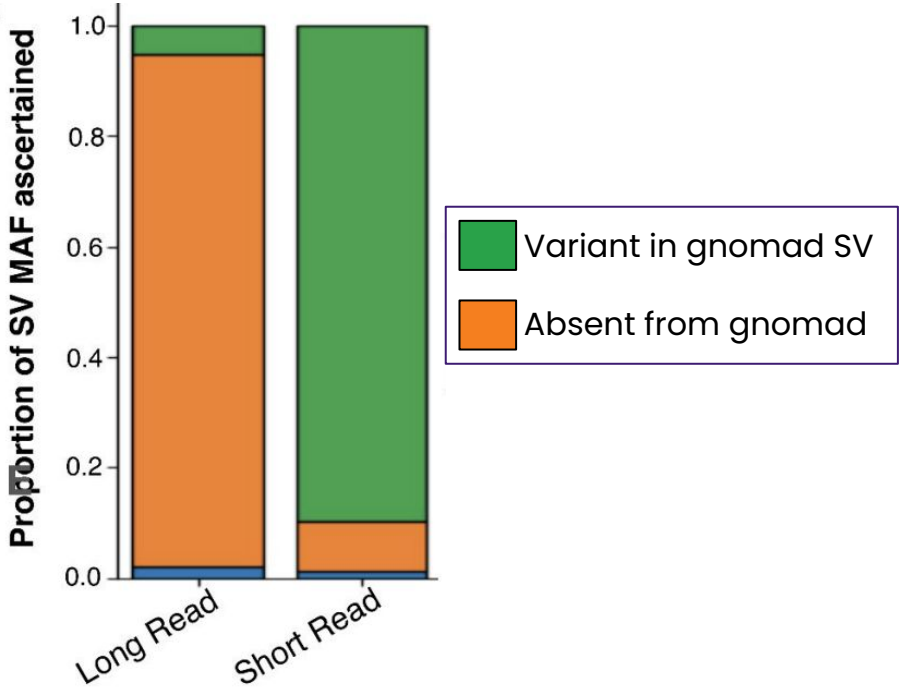
Nominating casual rare SVs with multi-omics outlier and machine learning

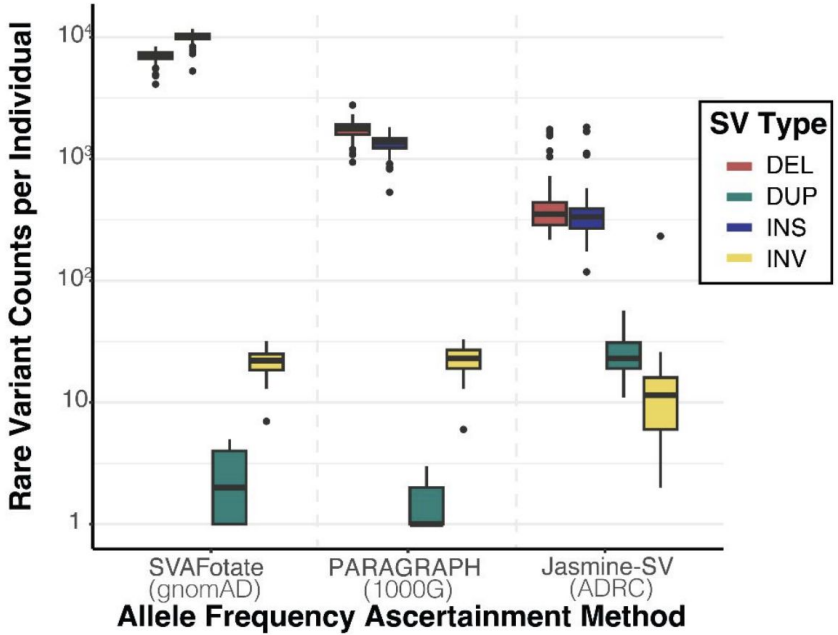
Defining rare SVs has been historically difficult because of low-ascertainment with short-read population resources



Jensen and Ni et al. 2025 Genome Research

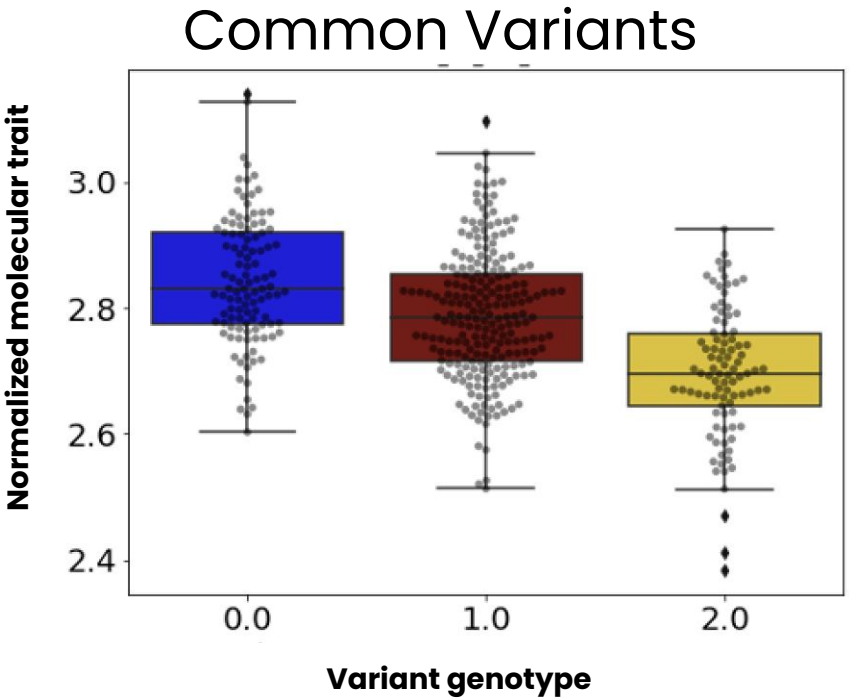
Long-read population resources are increasingly addressing this gap

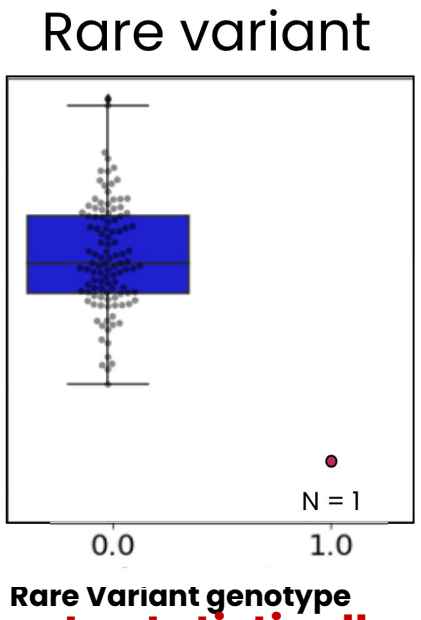




Jensen and Ni et al. 2025 Genome Research

Rare variants: where QTL analyses break down

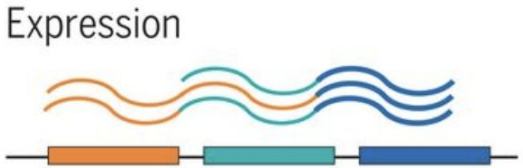


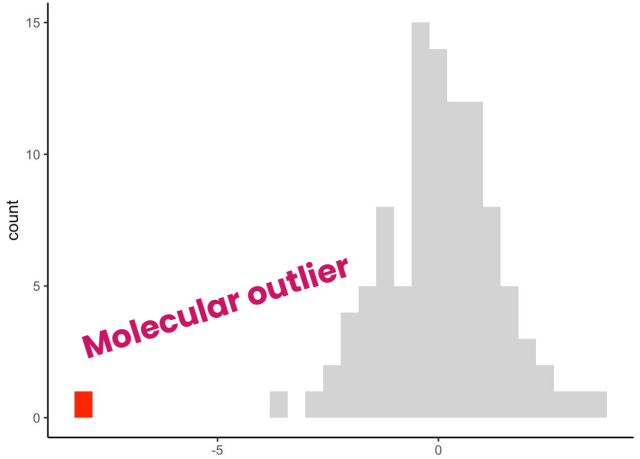


No way to statistically model

Switching approaches

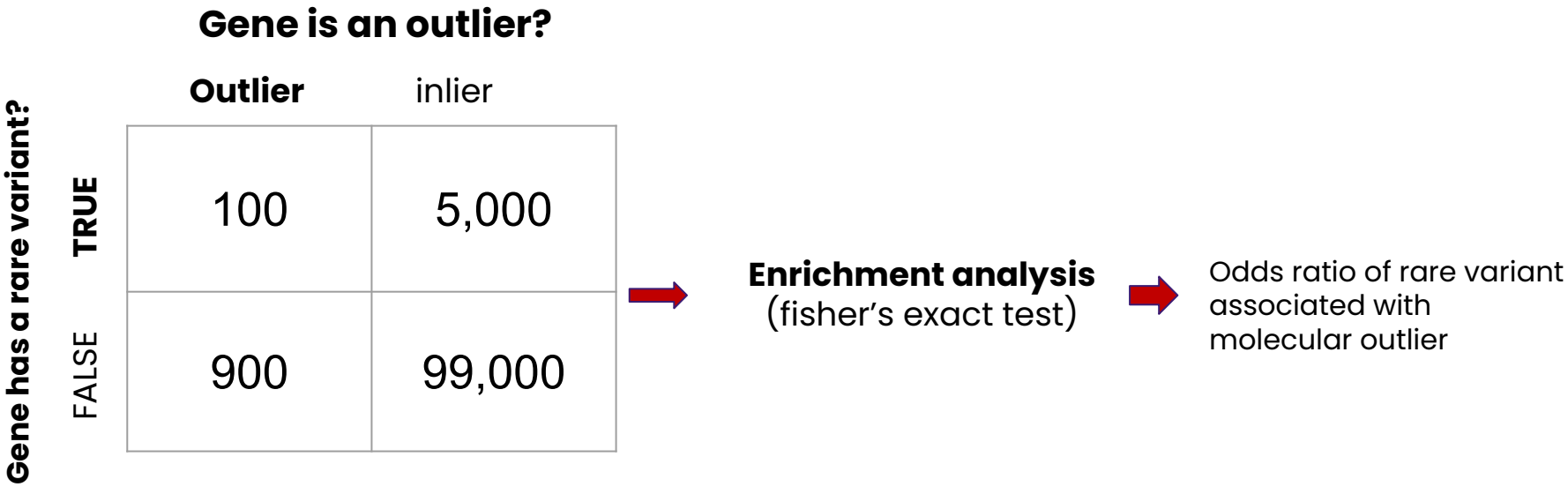
We can measure molecular outliers in distribution of each trait



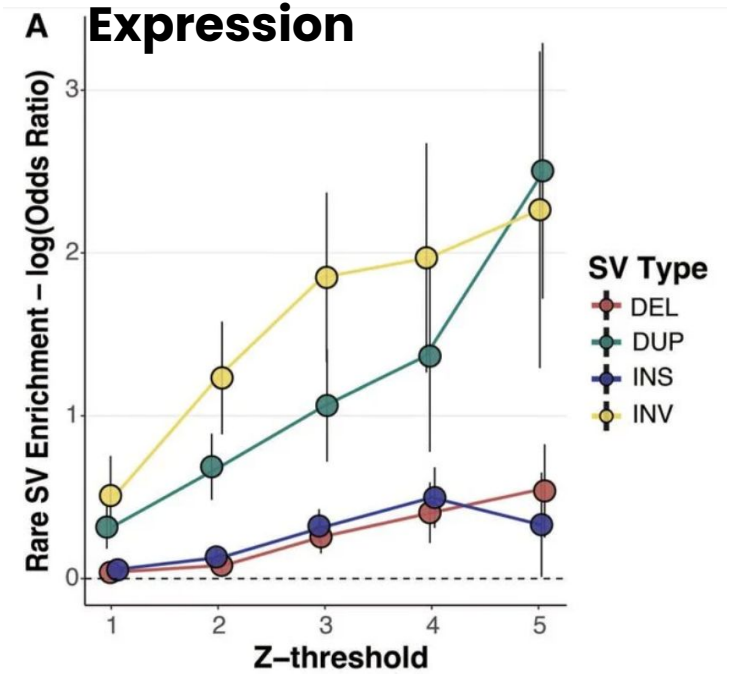


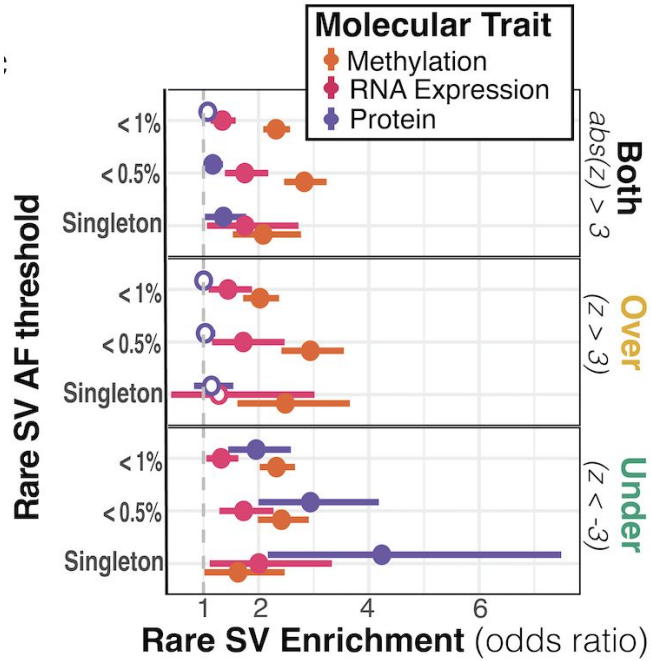
Expression outliers can detect aberrant transcription of genes

Across multiple events we can test enrichment of rare variants near these molecular outliers

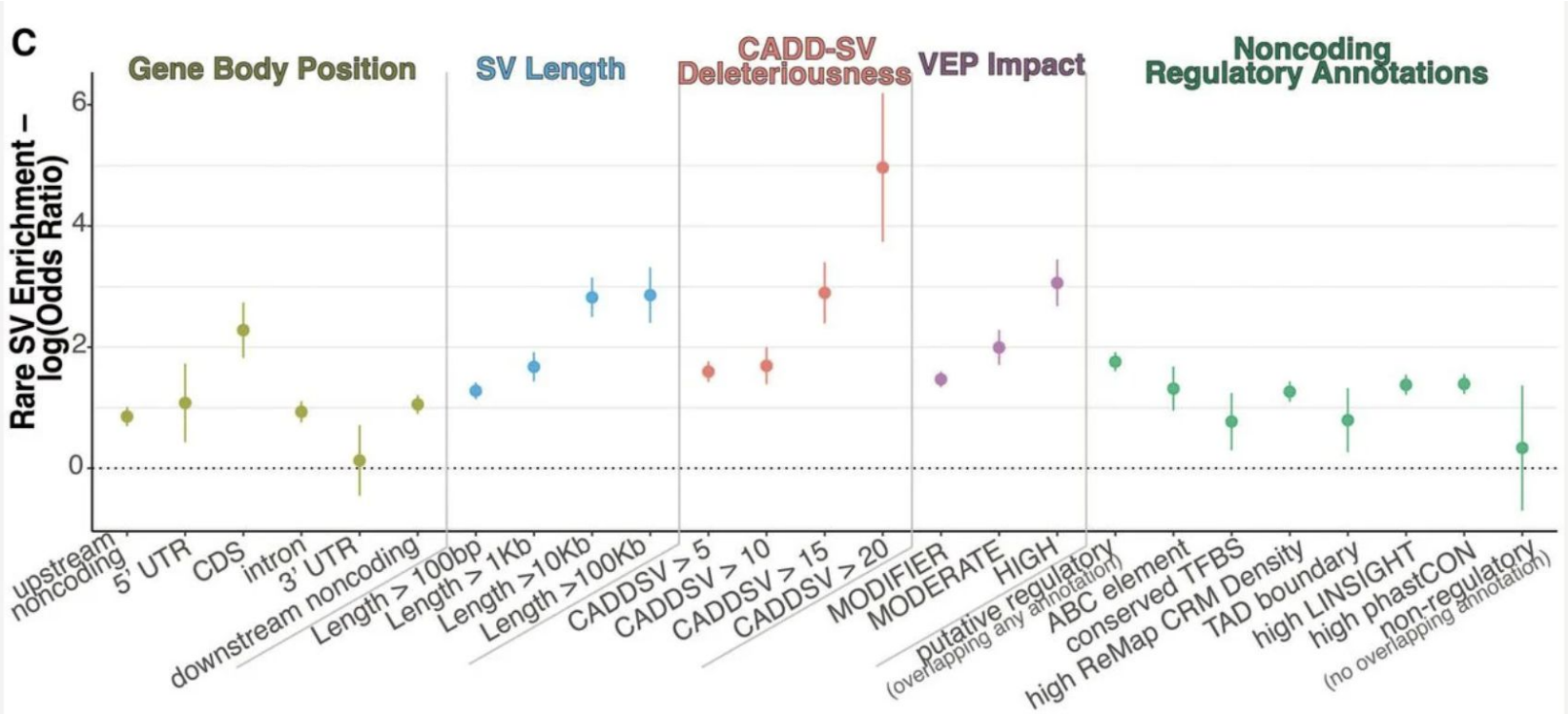


Rare SVs significantly enriched nearby molecular outliers across layers of regulation

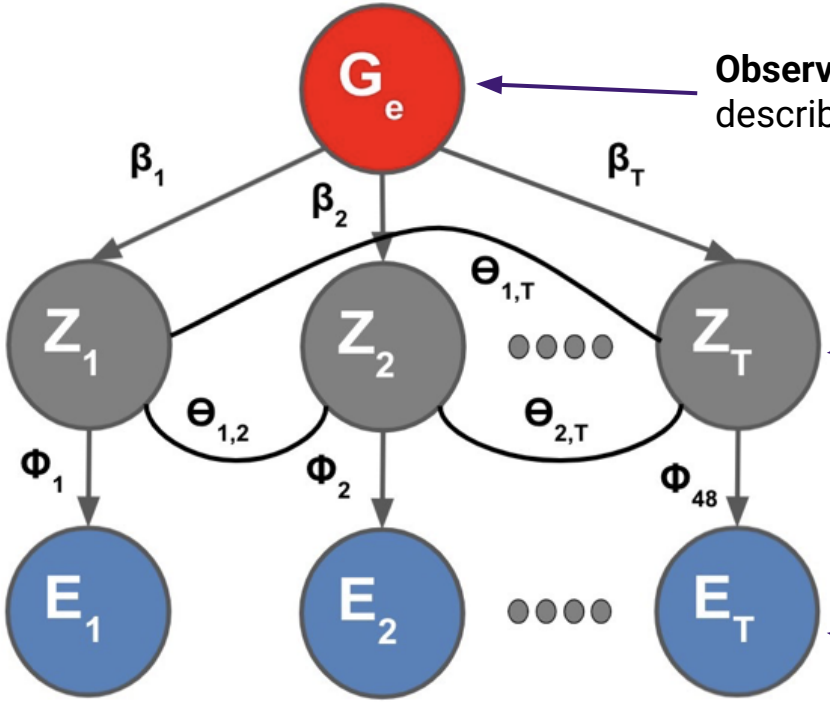




Genome annotations can inform what SVs are most likely to be functional



Watershed-SV: integrative variant prioritization with functional outlier signals + annotations



Observed: vector of genomic annotations describing SV's impact on nearby gene

Unobserved: functional status of variants to the nearby gene of interest

Observed: functional outlier signals

G_e category example	annotation example
summary score for SV	deleteriousness score	CADD, LINSIGHT, PhastCON, etc.
E_f genomic elements disrupted by SV	tissue-level regulatory elements	signal example ENCODE 25 states, multiple tissue resolutions for 49 GTEx tissues
	diverse functional signals	eOutlier, methylation outlier, spliceOutliers

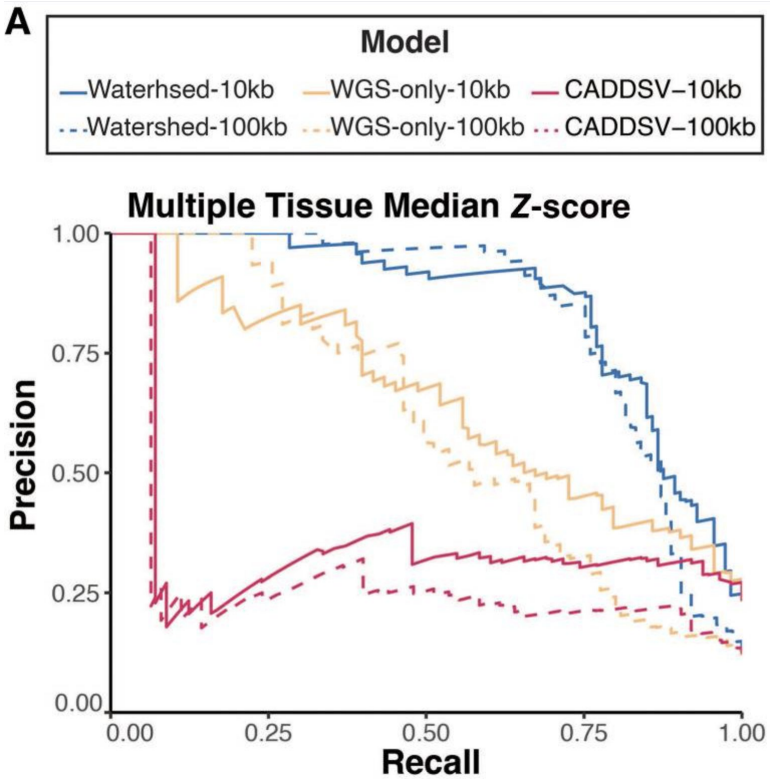
scores for prioritization:

$$P(Z_i | Z_{-i}, G_e, E)$$



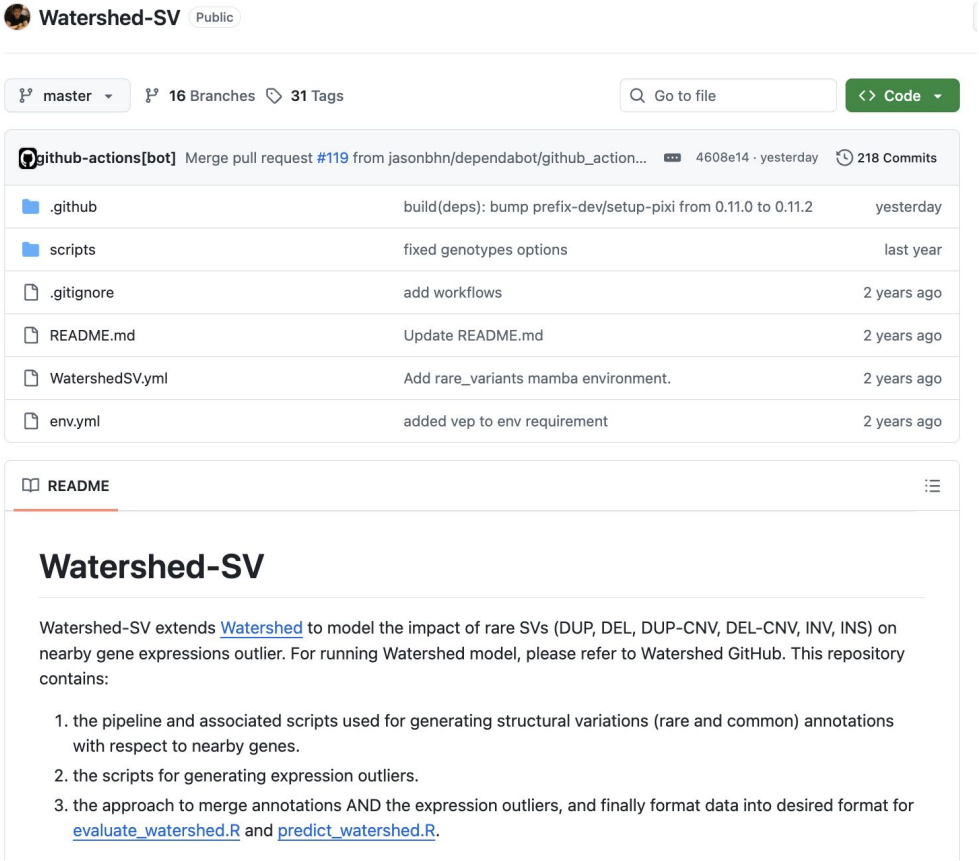
Bohan Ni, Johns Hopkins University

Benchmarked on GTEx data, integrating transcriptomics help Watershed-SV outperform competitors



Jensen and Ni et al. Genome Research 2025

https://github.com/jasonbhn/Watershed-SV



Watershed-SV Public

master · 16 Branches · 31 Tags · Go to file · Code

github-actions[bot] Merge pull request #119 from jasonbhn/dependabot/github_action... 4608e14 · yesterday · 218 Commits

.github	build(deps): bump prefix-dev/setup-pixi from 0.11.0 to 0.11.2	yesterday
scripts	fixed genotypes options	last year
.gitignore	add workflows	2 years ago
README.md	Update README.md	2 years ago
WatershedSV.yml	Add rare_variants mamba environment.	2 years ago
env.yml	added vep to env requirement	2 years ago

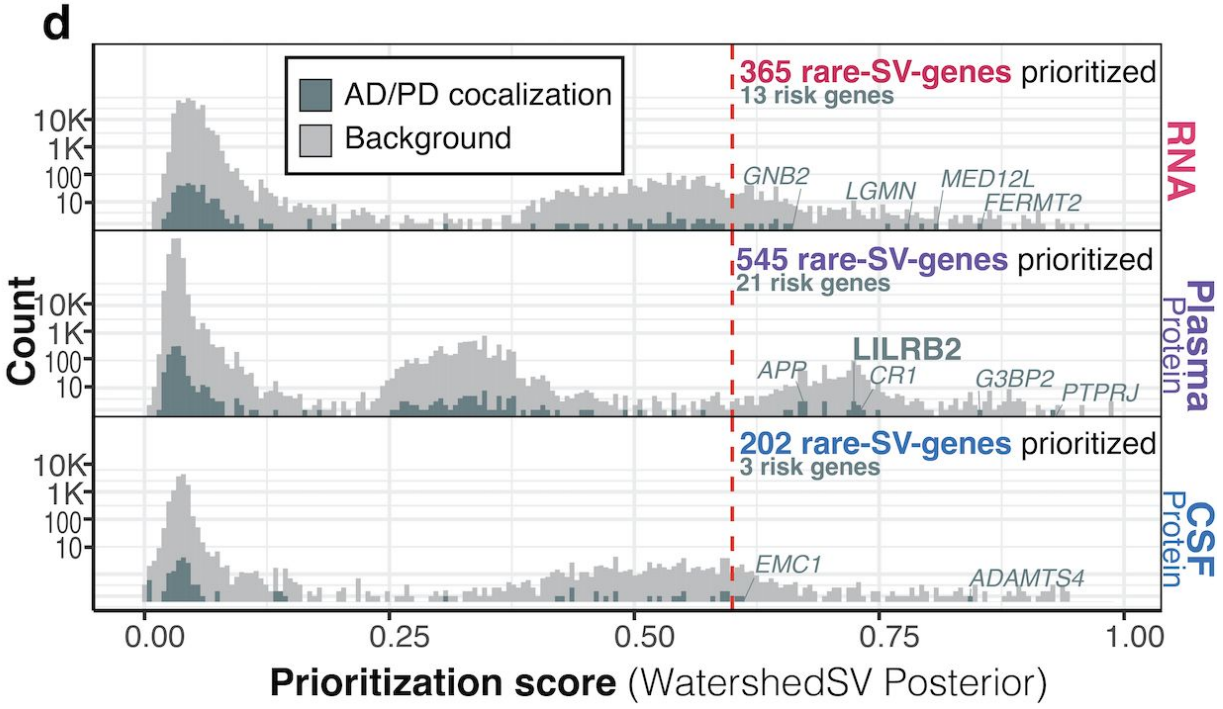
README

Watershed-SV

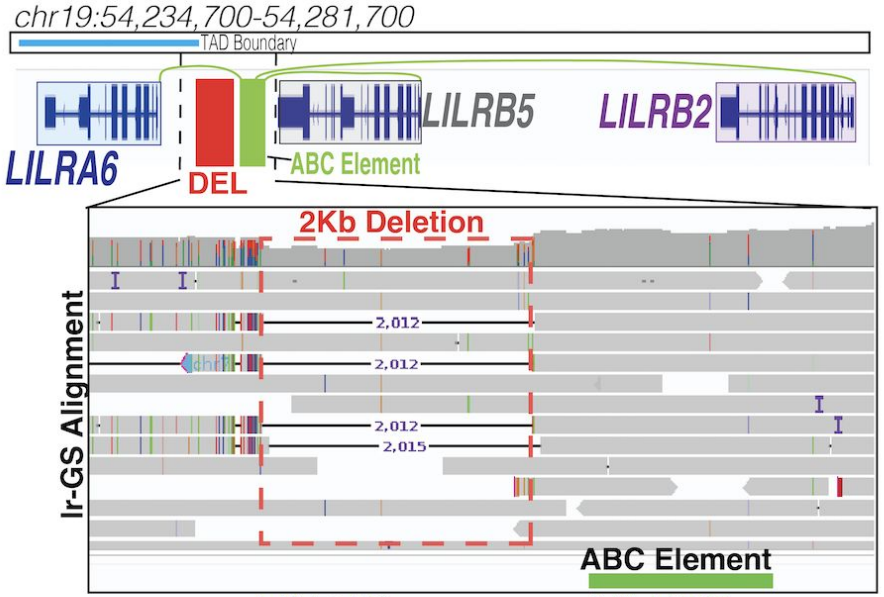
Watershed-SV extends Watershed to model the impact of rare SVs (DUP, DEL, DUP-CNV, DEL-CNV, INV, INS) on nearby gene expressions outlier. For running Watershed model, please refer to Watershed GitHub. This repository contains:

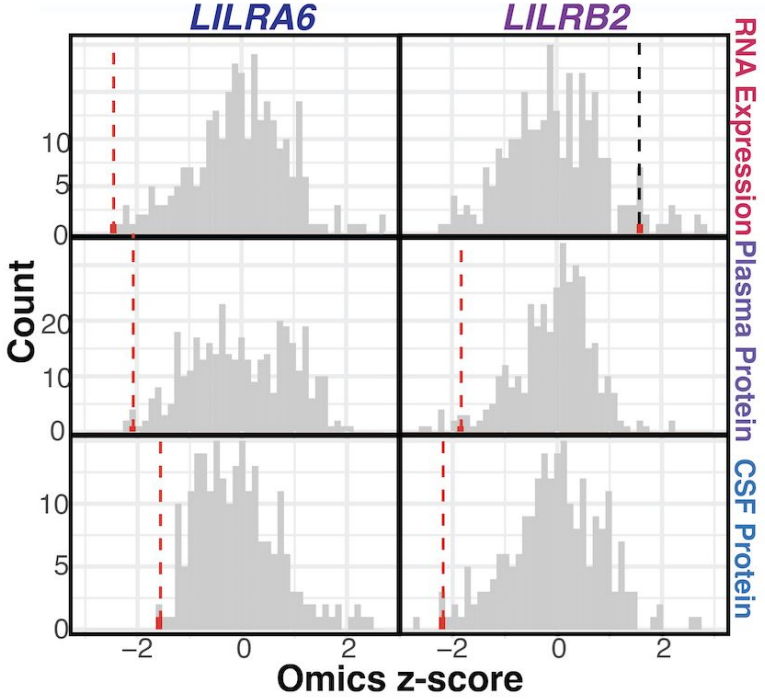
1. the pipeline and associated scripts used for generating structural variations (rare and common) annotations with respect to nearby genes.
2. the scripts for generating expression outliers.
3. the approach to merge annotations AND the expression outliers, and finally format data into desired format for evaluate_watershed.R and predict_watershed.R.

Overlapping watershed prioritized genes with coloc evidence helps prioritize rare SVs that could be involved in AD



WatershedSV helped identify non-coding variant linked to multi-omic outliers of risk genes





Workflow to prioritize rare variants

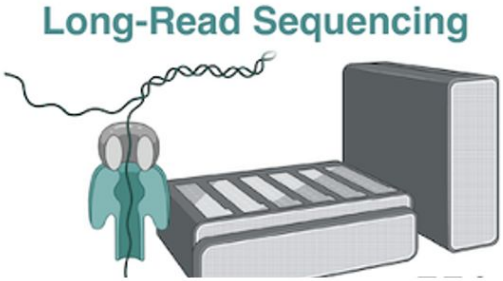
1. Annotate SVs with allele frequency using population resource (**Rare**)
2. Predict functional SVs with paired multi-omic data and WatershedSV (**High-impact**)
3. Overlap priortized genes to be phenotype relevant (**Phenotype-matched**)
    - Using colocalization genes for complex traits
    - Or phenotype matching with HPO terms etc

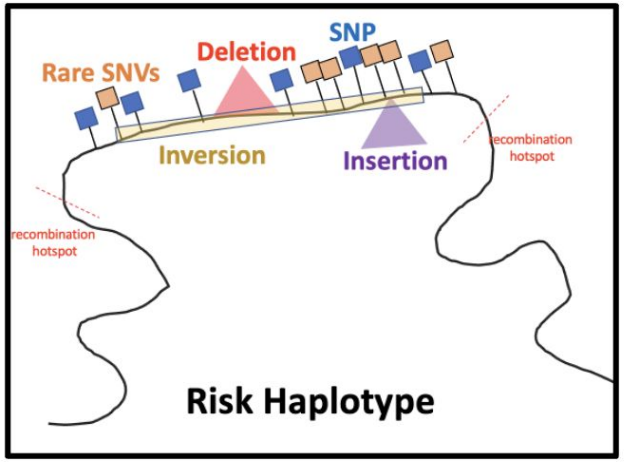
05

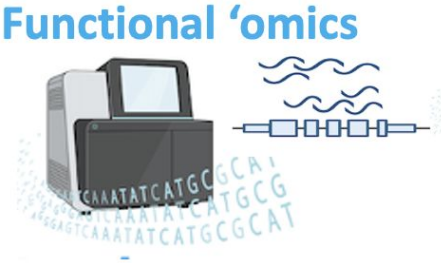
Conclusions and takeaways

Where does the field go from here?

We need matched multi-omics data for long-read sequencing to enable interpretation of SVs





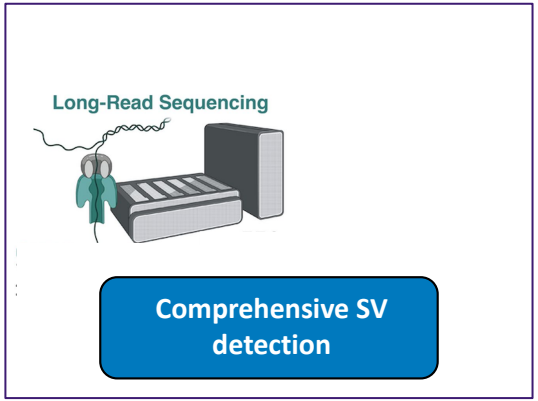


Integrating **Long-Read Sequencing** and **Multi-Omics** data allow us to pinpoint **functional SVs** driving disease risk

Conceptual workflow to find causal SVs from long-read sequencing in GWAS

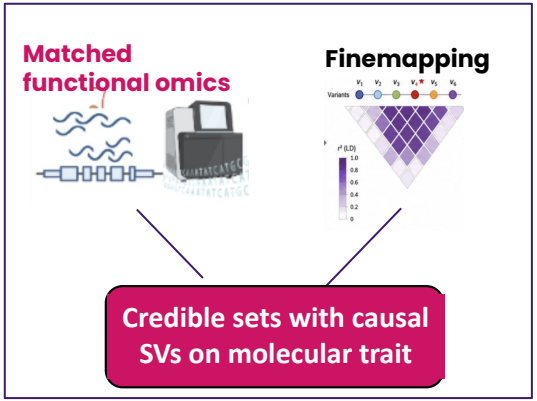
01

Long-read Cohort with Matched multi-omics



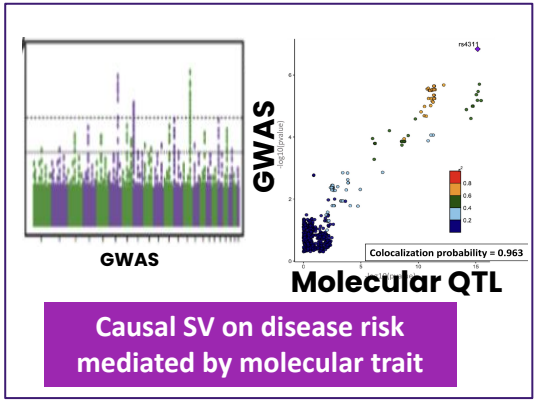
02

Joint statistical finemapping of SNPs + SVs with SuSiE



03

Coloc with GWAS over Proxy SNPs



We need matched multi-omics data for long-read sequencing to enable interpretation of SVs

- Large population cohorts are being sequencing with long-read technologies

- More effort should be done on generated matched omics data in across multiple .layers of regulation
  - RNA-seq
  - Proteomics
  - Methylation
  - More

- This data should be made easily accessible to the community

These matched multi-omic resources are becoming increasingly available

Resource	Platform	Long-read genomes	Tissue(s)	Available omics	Omics sample sizes
All of Us	PacBio HiFi	**14,521**	Blood; plasma	RNA-seq; proteomics	RNA: **8,980**; protein: **9,969**
1000 Chinese Pangenome (1KCP)	PacBio HiFi	**1,144**	Lymphoblastoid cell lines (LCLs)	RNA-seq; native DNA methylation	Matched RNA: **1,101**
Stanford ADRC/SAMS	ONT	**551**	Blood/PBMCs; plasma; CSF	Native DNA methylation; scRNA-seq; proteomics	RNA: **275**; plasma: **495**; CSF: **226**
NIH CARD — NABEC/HBCC	ONT	**351**	Brain cortex	Native DNA methylation; RNA-seq	Matched RNA: **281**
1000 Genomes + MAGE/GEUVADIS	ONT	**1,019**	LCLs	Native DNA methylation; RNA-seq	MAGE RNA: **731**

Context Matters!

Emerging consortia are also tackling this problem across a wide diversity of tissues and developmental states

dGTEX



Developmental GTEx

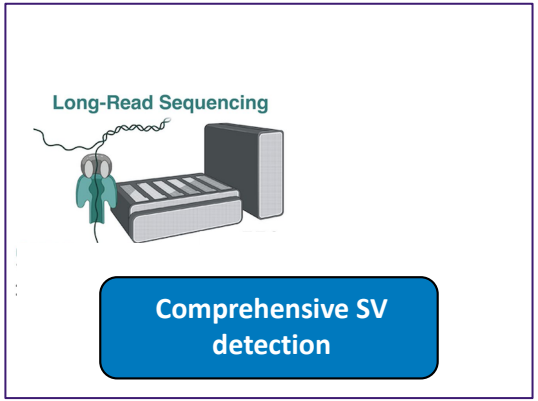
SMaHT

SMaHT
Somatic Mosaicism across Human Tissues

Conceptual workflow to find causal SVs from long-read sequencing in GWAS

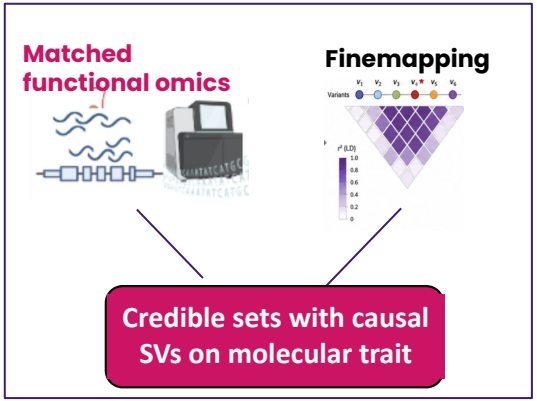
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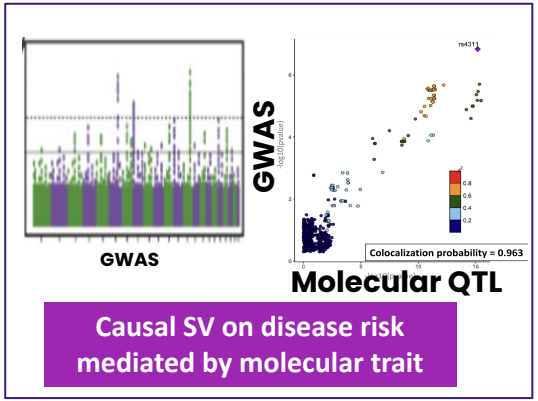
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Coloc with GWAS over Proxy SNPs



Thank You!

Shiny app

medRxiv
THE PREPRINT SERVER FOR HEALTH SCIENCES

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Tanner Jensen: tannerj@stanford.edu

https://github.com/Junkkkk/sv-finemapping-coloc-tutorial

https://qsushiny.shinyapps.io/adrc_lrs/